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NOTES

FOR

BSC 3RD YEAR

ZOLOGY

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ZOOLOGY PAPER III

Part 1 : MULTIPLE CHOICE QUESTIONS (MCQs)

- Q1. In many species of fishes, individuals swim together in large groups called:** (a) Pools (b) Groups (c) Schools (d) Animalcules **Answer - (c) Schools**
- Q2. Round dance is a common phenomenon among:** (a) Wasps (b) Termites (c) Butterflies (d) Honey bees **Answer - (d) Honey bees**
- Q3. Threat display is an act of:** (a) Courtship behaviour (b) Mating behaviour (c) Fighting behaviour (d) Social behaviour **Answer - (c) Fighting behaviour**
- Q4. Growth is a multifactorial process consisting of the following EXCEPT:** (a) Organ growth (b) Acellular growth (c) Sub-cellular growth (d) Cell growth **Answer - (b) Acellular growth**
- Q5. Organ which grows without cell division is found in:** (a) Coelenterates (b) Planarians (c) Annelids (d) Echinoderms **Answer - (b) Planarians**
- Q6. Which of the following have the highest ability for regeneration?** (a) Coelenterates (b) Planarians (c) Annelids (d) Echinoderms **Answer - (b) Planarians**
- Q7. Which of the following uses sonar system for the location of its prey?** (a) Fox (b) Bat (c) Wild cat (d) Firefly **Answer - (b) Bat**
- Q8. Which of the following does not make a nest of its own?** (a) Cuckoo (b) Sparrow (c) Crow (d) Parrot **Answer - (a) Cuckoo**
- Q9. Matching of body colour with its environment to avoid the predator is called:** (a) Camouflage (b) Mimicry (c) Warning coloration (d) Disguise **Answer - (a) Camouflage**
- Q10. Which of the following is NOT a defence mechanism evolved by a prey to avoid predators?** (a) Ejection of noxious chemicals (b) Possession of toxic hairs (c) Mimicry of inedible objects (d) Secretion of pheromones **Answer - (d) Secretion of pheromones**
- Q11. Those eggs which have a small amount of food reserve (yolk) are called:** (a) Telolecithal (b) Microlecithal (c) Mesolecithal (d) Macrolecithal **Answer - (b) Microlecithal**
- Q12. Endocytosis is also known as:** (a) Organ growth (b) Subcellular growth (c) Cell growth (d) Cellular ingestion **Answer - (d) Cellular ingestion**
- Q13. In a mammalian spermatozoon, mitochondria are located in the:** (a) Head (b) Middle piece (c) Tail (d) Not present at all **Answer - (b) Middle piece**
- Q14. The use of sonar system for locating prey is done in:** (a) Rats (b) Bats (c) Cats (d) Flies **Answer - (b) Bats**

PART 2: FILL IN THE BLANKS

- Q15. The opening at the apex of a gemmule is called _____. Answer - Micropyle**
- Q16. If the growth in an animal is in geometrical succession, the growth curve will be _____. Answer - Exponential (or Sigmoid)**
- Q17. _____ is a moulting hormone of insects. Answer - Ecdysone**
- Q18. Threat display is an act of _____ behaviour. Answer - Agonistic / Fighting**
- Q19. Polar bodies are formed during the process of _____. Answer - Oogenesis**
- Q20. The eggs which have a small amount of food reserve are called _____ eggs. Answer - Microlecithal**
- Q21. Development of an egg without fertilization is called _____. Answer -**

Parthenogenesis

Q22. Stereotyped behaviours are called _____ behaviour. Answer - Innate (or Instinctive)

Q23. Quick action of pheromones is designated as the _____ effect. Answer - Releaser

Q24. Building nest is an example of _____ behaviour. Answer - Innate / Instinctive

Q25. Fish that migrate from fresh water to sea water are called _____. Answer - Catadromous fish

Q26. Father of modern embryology is _____. Answer - Karl Ernst von Baer

Q27. The hormone responsible for moulting in insects is known as _____. Answer - Ecdysone

Q28. Leydig cells produce _____. Answer - Androgens (primarily Testosterone)

Q29. Metaplasia is also called _____. Answer - Cellular transformation

Q30. The separation of cuticle from the underlying epidermis in insects is known as _____. Answer - Apolysis

Q31. Sex attractants are known as _____. Answer - Pheromones

Q32. A fish which migrates from sea to fresh waters is called _____ fish. Answer - Anadromous

Q33. The cells which nourish the developing spermatozoa are known as _____ cells. Answer - Sertoli

Q34. The animal activities and behaviour related with distinct environmental frequencies are known as _____ activities. Answer - Circadian / Biological

Q35. Substances responsible for aging in animals are known as _____. Answer - Free radicals / Reactive Oxygen Species (ROS)

Q36. _____ is a phenomenon by which one type of cells (tissues) differentiate into another type of cells (tissues). Answer - Metaplasia

Q37. A developmental process in which morphogenesis gets out of control and ends up in a monster embryo is called _____. Answer - Teratogenesis

Q38. Malignant tumours found in connective tissue and muscles are known as _____. Answer - Sarcomas

Q39. Substances responsible for senescence in animals are known as _____. Answer - Free radicals

Q40. Word ethology is derived from Greek *ethos* meaning _____. Answer - Character or Habit

Q41. All animals that try to feed themselves primarily on other animals are called _____. Answer - Carnivores / Predators

Q42. Sex recognition and courtship are facilitated by the secretion of _____ in many fishes. Answer - Pheromones

Q43. Those animal activities and behaviours which are related to distinct environmental frequencies are known as _____. Answer - Biological rhythms

Q44. A frog tadpole during metamorphosis changes from ammonotelic to _____. Answer - Ureotelic

Q45. Vitelline membrane in mammal is called _____. Answer - Zona pellucida

Q46. Hippo and Rhino use _____ for chemical communication. Answer - Dung / Urine (Pheromones)

Q47. Killing own brother or sister is called _____. Answer - Fratricide

Q48. There were serious objections to the view point of _____ law of Haeckel. Answer - Biogenetic

Q49. The phenomenon by which one type of cell (tissue) differentiate into another type of

cell (tissue) is known as _____. **Answer - Metaplasia**

Q50. Strobilation is a kind of _____ reproduction found in Aurelia. Answer - Asexual

Q51. Deposition of yolk in an egg in the oviduct is called _____. Answer -

Vitellogenesis

Q52. Mobbing of crows around an eagle shows _____ behaviour. Answer - Agonistic / Mobbing

Q53. Matching of body colour with its environment to avoid the predator is called _____. Answer - Camouflage

Q54. One primary spermatocyte divides to form _____ primary spermatids. Answer - Four

Q55. Yolk in telolecithal egg is found in _____ pole. Answer - Vegetal

Q56. Polar bodies are formed during _____. Answer - Oogenesis

PART 3: SHORT QUESTIONS & DEFINITIONS

Q57. What is Metaplasia? Answer -

- **Definition:** Reversible replacement of one fully differentiated adult tissue type by another adult tissue type.
- **Significance:** Acts as an adaptive response to chronic physical or chemical stress.

Q58. What is Blastema? Answer -

- **Definition:** A mass of undifferentiated, pluripotent cells capable of growth and tissue regeneration.
- **Occurrence:** Formed at the site of organ/limb amputation in urodeles (like salamanders).

Q59. Mention the contribution of Karl von Frisch. Answer -

- **Discovery:** Austrian ethologist who decoded the communication system of honeybees.
- **Key Finding:** Explained how bees communicate food location via round and waggle dances.

Q60. What is Ad libitum sampling? Answer -

- **Definition:** An observational ethological method where all visible behavioral actions of subjects are recorded without formal rules or constraints.
- **Use:** Ideal for initial exploratory behavior studies.

Q61. What is Hilsa as a migratory fish? Answer -

- **Nature:** *Hilsa ilisha* is an anadromous migratory fish species.
- **Pattern:** Migrates from marine environments to freshwater rivers during breeding season to lay eggs.

Q62. What is Startle display? Answer -

- **Definition:** A sudden, striking defensive posture or color flash shown by prey animals upon confrontation.
- **Purpose:** Temporarily confuses or scares off potential predators to allow escape.

Q63. Define Learned Behaviour. Answer -

- **Definition:** Any permanent alteration in animal behavior resulting from individual past experience and learning.
- **Examples:** Classical conditioning, operant conditioning, and habituation.

Q64. What is Teratogenesis? Answer -

- **Definition:** The process leading to congenital developmental malformations or birth defects in an embryo.
- **Causes:** Triggered by physical, chemical, or biological agents called teratogens.

Q65. What is Capacitation? Answer -

- **Definition:** The final physiological maturation phase underwent by spermatozoa within the female reproductive tract.
- **Function:** Enables the sperm cell to penetrate and fertilize the egg.

Q66. What is Parthenogenesis? Answer -

- **Definition:** A form of asexual reproduction where an unfertilized egg develops into a complete individual.
- **Occurrence:** Natural in bees, wasps, aphids, and certain lizards.

Q67. What is Focal animal sampling? Answer -

- **Definition:** A quantitative sampling method focusing on observing a single individual for a specific time interval.
- **Function:** Used to accurately record specific behavioral frequencies and durations.

Q68. What is a Fixed Action Pattern (FAP)? Answer -

- **Definition:** An unlearned, highly stereotyped sequence of innate behavioral acts carried out to completion once triggered.
- **Trigger:** Activated by a specific environmental trigger called a sign stimulus.

Q69. What is a Sign stimulus? Answer -

- **Definition:** A external cue or signal that automatically triggers a specific Fixed Action Pattern in an animal.
- **Example:** Red belly of a male stickleback triggering aggression in rival males.

Q70. Define Filial Imprinting. Answer -

- **Definition:** Rapid, phase-sensitive innate learning occurring during a critical early window in young animals.
- **Example:** Newly hatched ducklings learning to identify and follow their mother.

Q71. Define Altruism. Answer -

- **Definition:** Behavioral actions performed by an individual that benefit another's survival at a personal cost.
- **Example:** Alarm calls given by worker honeybees or ground squirrels to warn kin.

Q72. Name two polyandrous birds. Answer -

- **Examples:** Jacanas and Phalaropes.
- **Pattern:** Females mate with multiple male partners while males incubate the eggs.

Q73. What is Neuroethology? Answer -

- **Definition:** The study of the neural and physiological pathways underlying natural animal behaviors.
- **Focus:** Links sensory processing in the brain directly to motor responses.

Q74. What is Embryogenesis? Answer -

- **Definition:** The developmental stage starting from zygote formation up to the establishment of major organ systems.
- **Processes:** Involves cleavage, blastulation, gastrulation, and organogenesis.

Q75. What is Cellular aging? Answer -

- **Definition:** The gradual loss of cell function, physiological capacity, and proliferative ability over time.
- **Causes:** Telomere shortening, free radical accumulation, and DNA damage.

Q76. What is Isolation Experiment? Answer -

- **Definition:** Raising an animal in complete isolation from species members from birth.
- **Purpose:** Differentiates genetically inherited (innate) behaviors from socially learned behaviors.

Q77. What is Induction? Answer -

- **Definition:** The embryonic mechanism where one cell group directs the development fate of adjacent target cells.
- **Example:** Neural tube induction by the chordamesoderm (notochord).

Q78. What is Testosterone? Answer -

- **Type:** Steroid androgen hormone secreted by interstitial Leydig cells in the testes.
- **Function:** Maintains male secondary sexual characters and regulates spermatogenesis.

Q79. What is Oogenesis? Answer -

- **Definition:** The differentiation process converting oogonia into functional, mature female haploid ova.
- **Yield:** Produces one mature ovum and small non-functional polar bodies per oocyte.

Q80. What is Appetitive behaviour? Answer -

- **Definition:** The initial, flexible, searching phase of a behavioral act driven by internal motivation.
- **Example:** Active searching for prey before the final attack and consumption.

Q81. What is Anadromous fish migration? Answer -

- **Definition:** Migration pattern where marine fishes swim upstream into freshwater rivers to spawn.
- **Examples:** Salmon, *Hilsa ilisha*.

Q82. What is Vitellogenesis? Answer -

- **Definition:** The synthesis and deposition process of yolk within the primary oocyte.
- **Hormonal control:** Governed mainly by estrogen in non-mammalian vertebrates.

Q83. What is Programmed cell death (Apoptosis)? Answer -

- **Definition:** Controlled, genetically regulated cell suicide required for physiological maintenance and development.
- **Role:** Removes webbed tissue between digits during vertebrate limb development.

Q84. What is Trial and Error learning? Answer -

- **Definition:** An operant conditioning process where an individual links a specific action with success or failure.
- **Mechanism:** Favorable outcomes reinforce actions, whereas unfavorable outcomes eliminate them.

Q85. What are Social hierarchies? Answer -

- **Definition:** Pecking-order or rank organization among individuals within a social group.
- **Function:** Reduces aggressive conflicts and organizes resource distribution (mates, food).

Q86. Mention the contribution of Konrad Lorenz. Answer -

- **Field:** Founding pioneer of modern Ethology.
- **Discovery:** Detailed early social bond development and **Imprinting** using geese.

Q87. What is Waggle dance? Answer -

- **Definition:** A figure-eight dance executed by returning worker bees inside the hive.
- **Function:** Communicates the precise direction and distance to a distant food source relative to the sun.

Q88. What is Hensen's node? Answer -

- **Definition:** The thickened organizer region located at the anterior tip of the primitive streak in birds and mammals.
- **Role:** Gives rise to the primary embryonic axis and notochord.

Q89. What is Homing behaviour? Answer -

- **Definition:** The innate capacity of an animal to navigate back to its original home site over unknown territory.

- **Examples:** Seen in homing pigeons, salmon, and sea turtles.

Q90. State the Biogenetic Law. Answer -

- **Proponent:** Ernst Haeckel ("Ontogeny recapitulates Phylogeny").
- **Statement:** An individual's embryonic developmental stages parallel the evolutionary developmental stages of its species.

Q91. What is an Innate Releasing Mechanism (IRM)? Answer -

- **Definition:** A neural network system in animals that recognizes sign stimuli and triggers an innate response.
- **Function:** Translates key environmental cues into immediate Fixed Action Patterns.

PART 4: DESCRIPTIVE / ESSAY QUESTIONS

Q92. Give a detailed account of teratogenesis and its inducing mechanisms. Answer -

- **Concept:** Study of congenital malformations arising during embryonic development.
- **Mechanisms:** Teratogens (radiations, drugs like Thalidomide, TORCH infections) disrupt cell proliferation, migration, gene expression, or induce excessive apoptosis, leading to severe structural birth defects.

Q93. Describe different forms of growth curves with suitable illustrations and their impact in animals' life cycle. Answer -

- **Types:** Sigmoid (S-shaped) curve showing lag, exponential, and stationary phases; and J-shaped curve showing rapid multiplication followed by a crash.
- **Impact:** Reflects environmental carrying capacity, nutrient availability, and life-history strategies (r-selection vs K-selection) of species.

Q94. Write short comparative accounts on: (a) Embryogenesis vs Blastogenesis (b) Vitellogenesis vs Spermatogenesis Answer -

- **(a) Embryogenesis vs Blastogenesis:** Embryogenesis arises sexually from a fertilized egg/zygote yielding genetic variation. Blastogenesis arises asexually from a multicellular blastema or bud, producing genetically identical clones.
- **(b) Vitellogenesis vs Spermatogenesis:** Vitellogenesis is the deposition of yolk in the female egg cell during oogenesis. Spermatogenesis is the complete developmental pathway transforming male germ cells into mature haploid spermatozoa.

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Q1. Define moulting. Discuss the role of hormones in moulting.

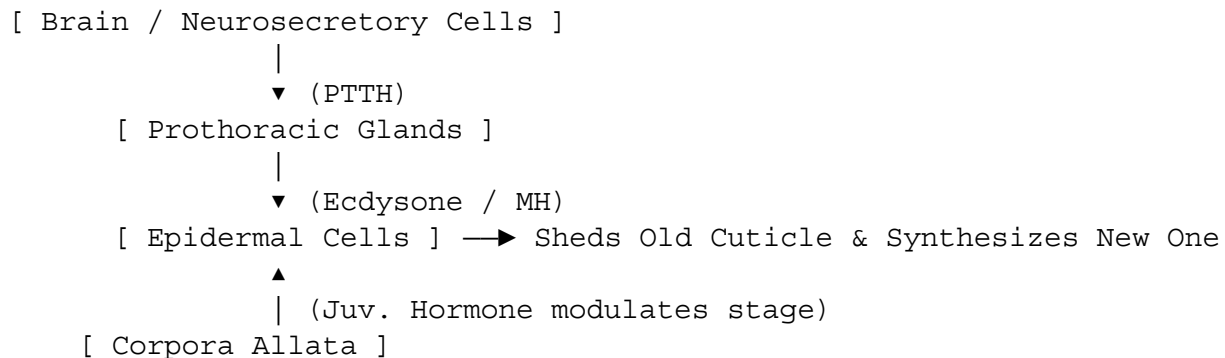
1. Definition

Moulting (or **ecdysis**) is the periodic shedding of the outer non-living exoskeleton or cuticle in arthropods (particularly insects and crustaceans), allowing the organism to increase in size and undergo development or metamorphosis.

Because the chitinous exoskeleton is rigid, growth cannot occur continuously. Shedding the old cuticle and synthesizing a larger one under hormonal control allows physical expansion.

2. Hormonal Control of Moulting in Insects

Moulting is regulated by an intricate neuroendocrine mechanism involving three main organs and their hormones:



A. Brain Hormone (Prothoracicotropic Hormone - PTTH)

- **Source:** Secreted by the neurosecretory cells of the *pars intercerebralis* in the protocerebrum and stored/released by the *corpora cardiaca*.
- **Function:** Environmental stimuli (light, temperature, stretch receptors) trigger PTTH release into the hemolymph. PTTH targets and stimulates the **prothoracic glands**.

B. Moulting Hormone (MH) / Ecdysone

- **Source:** Secreted by the **prothoracic glands**.
- **Function:**
 1. Ecdysone acts directly on epidermal cells to initiate apolysis (separation of the old cuticle from the epidermis).
 2. It stimulates cell division in the epidermis and secretion of moulting fluid containing chitinases and proteinases to digest the inner layer (endocuticle) of the old exoskeleton.
 3. It promotes the synthesis of the new epicuticle and exocuticle.

C. Juvenile Hormone (JH)

- **Source:** Secreted by the **corpora allata**.
- **Function:** Determines the *type* of moult:
 - **High JH + Ecdysone:** Keeps the insect in larval stage \rightarrow **Larval-to-Larval Molt**.
 - **Low JH + Ecdysone:** Triggers transformation \rightarrow **Larval-to-Pupal Molt**.
 - **Absence of JH + Ecdysone:** Leads to adult emergence \rightarrow **Pupal-to-Adult Molt**.

D. Eclosion Hormone (EH) & Bursicon

- **Eclosion Hormone:** Released by neurosecretory cells to initiate actual muscular movements and swallowing of air/water to crack open the old cuticle.
- **Bursicon:** Secreted immediately after shedding to induce **sclerotization** (hardening and darkening) of the newly formed cuticle.

3. Key Stages of the Moulting Process

1. **Apolysis:** Separation of epidermis from old cuticle.
2. **Secretion of Moulting Fluid:** Inactive fluid fills the space.
3. **New Epicuticle Formation:** Protects new tissue from digestive enzymes.
4. **Endocuticle Digestion:** Enzymes digest up to 90% of old endocuticle; nutrients are reabsorbed.
5. **Ecdysis:** Splitting and escaping from the old cuticle (*exuviae*).
6. **Sclerotization:** Hardening via bursicon action.

4. Conclusion

Moulting is an essential evolutionary adaptation in Ecdysozoans. It is tightly controlled by a cascade of neurohormones—primarily PTTH, Ecdysone, and JH—ensuring that growth and differentiation occur only at biologically appropriate times.

Q2. Mention types of asexual reproduction found in animals. Describe any one in detail.

1. Introduction

Asexual reproduction is a mechanism of reproduction involving a single parent, without the formation or fusion of gametes (syngamy). The resulting offspring are genetically identical to the parent and to each other, forming a **clone**. It is common among invertebrates (Protozoa, Porifera, Cnidaria, Platyhelminthes, Annelida) and lower chordates.

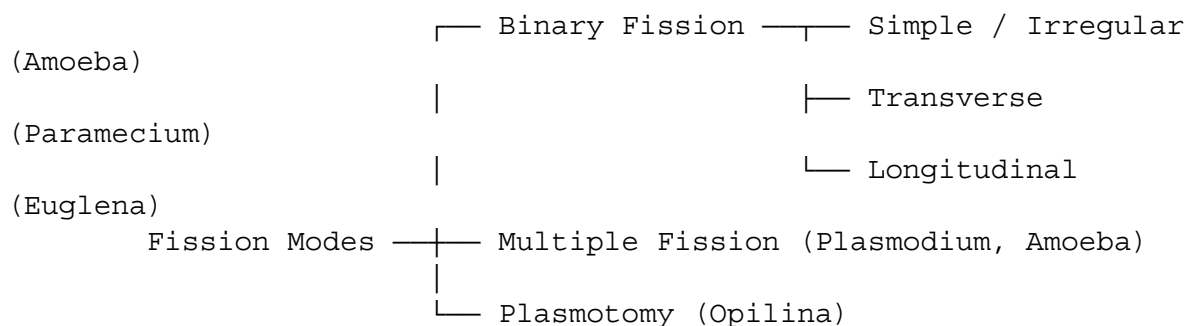
2. Types of Asexual Reproduction in Animals

Type	Mechanism	Examples
Fission	Division of parent body into two or more equal parts.	<i>Amoeba, Paramecium, Euglena</i>
Budding	Outgrowth (bud) develops on parent body, detaches, and forms new organism.	<i>Hydra, Sycon, Tunicates</i>
Fragmentation	Body breaks into multiple fragments; each regenerates missing parts.	<i>Planaria, Starfish, Syllis</i>
Gemmule Formation	Internal buds composed of archaeocytes enclosed in a protective capsule.	Freshwater sponges (<i>Spongilla</i>)
Sporulation	Spore formation inside a cyst during adverse conditions.	<i>Amoeba</i> , Sporozoans (<i>Plasmodium</i>)
Regeneration	Specialized asexual mode where broken body parts grow into full individuals.	<i>Planaria</i>

3. Detailed Account: Fission

Fission is the simplest and most widespread mode of asexual reproduction among unicellular and lower multicellular organisms. It involves nuclear division (mitosis or amitosis) followed by cytoplasmic division (cytokinesis).

Types of Fission:



A. Binary Fission

Parent individual divides into **two equal-sized daughter cells**.

1. **Simple / Irregular Binary Fission:** Cytokinesis can occur along any plane (e.g., *Amoeba*).
2. **Transverse Binary Fission:** Cytokinesis occurs perpendicular to the long axis.
 - *Example: Paramecium.* Micronucleus divides mitotically, macronucleus divides amitotically; constriction occurs transversely.
3. **Longitudinal Binary Fission:** Cytokinesis occurs along the longitudinal axis.
 - *Example: Euglena, Vorticella.* Division starts at the anterior end (flagellar pocket) and progresses posteriorly.

4. **Oblique Binary Fission:** Division plane is slanted (e.g., *Ceratium*).

B. Multiple Fission

Parent cell nucleus undergoes repeated mitotic divisions without immediate cytokinesis, producing numerous nuclei. Later, cytoplasm divides around each nucleus to yield many tiny daughter individuals simultaneously.

- **Examples:** *Plasmodium* (Schizogony/Sporogony), *Amoeba* during encystment.

C. Plasmotomy

Multinucleate protozoan divides into multinucleate daughter cells without prior nuclear division (e.g., *Opilina*, *Pelomyxa*).

4. Significance of Asexual Reproduction

- Rapid population expansion in favorable environments.
- No energy expenditure in finding a mate.
- Preserves successful genetic combinations intact.

Q3. What is growth? Explain various kinds of growth patterns in animals.

1. Definition

Growth is defined as an irreversible increase in the size, volume, dry weight, and cell number of an organism during its life cycle. It results from an excess of anabolic processes over catabolic processes and is accompanied by cellular differentiation and morphogenesis.

Mathematically, growth is expressed as:

$$\text{Growth} = \text{Anabolism} - \text{Catabolism} \quad (\text{where Anabolism} > \text{Catabolism})$$

2. Kinds of Growth Patterns in Animals

Growth in animals can be categorized based on cellular mechanisms, tissue localization, and overall body dynamics:

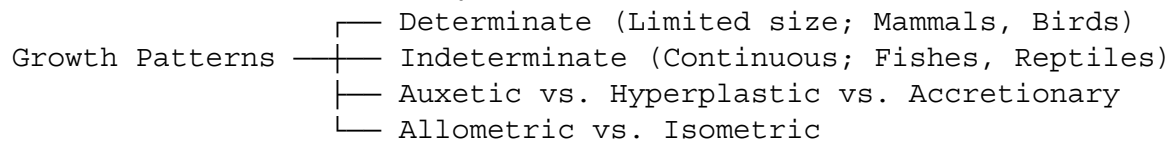
A. Classification Based on Cellular Mechanism

1. **Auxetic Growth:** Growth occurs strictly by an **increase in cell size** rather than cell number. The total cell count remains fixed throughout life (Eutely).
 - *Examples:* Nematodes (*Ascaris*), Rotifers, Tunicates.
2. **Hyperplastic Growth (Proliferative):** Growth occurs primarily by an **increase in cell number** via continuous cell division (mitosis).
 - *Examples:* Embryonic development of vertebrates, epithelial tissues.
3. **Accretionary Growth:** Growth driven by the accumulation of **extracellular materials** secreted by cells.

- *Examples:* Matrix deposition in bone and cartilage, shell growth in Molluscs.

B. Classification Based on Distribution/Symmetry

1. **Allometric Growth:** Different parts/organs of the body grow at **different rates** relative to the whole organism.
 - *Formula:* $y = bx^k$ (where y = organ size, x = body size, k = growth ratio).
 - *Example:* Human head grows slower than limbs during post-natal development; claw of male fiddler crab (*Uca*).
2. **Isometric Growth:** Body parts grow at **equal/proportional rates**, maintaining identical body proportions throughout development.
 - *Example:* Growth in many fishes and salamanders.



C. Classification Based on Limit of Growth

1. **Determinate Growth:** Organism grows to a characteristic adult size and then stops growing.
 - *Examples:* Insects, Birds, Mammals.
2. **Indeterminate Growth:** Organism continues to grow slowly throughout its lifetime as long as environmental conditions permit.
 - *Examples:* Invertebrates, Fishes, Reptiles, Amphibians.

D. Specialized Growth Patterns

- **Intermittent Growth:** Growth occurs in steps punctuated by shedding phases (moulting).
 - *Example:* Arthropods (Crustaceans and Insects).

3. Conclusion

Growth patterns reflect an evolutionary compromise between body design and functional requirements. Auxetic and determinate growth favor structural precision, while indeterminate and hyperplastic growth maximize environmental flexibility.

Q4. Define teratogenesis. Give an account of various types of terata with their causes of formation.

1. Definition

Teratogenesis (Greek: *Terato* = Monster; *Genesis* = Origin) is the production or development of congenital structural anomalies or malformations in an embryo or fetus. The agents that induce such abnormal developmental changes are called **teratogens**.

2. Types of Terata (Congenital Abnormalities)

Terata are broadly classified based on structural involvement into the following main categories:

	└─ Dysraphism (Spina Bifida, Anencephaly)
	└─ Duplication / Fusion (Conjoined twins, Cyclopia)
Types of Terata	└─ Agenesis / Aplasia (Phocomelia, Amelia)
	└─ Hypoplasia / Hyperplasia (Microcephaly, Polydactyly)
	└─ Atresia / Heterotopia (Ectopia cordis)

A. Neural Tube Defect Terata (Dysraphism)

Failure of the neural tube to close during early embryogenesis.

- **Anencephaly:** Absence of major portions of the brain and cranial vault.
- **Spina Bifida:** Incomplete closure of the vertebral column, leading to herniation of the spinal cord/meninges.

B. Limb and Skeletal Terata

- **Amelia:** Complete absence of one or more limbs.
- **Phocomelia:** "Seal-like limbs" where long bones are missing or severely reduced, and hands/feet attach near the trunk.
- **Polydactyly:** Presence of extra fingers or toes.
- **Syndactyly:** Fusion of digits.

C. Craniofacial Terata

- **Cleft Lip & Cleft Palate:** Non-fusion of embryonic facial processes.
- **Cyclopia:** Single central eye orbit due to failure of the forebrain to divide into two hemispheres (holoprosencephaly).

D. Duplication Anomalies

- **Conjoined Twins (Siamese Twins):** Incomplete separation of monozygotic twins (e.g., Thoracopagus, Dicephalus).

3. Causes of Teratogenesis (Teratogenic Agents)

Teratogenic causes are broadly categorized as **Genetic**, **Environmental**, or **Multifactorial**:

	└─ Genetic Factors (Chromosomal aberration / Gene mutation)
Causes of Teratogenesis	└─ Physical Agents (Radiation, Hyperthermia)
	└─ Chemical & Drug Agents (Thalidomide, Alcohol)
	└─ Infectious Pathogens (TORCH complex)
	└─ Maternal Metabolic Factors (Diabetes,

Malnutrition)

A. Genetic Factors

1. **Chromosomal Aberrations:**
 - Trisomy 21 → Down Syndrome (facial deformities, cardiac defects).
 - Monosomy X → Turner Syndrome.
2. **Gene Mutations:** Single-gene mutations causing achondroplasia (dwarfism) or osteogenesis imperfecta.

B. Environmental / External Teratogens

1. **Chemicals and Pharmaceuticals:**
 - **Thalidomide:** Prescribed in the 1950s for morning sickness; caused severe **phocomelia**.
 - **Alcohol:** Fetal Alcohol Spectrum Disorder (FASD) causing microcephaly, growth retardation, and facial anomalies.
 - **Retinoic Acid (Vitamin A derivatives):** Causes craniofacial and cardiovascular defects.
2. **Infectious Agents (TORCH Group):**
 - **Toxoplasma, Other (Syphilis), Rubella, Cytomegalovirus, Herpes simplex.**
 - *Rubella virus* during the 1st trimester causes microcephaly, cataracts, and congenital heart defects.
3. **Physical Agents:**
 - **Ionizing Radiation (X-rays, gamma rays):** Direct DNA damage leading to microcephaly and skeletal abnormalities.
 - **Hyperthermia:** High maternal fever during neural tube closure leads to spina bifida.
4. **Maternal Metabolic Factors:**
 - Uncontrolled maternal diabetes leads to macrosomia and neural defects.
 - Folate deficiency causes neural tube defects.

4. Principles of Teratogenesis (Wilson's Laws)

- **Critical Period:** Susceptibility to teratogenesis depends on the developmental stage at exposure (the **organogenesis phase**, weeks 3–8 in humans, is most sensitive).
- **Dose-Response Relationship:** Risk and severity increase with higher exposure levels.
- **Genotype Sensitivity:** Susceptibility depends on the genetic constitution of the mother and embryo.

Q5. Differentiate between morphallactic and epimorphic regeneration.

1. Introduction

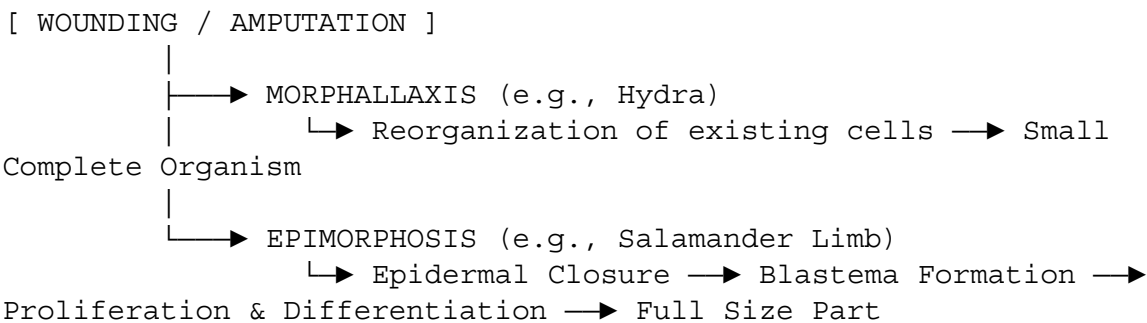
Regeneration is the capacity of an organism to restore or replace lost, damaged, or amputated body parts. In 1901, **Thomas Hunt Morgan** classified regenerative processes into two primary

modes: **Morphallaxis** and **Epimorphosis**.

2. Tabular Comparison

Characteristic	Morphallactic Regeneration (Morphallaxis)	Epimorphic Regeneration (Epimorphosis)
Definition	Reconstruction of the entire organism or organ mainly by re patterning existing tissues with minimal cell proliferation.	Restoration of a lost structure primarily through active cell proliferation and formation of a blastema.
Blastema Formation	Absent. No localized mass of undifferentiated proliferating cells is formed.	Present. A specialized mass of undifferentiated cells (blastema) forms at the wound site.
Cell Proliferation	Extremely limited or non-existent.	High rate of local cell proliferation (mitosis).
Original Body Size	Restored organism is initially much smaller than the original parent organism.	Regenerated organ/structure grows to the original size of the missing part.
Mechanism	Dedifferentiation and transdifferentiation of existing somatic cells across body axes.	Epidermal wound healing → Dedifferentiation → Blastema formation → Redifferentiation.
Involvement of Stem Cells	Uses resident pluripotent/totipotent stem cells spread through tissues (e.g., Neoblasts).	Uses dedifferentiated local tissue cells or lineage-restricted progenitor cells.
Classic Examples	<i>Hydra</i> , <i>Planaria</i> (whole-body regeneration from small fragments).	Amphibian limb regeneration (Axolotl/Salamander), Lizard tail regeneration.
Energy Source	Depends on internal structural reorganization and cell turnover.	Requires significant metabolic energy for sustained cell division.

3. Diagrammatic Overview of Pathways



4. Conclusion

While **Morphallaxis** prioritizes quick spatial repatterning and scale reduction without requiring cell division (characteristic of lower invertebrates), **Epimorphosis** relies on localized blastemal cell division to achieve true structural restoration to full scale (characteristic of higher invertebrates and vertebrates).

Q6. What is metamorphosis? Describe the hormonal control of metamorphosis in insects.

1. Definition of Metamorphosis

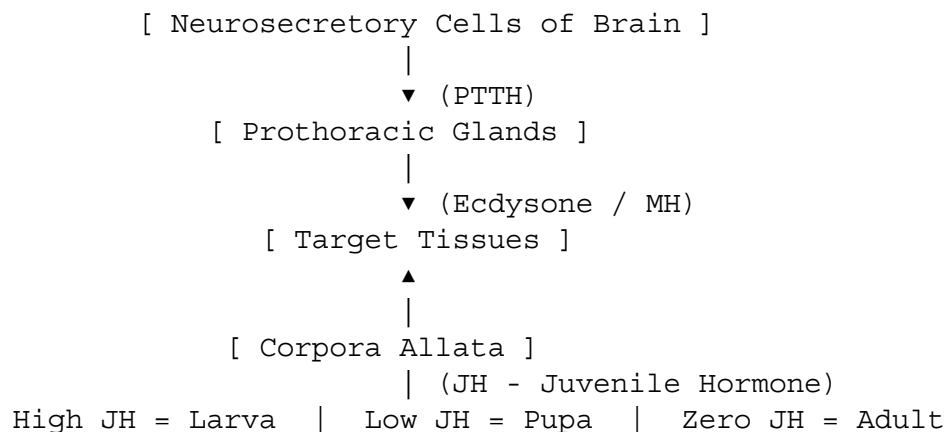
Metamorphosis (Greek: *Meta* = Change; *Morphe* = Form) is a post-embryonic developmental process in which an animal undergoes a rapid and radical transformation in structure, physiology, and behavior from a juvenile larval form to a sexually mature adult (*imago*). It is widespread in Insects, Amphibians, Molluscs, Echinoderms, and Tunicates.

2. Types of Metamorphosis in Insects

1. **Ametabolous Development:** Primitive insects with no metamorphosis; juvenile resembles adult except for size and sex maturity (e.g., *Lepisma* / Silverfish).
2. **Hemimetabolous Development (Incomplete):** Gradual transformation via nymphal stages. Nymphs lack wings and mature reproductive organs (e.g., Cockroach, Grasshopper). $\text{Egg} \rightarrow \text{Nymph} \rightarrow \text{Adult}$
3. **Holometabolous Development (Complete):** Dramatic transformation passing through a specialized pupal stage (e.g., Butterflies, Beetles, Flies). $\text{Egg} \rightarrow \text{Larva} \rightarrow \text{Pupa} \rightarrow \text{Adult}$

3. Hormonal Control of Insect Metamorphosis

Insect metamorphosis is governed by three primary hormones operating in a precise quantitative balance:



A. Prothoracicotropic Hormone (PTTH)

- **Origin:** Secreted by brain neurosecretory cells, stored and released by *corpora cardiaca*.
- **Role:** Activates the prothoracic gland to synthesize and release **Ecdysone**.

B. Ecdysone (Moulting Hormone)

- **Origin:** Secreted by **prothoracic glands** (located in the thorax).
- **Role:** Acts on target gene loci in cell nuclei to induce epidermal apolysis, cuticle secretion, and gene expression necessary for differentiation into pupal/adult structures.

C. Juvenile Hormone (JH)

- **Origin:** Secreted by the paired neuroendocrine glands, **corpora allata**.
- **Role:** Acts as a "**status quo**" hormone. It allows moulting to proceed while inhibiting the expression of adult-specific genes.
 - **High Titer of JH + Ecdysone:** Keeps the insect in larval form (Larva → Larva).
 - **Low Titer of JH + Ecdysone:** Allows commitment to pupal differentiation (Larva → Pupa).
 - **Absence/Trace Titer of JH + Ecdysone:** Triggers adult differentiation (Pupa → Adult/Imago).

D. Hormones Regulating Eclosion & Hardening

- **Eclosion Hormone (EH):** Released by brain cells to trigger stereotypic motor behavior that breaks the pupal case.
- **Bursicon:** Released shortly after emergence to direct wing expansion, tanning, and sclerotization (hardening) of the adult exoskeleton.

4. Genetic Mechanism (Puffing Pattern)

Ecdysone directly activates specific genes on polytene chromosomes (notably in Diptera like *Chironomus*), causing visible **chromosomal puffs** (Balbiani rings) representing active transcription of messenger RNA required for pupal and adult protein synthesis.

5. Conclusion

Insect metamorphosis is a finely tuned system where **Ecdysone provides the drive to moult and transform**, while **Juvenile Hormone functions as the developmental brake**, ensuring that transformation occurs only after complete larval growth.

Q7. Write an essay on vitellogenesis.

1. Introduction

Vitellogenesis (Latin: *Vita* = Life/Yolk; *Genesis* = Origin) is the biological process of yolk

2. Nature and Composition of Yolk

- **Proteins:** Primarily **Vitellogenin** (a lipophosphoprotein complex). In the egg, it is cleaved into **Lipovitellin** and **Phosvitin**.
- **Lipids:** Phospholipids, neutral fats, and cholesterol.
- **Carbohydrates:** Glycogen and mucopolysaccharides.
- **Inorganic Constituents:** Calcium, iron, and inorganic phosphorus.

(Estrogen)



1. **Hormonal Trigger:** Hypothalamic GnRH stimulates the pituitary to release **Gonadotropins (FSH/LH)**.
2. **Ovarian Secretion:** FSH acts on ovarian follicle cells (granulosa cells) to release **Estrogen** (Estradiol-17 β).
3. **Hepatic Synthesis:** Estrogen travels via blood to the **Liver**, binding to nuclear estrogen receptors in hepatocytes. This activates transcription of the **vitellogenin gene**.
4. **Secretion & Transport:** Liver cells synthesize **Vitellogenin (VTG)** and secrete it into the bloodstream.
5. **Receptor-Mediated Uptake:** VTG reaches the ovary, passes through capillary walls and follicle cell gaps, and binds to **vitellogenin receptors** on the oocyte membrane (oolemma).
6. **Endocytosis & Processing:** VTG is internalized via pinocytotic/clathrin-coated vesicles. Inside the oocyte cytoplasm, vesicles fuse with lysosomal processing enzymes, cleaving VTG into **Phosvitin** (protein/phosphorus-rich) and **Lipovitellin** (lipid-rich), which aggregate to form insoluble **yolk platelets**.

5. Role of Yolk in Embryogenesis

- Provides metabolic energy via beta-oxidation of fats.
- Supplies amino acids for protein synthesis during rapid cell division in blastulation and gastrulation.
- Influences the **cleavage pattern** during early development (e.g., Holoblastic in microlecithal eggs vs. Meroblastic in macrolecithal eggs).

6. Conclusion

Vitellogenesis is a pivotal step in reproduction. It integrates nervous, endocrine, metabolic, and cellular systems to equip the egg with the maternal investment required to support early embryonic life.

Q8. Give an account of oogenesis and explain how it differs from spermatogenesis?

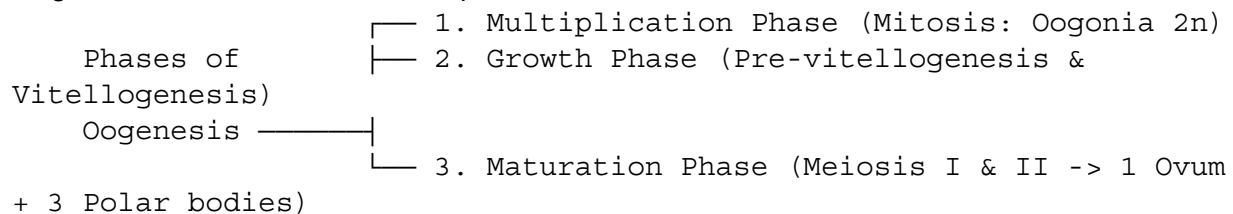
1. Introduction

Gametogenesis is the biological process by which diploid (2n) germ cells undergo meiosis to form haploid (1n) functional gametes.

- **Spermatogenesis:** Production of male gametes (spermatozoa) in the testes.
- **Oogenesis:** Production of female gametes (ova/eggs) in the ovaries.

2. Process of Oogenesis

Oogenesis occurs in three distinct phases:



A. Multiplication Phase

- Primary germ cells ($2n$) in the germinal epithelium of the ovary undergo repeated mitotic divisions to form millions of **Oogonia** ($2n$).
- Occurs primarily during embryonic development in humans/vertebrates.

B. Growth Phase

- Prolonged and dramatic phase compared to spermatogenesis. Oogonium transforms into a **Primary Oocyte** ($2n$).
- Subdivided into:
 1. **Pre-vitellogenesis:** Nuclear changes (lampbrush chromosome activity, gene amplification) and organelle duplication.
 2. **Vitellogenesis:** Active accumulation of yolk components.

C. Maturation Phase

- Involves two successive meiotic divisions characterized by **unequal cytoplasmic division**:
 1. **Meiosis I (Reductional):** Primary oocyte ($2n$) produces one large **Secondary Oocyte** ($1n$) and one tiny **First Polar Body** ($1n$).
 2. **Meiosis II (Equational):** Secondary oocyte divides into one functional **Ovum** ($1n$) and a **Second Polar Body** ($1n$).
- **Result:** One single oogonium produces **1 functional ovum** and **3 non-functional polar bodies** (which degenerate).

3. Detailed Comparison: Oogenesis vs. Spermatogenesis

Feature	Spermatogenesis	Oogenesis
Site of Occurrence	Seminiferous tubules of Testes.	Ovary (Follicles).
Yield per Primary Cell	1 Primary Spermatocyte yields 4 equal, functional Spermatozoa .	1 Primary Oocyte yields 1 functional Ovum and 3 non-functional Polar Bodies .

Feature	Spermatogenesis	Oogenesis
Continuity	Continuous process from puberty to old age.	Discontinuous process; initiated in embryonic stage, arrests at Prophase-I, resumes at puberty, ends at menopause.
Growth Phase	Short and limited in duration.	Extremely long and pronounced (months to years).
Yolk Synthesis	No yolk accumulation occurs.	Heavy accumulation of yolk (Vitellogenesis).
Cytokinesis	Equal cytoplasmic division.	Highly unequal cytoplasmic division (conserves nutrient-rich cytoplasm for ovum).
Motility & Size	Produces tiny, highly motile gametes (sperm).	Produces large, non-motile gametes (ovum).
Differentiation Phase	Spermiogenesis required to convert non-motile spermatids to flagellated sperm.	No distinct structural transformation step; ovum matures directly.

4. Conclusion

While spermatogenesis favors **quantity and motility** to maximize fertilization chances, oogenesis prioritizes **quality and resource accumulation**, ensuring the resulting single egg contains sufficient nutrients to drive early embryonic development.

Q9. Compare the phenomenon of blastogenesis and embryogenesis.

1. Introduction

In developmental biology, animals can originate through two fundamental developmental pathways depending on the mode of reproduction:

1. **Embryogenesis:** Development of an organism from a single fertilized egg (zygote) through sexual reproduction.
2. **Blastogenesis:** Development of an organism from an asexual bud, fragment, or gemmule through asexual reproduction.

2. Comprehensive Comparison Table

Feature	Embryogenesis	Blastogenesis
Originating Unit	Single diploid cell: Zygote (or unfertilized egg in parthenogenesis).	Multicellular unit: Blastema , Bud , or Gemmule .
Mode of Reproduction	Sexual reproduction (fusion of egg and sperm).	Asexual reproduction (budding, fission, gemmulation).
Genetic Variation	High genetic variability due to meiotic recombination and	Genetically identical to parent (Clones); no recombination.

Feature	Embryogenesis	Blastogenesis
	syngamy.	
Early Cleavage Phase	Undergoes systematic, synchronized cleavage divisions (holoblastic/meroblastic).	No cleavage stage. Development proceeds directly via somatic cell proliferation.
Morphogenetic Movements	Characterized by strict gastrulation movements (epiboly, emboly, invagination) to form germ layers.	No formal gastrulation phase. Tissue organization proceeds through local cell rearrangements and induction.
Germ Layer Formation	Definitive establishment of Ectoderm, Mesoderm, and Endoderm from blastomeres.	New tissues arise directly from pre-existing somatic tissues or totipotent stem cells (e.g., Archaeocytes, Neoblasts).
Polarity & Symmetry	Polarity determined early by sperm entry site, egg yolk distribution, and maternal gradients.	Polarity directly inherited from the axis of the parent body.
Larval Stages	Frequently involves complex, free-swimming larval forms undergoing metamorphosis.	Typically circumvents complex larval phases; directly forms miniature adult.
Rate of Development	Relatively slow, requiring complex cell differentiation sequences.	Generally rapid , enabling fast population growth in favorable conditions.
Examples	Zygotic development in Frog, Chick, Human, etc.	Budding in <i>Hydra</i> , Gemmules in <i>Spongilla</i> , Stolon development in Salps.

3. Key Conceptual Differences

EMBRYOGENESIS: Zygote (1 Cell) → Cleavage → Blastula → Gastrula → Organogenesis → Adult

BLASTOGENESIS: Parent Tissue (Multicellular Bud/Blastema) → Morphogenetic Sorting/Growth → Adult

- **Cell Source:** Embryogenesis relies on pluripotent/totipotent embryonic blastomeres. Blastogenesis relies on adult somatic stem cells (e.g., interstitial cells of *Hydra*) or redifferentiated local tissues.
- **Organogenesis:** In embryogenesis, organ primordia arise *de novo* from germ layers. In blastogenesis, organ structures are often continuous extensions of the parent's organ systems before detaching.

4. Conclusion

Embryogenesis is designed to generate **genetic diversity** and complex developmental lineage pathways. Blastogenesis is optimized for **rapid vegetative multiplication**, preserving parental

adaptations without the energetic investment of gametogenesis and fertilization.

Q10. Give an account of regeneration.

1. Definition & Scope

Regeneration is the ability of an organism to replace, repair, or restore lost, damaged, or amputated body parts or tissues. It spans a wide developmental spectrum, from cellular repair in mammals to complete whole-body restoration from tiny fragments in invertebrates like *Planaria* and *Hydra*.

2. Types / Modes of Regeneration

division; Hydra) — 1. Morphallaxis (Reorganization without
growth; Salamander) — 2. Epimorphosis (Blastema formation &
Modes of Regeneration — 3. Compensatory Regeneration
(Hypertrophy/Division without blastema; Human Liver)
— 4. Physiological Regeneration (Routine cell
turnover; Blood, Epithelium)

1. **Morphallaxis:** Reconstitution of a complete, smaller organism by re-patterning existing somatic cells without significant proliferation (e.g., *Hydra*, *Planaria*).
2. **Epimorphosis:** Restoring missing organs by forming a mass of proliferating, undifferentiated cells called a **blastema** at the wound site (e.g., Amphibian limb regeneration).
3. **Compensatory Regeneration:** Mature cells divide while retaining their differentiated functions to restore original tissue mass without forming a blastema (e.g., mammalian liver regeneration after partial hepatectomy).
4. **Physiological Regeneration:** Replacement of cells lost during normal daily wear and tear (e.g., shedding of human epidermis, intestinal epithelium turnover, erythrocyte replacement).

3. Mechanism of Epimorphic Limb Regeneration (e.g., Salamander)

Epimorphic regeneration proceeds through five well-defined sequential stages:

[Wound Healing] → [Apical Epidermal Cap] → [Dedifferentiation]
→ [Blastema Formation] → [Redifferentiation & Outgrowth]

1. **Wound Healing:** Epidermal cells migrate within hours to cover the cut surface without scar formation, creating a thick multi-layered **Apical Epidermal Cap (AEC)**.
2. **Dedifferentiation:** Tissues immediately beneath the cap (bone, muscle, cartilage, connective tissue) break down their extracellular matrix. Cells lose their mature characteristics and revert to a stem-cell-like embryonic state.
3. **Blastema Formation:** Dedifferentiated cells accumulate under the AEC, forming a cone of rapidly proliferating mesenchymal cells called the **Regeneration Blastema**.

4. **Proliferation & Patterning:** The AEC acts similarly to the Apical Ectodermal Ridge (AER) in embryonic limb buds, secreting growth factors (e.g., FGFs) that drive blastema expansion along proximal-distal axes.
5. **Redifferentiation & Morphogenesis:** Blastema cells differentiate into chondrocytes, myocytes, and osteocytes, perfectly reconstructing missing limb parts in proper spatial orientation.

4. Factors Influencing Regeneration

- **Nerve Supply:** Epimorphic limb regeneration is **neurotrophic**; if nerves supplying the limb stump are cut, blastema proliferation halts (driven by NAG - Newt Anterior Gradient protein).
- **Age:** Regenerative capacity generally decreases as developmental complexity increases and organisms age.
- **Hormones:** Adrenal steroids, thyroid hormone, and pituitary growth hormone play permissive roles.
- **Temperature & Metabolic Rate:** Higher metabolic rates accelerate regenerative speed up to a physiological threshold.

5. Systematic Distribution across Animal Phyla

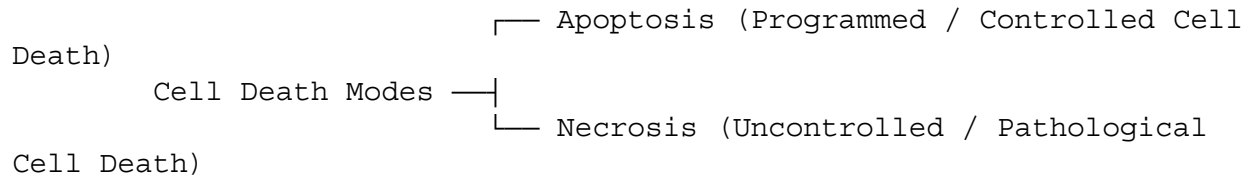
- **Protozoa:** High capacity (e.g., *Stentor*, *Amoeba* can regenerate if nucleus is intact).
- **Porifera & Cnidaria:** Exceptional capacity (*Sponges*, *Hydra*).
- **Platyhelminthes:** High capacity (*Planaria* can regenerate from 1/279th of a body piece via **Neoblasts**).
- **Annelida & Echinoderms:** High capacity (*Earthworms*, *Starfish* arms).
- **Vertebrates:**
 - *Urodeles (Newts/Salamanders)*: Full limb, tail, and lens regeneration.
 - *Anurans (Frogs)*: Regenerate limbs as tadpoles, lose ability after metamorphosis.
 - *Reptiles*: Autotomy and tail regeneration in lizards (imperfect; cartilaginous rod replaces bone).
 - *Mammals*: Restricted mainly to tissue repair, physiological turnover, and compensatory liver growth.

Regeneration blends embryonic developmental programs with adult tissue response. Unlocking the molecular mechanisms that restrict epimorphic blastema formation in higher vertebrates remains a core focus of modern regenerative biology and medicine.

Q11. Write notes on: a) Cell death b) Teratogenesis

a) Cell Death

Cell death is a fundamental biological process essential for maintaining tissue homeostasis, embryonic development, and removing damaged or potentially harmful cells. It occurs via two primary distinct pathways: **Apoptosis** and **Necrosis**.



1. Apoptosis (Programmed Cell Death)

- **Definition:** A highly regulated, energy-dependent (ATP-requiring) physiological cell death mechanism.
- **Characteristics:**
 1. Shrinkage of cell volume and nuclear chromatin condensation (pyknosis).
 2. Plasma membrane blebbing without loss of membrane integrity.
 3. DNA fragmentation by endonucleases into regular ladder-like fragments (~180–200 bp).
 4. Formation of **apoptotic bodies**, which are rapidly phagocytosed by macrophages without eliciting an inflammatory response.
- **Molecular Pathway:** Driven by a cascade of cysteine-aspartic proteases known as **Caspases** (Initiator caspases: Caspase-8, 9; Executioner caspases: Caspase-3, 6, 7).
- **Biological Significance:** Essential during metamorphosis (e.g., tadpole tail resorption), interdigital membrane regression in embryonic digits, and elimination of autoreactive immune cells.

2. Necrosis

- **Definition:** Accidental, uncontrolled cell death resulting from acute environmental injury, mechanical trauma, or toxic exposure.
- **Characteristics:** Cell swelling, organelle rupture, plasma membrane breakdown, and release of intracellular contents, triggering a **severe local inflammatory response**.

b) Teratogenesis

(For a complete discussion on Teratogenesis, see Question 4).

- **Definition:** The induction of structural, functional, or developmental anomalies in a developing embryo or fetus by external agents during pregnancy.
- **Teratogenic Agents (Teratogens):**
 1. **Drugs & Chemicals:** Thalidomide (causes *phocomelia*), Alcohol (causes *Fetal*

- Alcohol Spectrum Disorder*), Retinoic acid.
2. **Infectious Pathogens (TORCH):** Rubella virus, Cytomegalovirus, *Toxoplasma gondii*.
 3. **Physical Agents:** Ionizing radiation (X-rays), hyperthermia.
- **Critical Period:** Embryos are most susceptible during the **organogenesis phase** (weeks 3 to 8 in human gestation), where rapid cell differentiation and organ formation take place.

Q12. Give an illustrated account of different types of growth curves.

1. Introduction

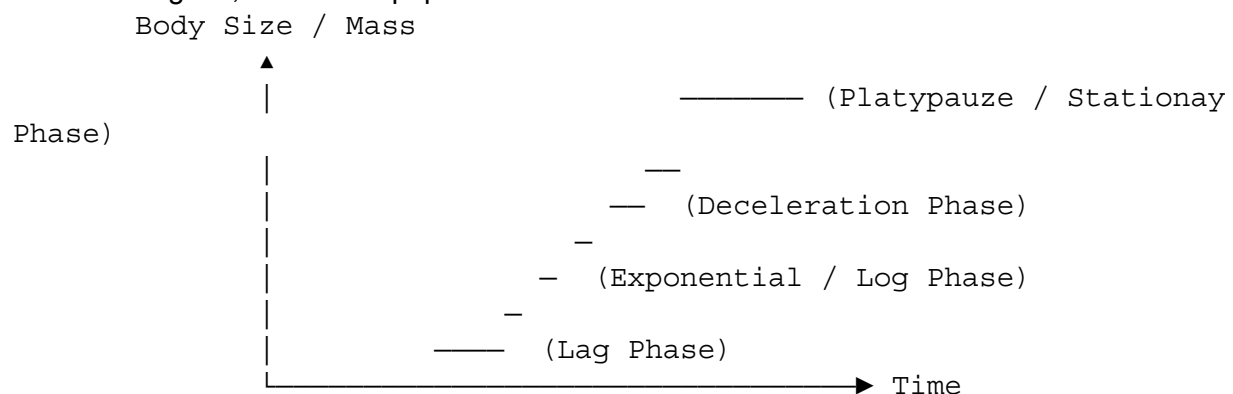
A **Growth Curve** is a graphical representation depicting how a biological metric (such as cell number, body mass, height, or population size) changes over time. Growth is rarely constant; it exhibits distinct acceleration and deceleration phases dictated by physiological and environmental constraints.

2. Major Types of Growth Curves

- Curve
- 1. Sigmoid (S-shaped) / Absolute Growth
 - 2. Intermittent / Stepped Growth Curve
- (Arthropods)
- Types of Growth Curves
- 3. Exponential (J-shaped) Curve
 - 4. Relative / Relative Rate Growth Curve

A. Sigmoid (S-Shaped) Growth Curve

The sigmoid curve is the standard growth pattern displayed by most higher multicellular animals, individual organs, and stable populations.



It consists of four distinct phases:

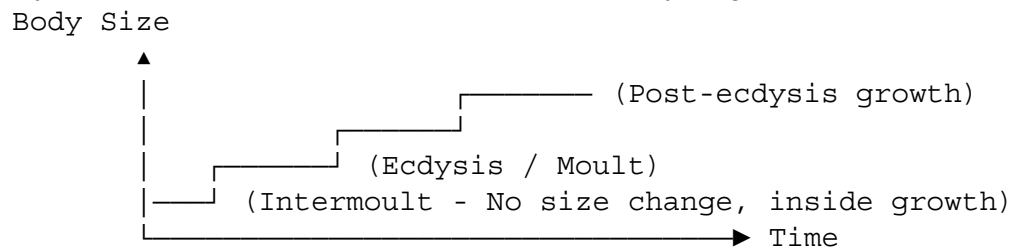
1. **Lag Phase:** Initial slow phase characterized by cell adaptation and low metabolic turnover.
2. **Exponential (Log) Phase:** Rapid acceleration phase where maximum growth rate occurs

due to intense cell division (hyperplasia) and expansion (auxesis).

3. **Deceleration Phase:** Growth slows down due to internal developmental signals, aging, or resource limits.
4. **Stationary (Plateau) Phase:** Growth rate drops to zero ($\text{Anabolism} = \text{Catabolism}$). Maximum adult size or carrying capacity (K) is reached.

B. Intermittent (Stepped / Staircase) Growth Curve

Exhibited by **Arthropods** (insects, crustaceans) protected by a rigid chitinous exoskeleton.



- **Mechanism:** True physical growth in overall body volume occurs in rapid steps immediately after shedding the old cuticle (**Ecdysis**), while the new cuticle is still soft and stretchable.
- During the **intermoult phase**, dry weight increases internally, but external dimensions remain completely constant, resulting in a step-like curve.

C. Exponential (J-Shaped) Growth Curve

Observed in lower organisms or populations undergoing unrestricted growth in an ideal environment with unlimited resources.

- **Equation:** $\frac{dN}{dt} = rN$
- Growth accelerates continuously until environmental limits cause a sudden crash.

3. Conclusion

Sigmoid curves highlight resource-limited, self-regulating biological growth typical of higher vertebrates, while intermittent curves illustrate how physical structures like exoskeletons constrain physiological expansion.

Q13. What do you understand by aging? Discuss different theories of aging.

1. Definition of Aging

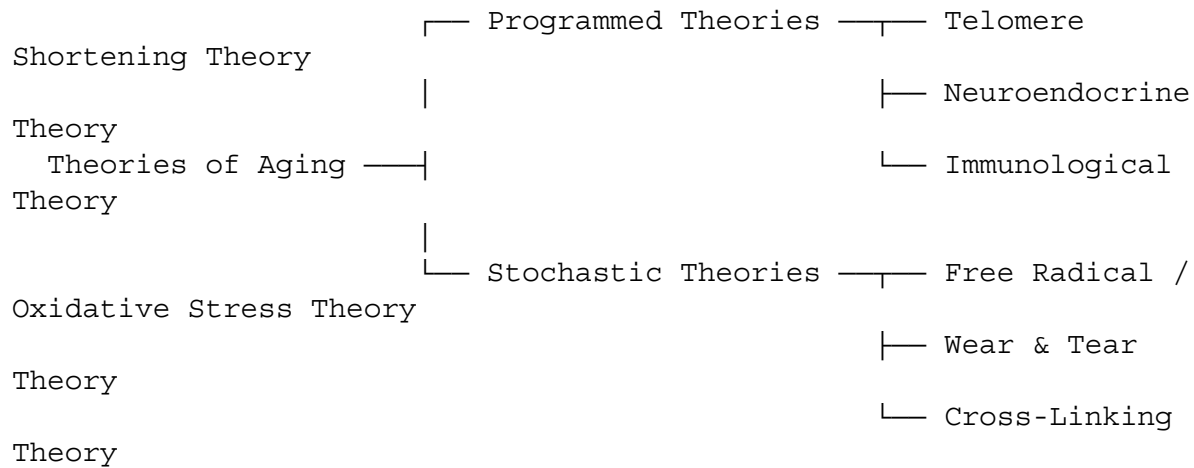
Aging (or **senescence**) is defined as the progressive, irreversible physiological decline in structural integrity and functional capacity of an organism over time. It increases the organism's vulnerability to disease and exponentially increases the probability of death.

Mathematically, mortality rate due to aging follows the **Gompertz-Makeham Law**:

$$\mu(t) = A + R e^{\alpha t}$$

2. Major Theories of Aging

Theories of aging are broadly divided into **Programmed (Genetic) Theories** and **Damage/Error (Stochastic) Theories**:



A. Programmed (Genetic) Theories

Aging is driven by a built-in biological clock controlled by regulated gene expression.

1. Telomere Shortening Theory (Hayflick Limit):

- Normal somatic cells have a finite division capacity (e.g., 50 divisions for human fibroblasts, known as the *Hayflick Limit*).
- With each round of replication, ends of chromosomes (**telomeres**) progressively shorten due to the end-replication problem. When telomeres reach a critically short length, cell division ceases, triggering cellular senescence.

2. Neuroendocrine Theory:

- Proposes that aging is governed by functional changes in the **Hypothalamic-Pituitary Axis**, leading to altered hormone levels (e.g., decline in growth hormone, DHEA, estrogen/testosterone) that reduce metabolic efficiency.

3. Immunological Theory:

- The immune system's efficiency declines over time (**immunosenescence**). Thymic involution reduces T-cell production, making the body vulnerable to infections and autoimmune reactions.

B. Stochastic (Damage/Error) Theories

Aging is caused by the cumulative accumulation of random environmental and metabolic damage over time.

1. Free Radical / Oxidative Stress Theory (Harman, 1956):

- **Reactive Oxygen Species (ROS)** such as superoxide radical (O_2^-), hydrogen peroxide (H_2O_2), and hydroxyl radicals (OH^-) generated as metabolic byproducts in mitochondria damage lipids, proteins, and nuclear/mitochondrial DNA over time.

2. Cross-Linking Theory:

- With age, advanced glycation end-products (AGEs) form covalent cross-links between proteins (notably collagen and elastin), causing tissue stiffness, loss of

elasticity in blood vessels, and reduced organ performance.

3. Error-Catastrophe Theory (Orgel):

- Random mutations or transcription/translation errors lead to defective enzymes involved in DNA replication and repair. Accumulation of faulty machinery amplifies error rates until a lethal "error catastrophe" occurs.

3. Conclusion

Aging is a multifactorial phenomenon. It represents an interplay between evolutionary programmed genomic stability (telomere dynamics) and cumulative environmental damage (oxidative damage to cells).

Q14. Describe the morphogenetic process associated with the metamorphosis of frog.

1. Introduction

Metamorphosis in amphibians (e.g., *Rana tigrina*) is a dramatic post-embryonic process converting an aquatic, herbivorous, gill-breathing **tadpole larva** into a terrestrial, carnivorous, lung-breathing **adult frog**.

This transition involves major **morphogenetic processes**: **Progression/Growth** of adult-specific structures, **Regression/Resorption** of larval structures, and **Remodeling** of existing tissues.

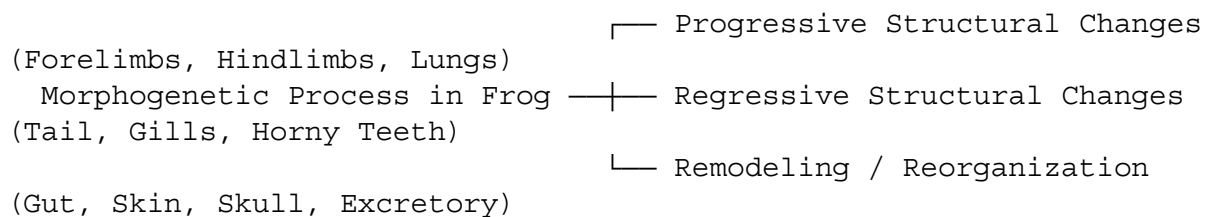
2. Hormonal Control

The entire morphogenetic cascade is orchestrated by the **Thyroid Gland**:

Hypothalamus (CRH)→Anterior Pituitary (TSH)→Thyroid Gland (T3 and T4)

- **Triiodothyronine (T3)** is the active tissue-level morphogenetic signal.

3. Morphogenetic Changes during Frog Metamorphosis



A. Progressive Changes (Development of New Adult Structures)

1. **Limb Development:** Hindlimbs emerge first, followed by forelimbs (which erupt through the opercular membrane).
2. **Respiratory Shift:** Internal gills regress; lungs differentiate, and vascularization develops to support pulmonary and cutaneous respiration.
3. **Sensory & Cranial Adaptations:** Eyes protrude, develop nictitating membranes, and

move dorsally to support stereoscopic vision. The middle ear apparatus (tympanic membrane and columella auris) forms.

B. Regressive Changes (Destruction of Larval Structures)

1. **Tail and Fin Resorption:** The long muscular tail and fin folds undergo complete regression via **autophagy and apoptotic cell death**, mediated by local lysosomal hydrolytic enzymes (cathepsins, collagenases). Macrophages phagocytose the broken-down tissue, recycling nutrients to other organs.
2. **Loss of Larval Mouthparts:** Horny teeth and the larval horny beak are shed.
[Larval Tail Tissue] —(T3 signal)—> [Apoptosis & Cathepsin activation] —> [Phagocytosis by Macrophages] —> [Nutrients Recycled]

C. Remodeling and Physiological Changes

1. **Digestive Tract Overhaul:** The long, coiled intestine suitable for digesting algae shrinks into a short, straight intestine adapted for a carnivorous diet.
2. **Excretory Switch (Biochemical Metamorphosis):**
 - Tadpoles are **ammonotelic** (excrete toxic ammonia through gills).
 - Adult frogs undergo biochemical induction of urea-cycle liver enzymes (carbamoyl phosphate synthetase) to become **ureotelic** (excrete urea).
3. **Hemoglobin Switch:** Larval hemoglobin (high O₂ affinity) is replaced by adult hemoglobin (lower O₂ affinity, higher oxygen unloading efficiency to tissues).

4. Conclusion

Frog metamorphosis is a textbook example of coordinated morphogenetic execution, where a single systemic endocrine trigger (T3) directs cell death in larval tissues while simultaneously promoting cell division and differentiation in adult organ primordia.

Q15. Describe the process of fertilization.

1. Definition

Fertilization is the complex physicochemical process involving the fusion of a haploid male gamete (spermatozoon) and a haploid female gamete (ovum) to form a diploid zygote (2n). It restores diploidy, determines sex, and activates the egg to initiate cleavage.

2. Events/Phases of Fertilization

- | | |
|-------------------------|--|
| | ┌ 1. Gamete Contact & Chemotaxis |
| | ├ 2. Acrosome Reaction & Sperm Penetration |
| Phases of Fertilization | ├ 3. Cortical Reaction & Block to |
| Polyspermy | └ 4. Egg Activation & Metabolic Awakening |

— 5. Amphimixis (Karyogamy / Fusion of Pronuclei)

Phase 1: Chemotaxis and Gamete Binding

1. **Species-Specific Attraction:** The egg jelly envelope secretes species-specific chemotactic peptides (e.g., *Resact* in sea urchins) to attract sperm.
2. **Sperm-Zona Binding:** In mammals, sperm bind to specific glycoprotein receptors (**ZP3 / ZP2**) on the Zona Pellucida using membrane receptors (e.g., Sed1, GalT).

Phase 2: Acrosome Reaction and Penetration

- Contact with egg envelopes triggers calcium influx (Ca^{2+}) into the sperm head, inducing fusion of the acrosomal membrane with the sperm plasma membrane.
- Lytic enzymes (**hyaluronidase, acrosin, lysins**) are released via exocytosis, digesting a path through the corona radiata and zona pellucida/vitelline envelope.

[Sperm Contact with ZP3] $\rightarrow$ [Ca^{2+} Influx] $\rightarrow$ [Acrosomal Exocytosis] $\rightarrow$ [Enzymatic Digestion of Envelopes]

Phase 3: Prevention of Polyspermy

Fertilization by a single sperm (**monospermy**) is crucial; fusion with multiple sperm (**polyspermy**) causes abnormal cleavage and embryonic death.

1. **Fast Block to Polyspermy (Transient):**
 - Sperm fusion triggers an instantaneous influx of Na^{+} ions into the egg cytoplasm, shifting resting membrane potential from **-70mV to +20mV**. Positive potential prevents further sperm fusion for ~60 seconds.
2. **Slow Block to Polyspermy (Permanent - Cortical Reaction):**
 - Ca^{2+} released from the endoplasmic reticulum causes **cortical granules** to fuse with the oolemma.
 - Cortical enzymes (mucopolysaccharides, proteases) enter the perivitelline space, cleave ZP3 receptors, and harden the vitelline membrane into an impenetrable **Fertilization Membrane**.

[Ca^{2+} Wave] $\rightarrow$ [Cortical Granule Exocytosis] $\rightarrow$ [Receptor Cleavage & Membrane Hardening]

Phase 4: Egg Activation

Ca^{2+} release induces resumption and completion of Meiosis II in the egg, extrusion of the second polar body, a surge in protein synthesis, and increased oxygen consumption (metabolic awakening).

Phase 5: Amphimixis (Pronuclear Fusion)

- The sperm head rotates 180° inside the egg cytoplasm; its chromatin decondenses to form

the **Male Pronucleus**.

- The egg nucleus completes meiosis to form the **Female Pronucleus**.
- Both pronuclei move toward each other guided by microtubules (aster rays), lose their nuclear envelopes, and mix their chromosomes (**Amphimixis**), forming a diploid (2n) **zygote nucleus**.

3. Conclusion

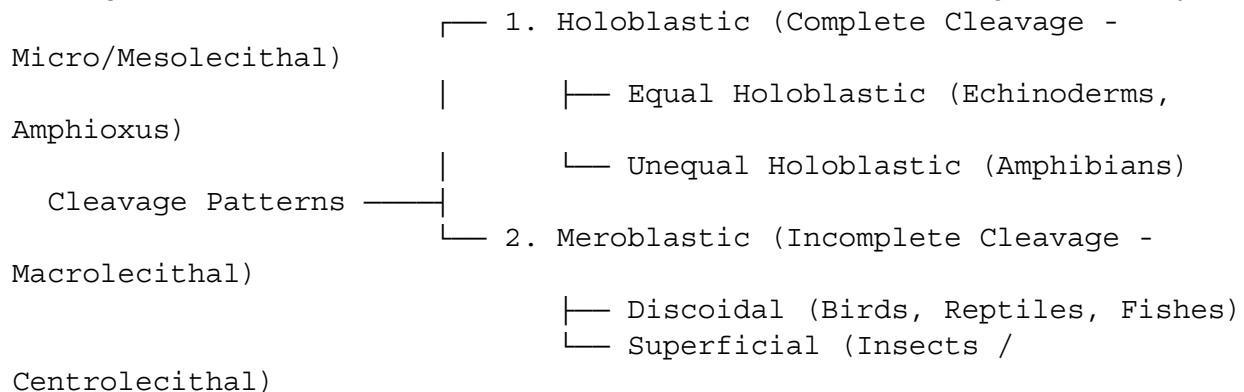
Fertilization is a multi-step recognition and fusion event that activates maternal developmental programs while combining parental genetic lines to launch embryonic life.

Q16. Write short notes on: A) Classification of cleavage B) Cleidoic egg characteristics C) Growth curves

A) Classification of Cleavage

Cleavage refers to the rapid sequence of mitotic cell divisions undergone by the zygote, converting the single-celled zygote into a multicellular blastula without increasing total cytoplasmic volume.

Cleavage patterns are classified based on the **amount and distribution of yolk** (lecithality):



1. Holoblastic (Total / Complete) Cleavage:

- Cleavage furrows pass through the entire egg.
- **Equal Holoblastic:** Divides zygote into equal blastomeres (occurs in microlecithal/alecithal eggs, e.g., *Amphioxus*, placental mammals).
- **Unequal Holoblastic:** Divides zygote into smaller **micromeres** at the animal pole and larger yolk-rich **macromeres** at the vegetal pole (occurs in mesolecithal eggs, e.g., Frog).

2. Meroblastic (Partial / Incomplete) Cleavage:

- Cleavage furrows cannot penetrate the heavy yolk mass; restricted to yolk-free region.
- **Discoidal Cleavage:** Division is confined to a small cap of cytoplasm at the animal pole called the blastodisc (occurs in macrolecithal eggs, e.g., Birds, Reptiles).
- **Superficial Cleavage:** Nuclear division occurs without immediate cytokinesis;

nuclei migrate to the peripheral cytoplasm (occurs in centrolecithal eggs, e.g., Insects).

B) Cleidoic Egg Characteristics

A **Cleidoic egg** (Greek: *Kleis* = Closed/Box) is a self-contained, terrestrial egg enclosed within a protective shell, representing a key evolutionary adaptation that allowed vertebrates (reptiles, birds, egg-laying mammals) and arthropods to transition from aquatic environments to land.

[Porous Shell] → [Albumen (Water/Protein)] → [Yolk (Nutrients)] → [Extra-embryonic Membranes]

Key Characteristics:

1. **Calcareous or Leathery Shell:** Provides physical protection and structural support while allowing gaseous exchange (O₂ and CO₂) through microscopic pores.
2. **Heavy Yolk (Macrolecithal & Telolecithal):** Contains abundant nutrient reserves to nourish the embryo throughout development.
3. **Albumen Layer (Egg White):** Supplies water, proteins, and cushions against mechanical shock.
4. **Presence of Extra-embryonic Membranes:**
 - **Amnion:** Encloses an amniotic fluid-filled cavity, providing a fluid environment to prevent desiccation.
 - **Yolk Sac:** Surrounds yolk and absorbs nutrients via vitelline circulation.
 - **Allantois:** Serves as a storage sac for nitrogenous wastes (uric acid) and assists in respiration.
 - **Chorion:** Lies beneath the shell, functioning in gas exchange alongside the allantois.
5. **Uricotelic Excretion:** Waste products are converted into non-toxic, insoluble **uric acid** crystals stored safely inside the allantois.

C) Growth Curves

(For a complete discussion on Growth Curves, see Question 12).

- **Definition:** Plots of body size, mass, or cell count against time, reflecting developmental dynamics.
- **Sigmoid (S-shaped) Curve:** Typical of vertebrates; comprises **Lag phase**, **Log (Exponential) phase**, **Deceleration phase**, and **Stationary (Plateau) phase**.
- **Intermittent (Stepped) Curve:** Characteristic of arthropods, where growth jumps in discrete steps post-ecdysis.

Q17. Differentiate between any three of the following:

A) Epiboly and Emboly

Feature	Epiboly	Emboly
Definition	The morphogenetic movement	The inward movement of

Feature	Epiboly	Emboly
	where an outer layer of blastomeres stretches and spreads downward over lower cells.	blastomeres from the blastula surface into the interior to form internal germ layers.
Cell Types Involved	Primary ectodermal cells / micromeres at the animal pole.	Mesodermal and Endodermal cells / macromeres at the vegetal pole.
Direction of Movement	Spreading externally over the outer surface .	Inward movement into the internal cavity (blastocoel) .
Primary Outcome	Encloses the embryo with an outer Ectodermal layer .	Establishes the Endoderm , Mesoderm , and internal Archenteron gut cavity.
Mechanisms	Radial intercalation, cell flattening, and cell division.	Invagination, Involution, Ingression, and Delamination.
Example	Micromeres growing over macromeres in Frog gastrulation.	Invagination of vegetal macromeres during <i>Amphioxus</i> / Frog gastrulation.

B) Epimorphosis and Morphallaxis

Feature	Epimorphosis	Morphallaxis
Definition	Regeneration of lost body parts through active cell proliferation and blastema formation.	Regeneration of a whole body primarily by repatterning existing tissue with minimal cell division.
Blastema Formation	Present. A specialized mass of undifferentiated proliferating cells (blastema) is formed.	Absent. No localized proliferating blastema mass is formed.
Role of Mitosis	High rate of local cell proliferation (mitotic divisions).	Extremely low or non-existent cell proliferation.
Resulting Size	Regenerates lost structure to its original full size .	Reconstructed organism is initially much smaller than the original parent.
Classic Examples	Amphibian limb regeneration (Axolotl), Lizard tail regeneration.	Whole-body regeneration in <i>Hydra</i> and <i>Planaria</i> .

C) Complete and Incomplete Metamorphosis

Feature	Complete Metamorphosis (Holometabolous)	Incomplete Metamorphosis (Hemimetabolous)
Developmental Stages	4 distinct stages: Egg → Larva → Pupa → Adult .	3 stages: Egg → Nymph → Adult .
Pupal Stage	Present. Includes a quiescent pupal stage where dramatic body reconstruction takes	Absent. No pupal resting stage.

Feature	Complete Metamorphosis (Holometabolous)	Incomplete Metamorphosis (Hemimetabolous)
	place.	
Immature Stage	Called a Larva (caterpillar, maggot, grub); looks completely different from the adult.	Called a Nymph ; closely resembles the adult in form and habitat (except lacks wings and mature gonads).
Ecological Niche	Larva and adult usually occupy different habitats and eat different food , avoiding competition.	Nymph and adult usually share the same habitat and feed on similar food sources .
Examples	Butterflies, Moths, Beetles, Houseflies, Bees.	Cockroaches, Grasshoppers, Dragonflies, Bugs.

Question 18

Q. Describe the phenomenon of metamorphosis in insects. What is the role of endocrine glands in it?

1. Introduction

Metamorphosis (Greek: *meta* = change, *morphe* = form) is a post-embryonic developmental process in insects involving radical morphological, physiological, anatomical, and biochemical transformations from a juvenile larval/nymphal stage into a sexually mature adult (imago).

2. Types of Metamorphosis in Insects

Insects exhibit four principal patterns of metamorphosis based on the degree of structural transformation:

- **Ametabolous Development (No Metamorphosis):**
 - **Features:** Primitive wingless insects show direct development without distinct larval forms. Young hatchlings (nymphs) resemble adults except in size and sexual maturity.
 - **Examples:** *Lepisma* (Silverfish), *Thermobia*.
- **Paurometabolous Development (Gradual Metamorphosis):**
 - **Features:** Life cycle includes Egg → Nymph → Adult. Nymphs occupy the same habitat as adults, possess compound eyes, and gradually develop external wing pads (exopterygote).
 - **Examples:** *Periplaneta americana* (Cockroach), Grasshoppers, Bugs.
- **Hemimetabolous Development (Incomplete Metamorphosis):**
 - **Features:** Life cycle consists of Egg → Naiad (aquatic) → Adult (terrestrial/aerial). Naiads possess specialized adaptations like tracheal gills.
 - **Examples:** Dragonflies, Mayflies, Damselflies.
- **Holometabolous Development (Complete Metamorphosis):**
 - **Features:** Involves four distinct stages: Egg → Larva → Pupa → Adult. The larva (e.g., caterpillar, maggot, grub) undergoes extensive tissue restructuring during the quiescent pupal stage via histolysis and histogenesis from imaginal discs.
 - **Examples:** *Bombyx mori* (Silkworm), *Drosophila*, Butterflies, Honeybees.

3. Endocrine Glands Involved in Metamorphosis

Insect metamorphosis is regulated by a neuroendocrine mechanism involving four primary endocrine centers:

1. **Neurosecretory Cells (NSCs) of Brain:**
 - Located in the intercerebralis region of the protocerebrum.
 - Secrete **Prothoracicotropic Hormone (PTTH)** (also called Activation Hormone).
2. **Corpora Cardiaca (CC):**
 - Paired neurohemal organs situated behind the brain.
 - Store and release PTTH into the hemolymph.
3. **Prothoracic Glands (PTG):**

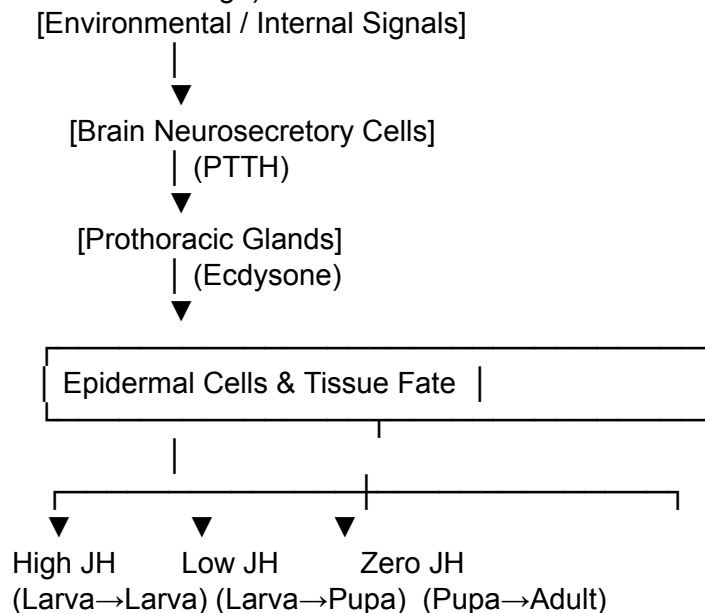
- Paired thoracic glands located near the anterior spiracles.
- Synthesize and secrete **Ecdysone** (moulting hormone), a steroid derived from cholesterol.

4. **Corpora Allata (CA):**

- Paired epithelial endocrine glands located posterior to corpora cardiaca.
- Synthesize and release **Juvenile Hormone (JH)** (a sesquiterpenoid).

4. Hormonal Control Mechanism

- **Trigger & PTTH Secretion:** Environmental stimuli (photoperiod, temperature, nutrition) stimulate the brain NSCs to release PTTH into the hemolymph via Corpora Cardiaca.
- **Ecdysone Secretion:** PTTH acts on the Prothoracic Glands, inducing the release of **Ecdysone** (α -ecdysone), which is converted in peripheral tissues to the active form, **20-hydroxyecdysone**.
- **Role of Juvenile Hormone (JH):**
 - **High JH + Ecdysone:** Promotes larval-to-larval moulting (maintains juvenile structures).
 - **Low JH + Ecdysone:** Triggers larval-to-pupal moulting in holometabolous insects.
 - **Absence of JH + Ecdysone:** Initiates pupal-to-adult transformation (emergence of imago).



Question 19

Q. Compare the following phenomena: (A) Blastogenesis vs. Embryogenesis (B) Oogenesis vs. Spermatogenesis

(A) Blastogenesis vs. Embryogenesis

Parameter	Blastogenesis	Embryogenesis
Definition	Development of a new organism from a somatic blastema/bud via asexual reproduction.	Development of an embryo from a fertilized egg (zygote) via sexual reproduction.
Origin Cell	Somatic cells / Blastema cells (multicellular origin).	Single-celled diploid zygote (unicellular origin).
Genetic Variation	Produces genetically identical clones (no recombination).	Shows genetic variation due to meiotic crossing-over and fertilization.
Cleavage Phase	Lacks standard cleavage pattern; involves mitotic cell proliferation.	Exhibits characteristic cleavage (holoblastic/meroblastic).
Germ Layers	Germ layer differentiation is often altered or absent.	Strict formation and differentiation of 3 primary germ layers (Ecto-, Meso-, Endoderm).
Polarity & Axis	Axis determined by the parent organism's body symmetry.	Axis established by egg polarity, sperm entry point, and morphogenetic gradients.
Occurrence	Sponges, Coelenterates, Tunicates, Bryozoans.	Found in all sexually reproducing multicellular animals.

(B) Oogenesis vs. Spermatogenesis

Feature	Oogenesis	Spermatogenesis
Definition	Process of formation and maturation of functional female gametes (ova).	Process of formation and differentiation of functional male gametes (spermatozoa).
Site of Action	Ovaries (Graafian follicles/ovarian follicles).	Testes (Seminiferous tubules).
Yield per Primary Cell	1 Primary Oocyte yields 1 functional Ovum + 2 to 3 polar bodies.	1 Primary Spermatocyte yields 4 functional Spermatozoa .
Cytokinesis	Unequal cytokinesis (retains maximum cytoplasm in the egg).	Equal cytokinesis (cytoplasm distributed equally).
Growth Phase	Extensively prolonged growth phase (vitellogenesis/yolk synthesis).	Relatively short growth phase without yolk synthesis.
Motility & Size	Ovum is large, non-motile, nutrient-rich.	Sperm is minute, highly motile, streamlined.
Spermiogenesis Phase	Absent.	Present (transformation of spermatid to flagellated spermatozoon).
Continuity	Arrested at Prophase-I	Continuous process from

Feature	Oogenesis	Spermatogenesis
	(Dictyotene stage) in human females until puberty; stops at menopause.	puberty throughout life in males.

Question 20

Q. Write short notes on: (A) Ability of regeneration in different animal groups (B) Kinds of asexual reproduction (C) Biogenetic law (D) Tissue reactivity and induction process

(A) Ability of Regeneration in Different Animal Groups

Regeneration is the capacity of an organism to replace lost, damaged, or amputated body parts. The regenerative potential varies inversely with evolutionary complexity:

- **Protozoa:** Show high morphallactic capability (e.g., *Stentor*, *Amoeba* retain regeneration if a portion of the nucleus is intact).
- **Porifera & Cnidaria:** Possess extreme regenerative powers. *Hydra* can regenerate an entire functional organism from a small fragment of body tissue containing both ectoderm and endoderm through morphallaxis.
- **Platyhelminthes:** *Planaria* demonstrates exceptional epimorphic regeneration driven by totipotent adult stem cells called **Neoblasts**. A piece as small as $1/279^{\text{th}}$ of the body can regenerate a complete worm.
- **Annelida & Echinodermata:** Starfish (*Asterias*) regenerate lost arms and central disc; Earthworms (*Pheretima*) regenerate anterior/posterior segments.
- **Vertebrates:**
 - *Urodela (Amphibians):* Axolotl and salamanders undergo complete limb and tail regeneration via blastema formation.
 - *Reptiles:* Lizards display caudal autotomy followed by tail regeneration (lacks true vertebral column; supported by a cartilaginous rod).
 - *Mammals:* Restricted to compensatory hyperplasia (liver regeneration) and localized tissue repair (skin, bone healing).

(B) Kinds of Asexual Reproduction

Asexual reproduction involves a single parent without gametic fusion, resulting in genetically identical offspring (clones):

1. **Binary Fission:** Division of parent body into two equal daughter individuals (e.g., *Amoeba* - irregular; *Paramecium* - transverse; *Euglena* - longitudinal).
2. **Multiple Fission:** Parent nucleus divides repeatedly followed by cytoplasmic division into numerous daughter cells (e.g., *Plasmodium* schizogony).
3. **Budding (Gemmation):** Outgrowth (bud) develops on parent body, grows, and detaches:
 - *Exogenous Budding:* External buds in *Hydra*.
 - *Endogenous Budding:* Internal buds called **Gemmules** in freshwater sponges (*Spongilla*).
4. **Fragmentation:** Body breaks into multiple pieces, each developing into a full organism (e.g., *Spirogyra*, Planarians).
5. **Plasmotomy:** Division of a multinucleate protozoan into multinucleate offspring without mitosis (e.g., *Opalina*).

6. **Sporulation:** Formation of resistant spores under adverse conditions (e.g., *Amoeba* under encystment).

(C) Biogenetic Law

Formulated by **Ernst Haeckel** (1866), the Biogenetic Law states: "**Ontogeny recapitulates Phylogeny.**"

- **Concept:** Ontogeny (individual embryonic development) serves as a brief, condensed repetition of Phylogeny (evolutionary history of the species).
- **Classic Examples:**
 - Human embryonic development exhibits transient pharyngeal gill slits and a tail, reflecting aquatic fish-like and ancestral vertebrate stages.
 - Frog tadpole larva possesses gills and a two-chambered heart (fish-like), which transforms into a lung-breathing, three-chambered adult frog.
- **Modern Re-evaluation (Von Baer's Laws):** Karl Ernst von Baer modified this view, stating that embryos reveal general characters of the animal group early in development, while specialized characters appear later. Embryos do not pass through adult stages of ancestral forms, but rather through embryonic stages of ancestors.

(D) Tissue Reactivity and Induction Process

- **Embryonic Induction:** The process by which one embryonic tissue (the **Inductor** or **Organizer**) acts upon an adjacent responsive tissue to direct its developmental pathway.
 - *Spemann-Mangold Experiment (1924):* Transplanted dorsal lip of blastopore in newt embryos induced a secondary neural tube, establishing the dorsal lip as the **Primary Organizer**.
- **Tissue Reactivity (Competence):** The biochemical and structural physiological ability of a responder tissue to recognize, process, and execute an inductive signal.
 - Competence is stage-specific and time-bound. For instance, ectoderm is competent to respond to optic vesicle signals to form the lens only during a specific temporal window in early development.

Question 21

Q. Define growth? Give an account of the types and mechanism of growth.

1. Definition of Growth

Growth is defined as a fundamental, irreversible, quantitative increase in the total mass, volume, cell number, and protoplasmic substance of an organism over time, accompanied by metabolic processes.

2. Types of Growth

1. **Auxetic Growth:** Growth occurring purely by an increase in cell size without an increase in cell number (e.g., nematodes, rotifers, muscle cells).
2. **Hyperplastic (Proliferative) Growth:** Growth driven by continuous mitotic cell divisions increasing total cell number (e.g., early embryonic cleavage, meristematic tissues).

3. **Accretionary Growth:** Increase in extracellular matrix, non-living materials, or intercellular structural substances (e.g., bone matrix deposition, shell formation in mollusks).
4. **Differential Growth:** Unequal growth rates in different body organs leading to changes in body proportions during development (Allometric growth).
5. **Limited (Definite) vs. Unlimited (Indefinite) Growth:**
 - *Definite:* Animals reach a specific body size at maturity (e.g., birds, mammals).
 - *Indefinite:* Organisms grow continuously throughout life (e.g., fishes, reptiles, plants).

3. Mechanisms of Growth

The process of growth operates through three interconnected cellular phases:

Cell Division (Hyperplasia) → Cell Enlargement (Auxesis) → Maturation & Matrix Deposition (Accretion)

- **Cellular Phase (Hyperplasia):** Active mitotic cycles increase embryonic blastomere population. Controlled by cyclins, cyclin-dependent kinases (CDKs), and growth factors (e.g., EGF, FGF).
- **Protoplasmic Accumulation:** Anabolism exceeds catabolism ($A > C$). Protein synthesis rates rise significantly, expanding cytoplasm and organelle numbers.
- **Water and Solute Influx:** Osmotic uptake of water and cellular nutrient deposition expand individual cell boundaries.
- **Hormonal Control of Growth:**
 - **Growth Hormone (Somatotropin):** Stimulates cell division and protein synthesis via IGF-1 (Insulin-like Growth Factor-1).
 - **Thyroxine ($\mathbf{T_3/T_4}$):** Essential for tissue differentiation and basal metabolic regulation.
 - **Steroid Hormones (Androgens/Estrogens):** Regulate tissue maturation and epiphysis fusion in bones.

Question 22

Q. Describe morphogenetic processes.

1. Definition

Morphogenetic processes (Morphogenesis: *morphe* = form, *genesis* = origin) encompass the coordinated spatial organization, cell movements, structural transformations, and tissue folding patterns that give rise to the characteristic three-dimensional anatomical shape of an embryo.

2. Primary Morphogenetic Movements

During gastrulation, individual cells and tissue sheets undergo dynamic morphogenetic movements categorized into two broad types:

A. Epibolic Movements (Epiboly)

- Ectodermal tissue sheets spread externally over the embryo surface, enclosing inner

layers. Driven by radial intercalation and cell flattening.

B. Embolic Movements (Emboly)

Movement of presumptive mesodermal and endodermal cells from the exterior to the interior of the embryo:

1. **Invagination:** Infolding of a cell sheet into the blastocoel cavity (e.g., endoderm formation in *Amphioxus*).
2. **Involution:** Inturning of an outer expanding cell layer over an internal basal surface (e.g., amphibian dorsal lip blastopore invagination).
3. **Ingression:** Migration of individual blastomeres from the surface layer into the central cavity (e.g., primary mesenchyme cells in sea urchin gastrula).
4. **Delamination:** Splitting of a single sheet into two parallel layers (e.g., hypoblast formation in avian embryos).
5. **Convergent Extension:** Intercalation of cells along a narrow axis, causing the tissue sheet to elongate along one vector while narrowing along another (e.g., chordamesoderm axis elongation).

3. Cellular & Molecular Basis of Morphogenesis

- **Cell Adhesion Molecules (CAMs):** Cadherins (E-cadherin, N-cadherin) mediate cell-cell recognition and differential sorting, allowing similar cell types to group together.
- **Cytoskeletal Dynamics:** Actin filament contraction, apical constriction, and tubulin polymerization drive changes in cell shape (e.g., neural tube closure).
- **Extracellular Matrix (ECM) Interactions:** Fibronectin, collagen, and laminin guide migrating cells along specific developmental paths.
- **Morphogen Gradients:** Soluble signaling ligands (e.g., Activin, Sonic Hedgehog, BMPs, Wnt) diffuse through tissue, directing distinct spatial cell fates based on local concentration thresholds.

Question 23

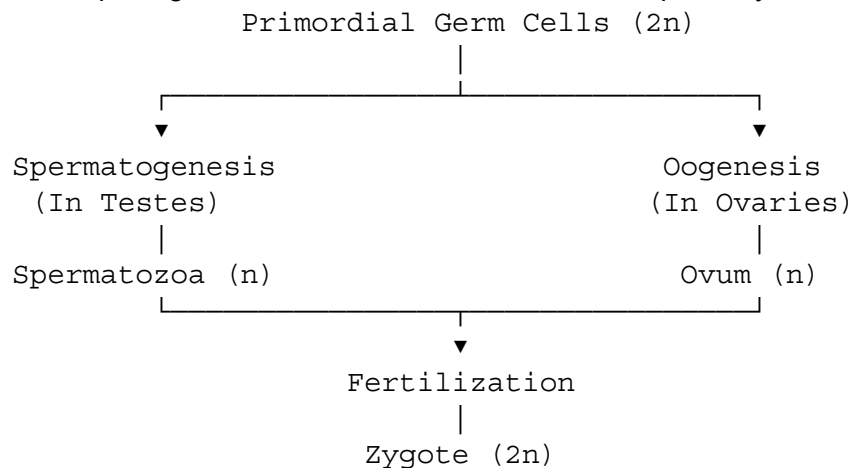
Q. What is sexual reproduction? Give an account of gametogenesis.

1. Definition of Sexual Reproduction

Sexual reproduction is a fundamental biological process involving the fusion of two haploid (n) specialized reproductive cells called gametes (sperm and ovum) through fertilization to form a single-diploid ($2n$) zygote. This zygote then undergoes mitotic divisions and differentiation to develop into a genetically distinct organism.

2. Overview of Gametogenesis

Gametogenesis is the process by which primordial germ cells (PGCs) undergo mitotic proliferation, meiotic reduction divisions, and functional differentiation in the gonads (testes and ovaries) to form haploid gametes. It is divided into two distinct pathways:



3. Comparative Stages of Gametogenesis

A. Spermatogenesis (Male Gametogenesis)

Occurs in the seminiferous tubules of testes and consists of two major phases:

1. Spermatocytogenesis (Formation of Spermatids):

- **Multiplication Phase:** Spermatogonia ($2n$) undergo repeated mitoses to form Type A (stem cells) and Type B spermatogonia.
- **Growth Phase:** Type B spermatogonia enlarge to become Primary Spermatocytes ($2n$).
- **Maturation Phase:** Primary spermatocytes undergo Meiosis-I to form two secondary spermatocytes (n). Each secondary spermatocyte undergoes Meiosis-II, producing four equal, haploid Spermatids (n).

2. Spermiogenesis (Spermatoleiosis):

- Transformation of non-motile, spherical spermatids into functional, flagellated Spermatozoa (n).

- **Key Changes:** Formation of acrosome from Golgi apparatus, condensation of nuclear chromatin, axial filament formation from centrioles, condensation of mitochondria into a spiral ring (**Nebenkern**), and shedding of excess cytoplasm as residual bodies.

B. Oogenesis (Female Gametogenesis)

Occurs in the ovarian cortex and proceeds through three stages:

1. **Multiplication Phase:** Oogonia ($2n$) multiply by mitosis during fetal development.
2. **Growth Phase:** Prolonged phase divided into:
 - *Pre-vitellogenesis:* Synthesis of ribosomal RNA, tRNA, mRNA, and organelle proliferation.
 - *Vitellogenesis:* Synthesis and accumulation of yolk (vitellin) in the egg cytoplasm (driven by liver-derived vitellogenin under estrogen regulation).
3. **Maturation Phase:**
 - Primary oocyte ($2n$) completes Meiosis-I, undergoing highly unequal cytokinesis to produce one large **Secondary Oocyte** (n) and a small **First Polar Body** (n).
 - Secondary oocyte undergoes Meiosis-II (often arrested at Metaphase-II until fertilization), yielding one large functional **Ovum** (n) and a **Second Polar Body** (n).

Question 24

Q. Write short notes on: (A) Cellular death (B) Polarity and gradient (C) Regeneration in Hydra (D) Maturation of gametes

(A) Cellular Death

Cellular death is an essential biological process maintaining tissue homeostasis, morphogenesis, and eliminating damaged or unwanted cells. It occurs via two primary pathways:

- **Apoptosis (Programmed Cell Death - PCD):**
 - Gene-directed, energy-dependent physiological process.
 - *Features:* Cell shrinkage, chromatin condensation, DNA fragmentation by endonucleases (laddering), membrane blebbing, and formation of apoptotic bodies phagocytosed by macrophages without inducing inflammation.
 - *Role:* Digit separation in tetrapod limbs, larval tail resorption during amphibian metamorphosis, neural pruning.
- **Necrosis (Accidental Cell Death):**
 - Pathological, unprogrammed cell death caused by physical trauma, hypoxia, toxins, or infection.
 - *Features:* Loss of membrane integrity, cell swelling (oncosis), lysis of organelles, leakage of cytoplasmic contents, leading to localized tissue inflammation.

(B) Polarity and Gradient

- **Polarity:** Refers to the spatial orientation and structural/functional differences established along specific body axes of an egg, embryo, or tissue (e.g., Anterior-Posterior, Dorsal-Ventral, Animal-Vegetal axes).

- *Animal Pole*: Cytoplasm-rich, contains the nucleus, site of polar body extrusion.
- *Vegetal Pole*: Yolk-rich, lower metabolic activity.
- **Gradient Theory (Child's Metabolic Gradient Theory)**:
 - Proposed by C.M. Child. Physiological and morphogenetic processes are controlled by quantitative differences in metabolic activity along an axis.
 - The region with the highest metabolic rate acts as the dominant regional organizer (e.g., apical region in *Hydra*), establishing a concentration gradient of morphogens (e.g., Bicoid gradient in *Drosophila* establishing anterior structures).

(C) Regeneration in Hydra

- *Hydra* displays extraordinary regenerative capacities via **Morphallaxis**—reorganization of existing somatic tissues without substantial cell proliferation.
- **Mechanism**:
 - If a *Hydra* is bisected horizontally, the upper half regenerates a base (peduncle/basal disc), while the lower half regenerates a head (hypostome and tentacles).
 - Driven by interstitial stem cells (i-cells) and epithelial stem cell layers.
 - **Head Activator and Head Inhibitor Gradients**: Head formation is governed by a high concentration of Head Activator peptide at the apical tip. The hypostome acts as an organizer, secreting a Head Inhibitor factor that prevents secondary head formation nearby.

(D) Maturation of Gametes

- Maturation is the final nuclear and cytoplasmic process converting immature germ cells into functionally competent gametes capable of fertilization.
- **Sperm Maturation**:
 - *Epididymal Transit*: Sperm acquire progressive motility and surface glycoprotein modifications in the epididymis.
 - *Capacitation*: Occurs inside the female reproductive tract; involves removal of decapacitating membrane cholesterol/glycoproteins, hyperactivation of motility, and priming for the acrosome reaction.
- **Oocyte Maturation**:
 - *Meiotic Resumption*: Driven by **Maturation Promoting Factor (MPF)** (Cyclin B + Cdk1). Luteinizing Hormone (LH) surge overcomes cAMP-mediated meiotic arrest at Prophase-I (Diplotene stage).
 - *Cytoplasmic Maturation*: Reorganization of cortical granules, synthesis of proteins necessary for sperm chromatin decondensation, and block to polyspermy.

Question 25

Q. Write an essay on environmental and genetic teratogenicity. Also mention the causes of teratogenesis.

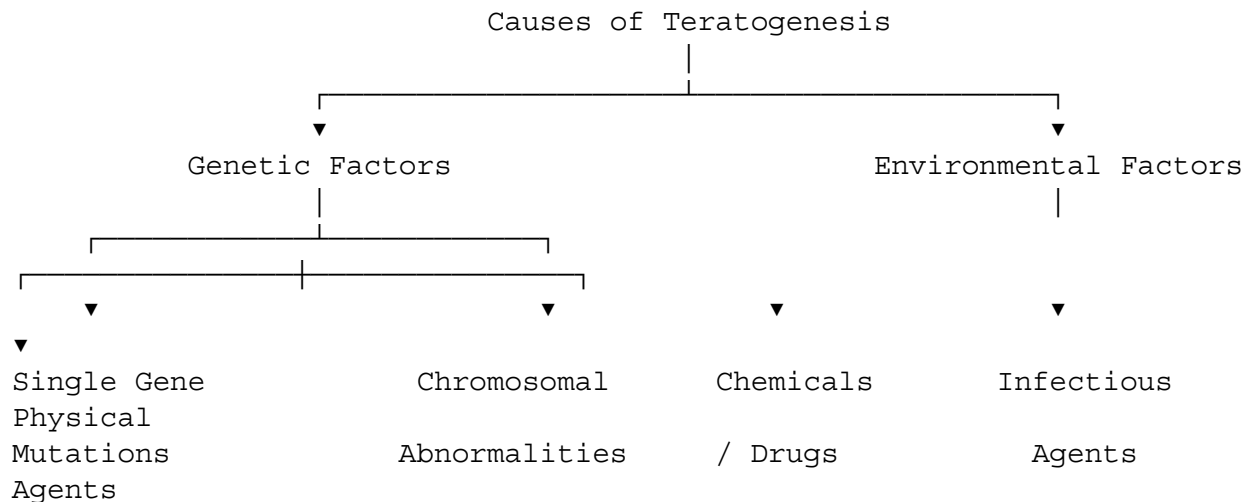
1. Introduction

Teratogenesis (Greek: *terato* = monster, *genesis* = origin) is the structural, functional, or

developmental deformation induced in an embryo or fetus during prenatal development, leading to congenital malformations. The agents that cause these defects are known as **teratogens**.

2. Causes of Teratogenesis

Teratogenic etiology is broadly classified into genetic factors, environmental factors, and their complex interactions.



A. Genetic Factors (Genetic Teratogenicity)

1. Gene Mutations:

- Mutations in critical developmental homeobox genes (e.g., *HOX*, *PAX*, *SHH*) disrupt structural patterning.
- *Example: Sonic Hedgehog (SHH) gene mutation causes Holoprosencephaly (failure of forebrain cleavage).*

2. Chromosomal Aberrations:

- *Aneuploidy: Trisomy 21 (Down Syndrome causing craniofacial and cardiac defects), Trisomy 18 (Edwards Syndrome).*
- *Structural Chromosomal Abnormalities: Deletions (e.g., Cri-du-chat syndrome: 5p deletion), translocations, and duplications.*

B. Environmental Factors (Environmental Teratogenicity)

1. Chemical Agents and Drugs:

- **Thalidomide:** Infamous cause of **Phocomelia** (severe reduction/absence of long limb bones) when ingested during weeks 4–8 of gestation.
- **Alcohol:** Causes **Fetal Alcohol Syndrome (FAS)**, characterized by microcephaly, facial dysmorphism, and intellectual disability.
- **Anti-epileptics (Valproic Acid):** Leads to neural tube defects (spina bifida).
- **Heavy Metals:** Methylmercury causes Minamata disease (neurological damage, cerebral palsy).

2. Infectious Agents (TORCH Complex):

- **Toxoplasma, Other (Syphilis, Varicella), Rubella, Cytomegalovirus, Herpes Simplex**

- Virus.
- *Rubella virus*: Causes congenital Rubella syndrome (cataracts, cardiac malformations, deafness).
- 3. **Physical Agents:**
 - **Ionizing Radiation (X-rays, Gamma rays)**: Induces chromosomal breaks, cell death, leading to microcephaly and central nervous system defects.
 - **Hyperthermia**: Maternal fever during early pregnancy can cause neural tube closure defects.
- 4. **Maternal Metabolic Factors:**
 - Uncontrolled maternal diabetes mellitus increases risk of congenital heart defects and Sacral Agenesis.

3. Principles of Teratogenesis (Wilson's Laws)

1. **Critical Period Concept**: Susceptibility to teratogens depends on the developmental stage at exposure. Organogenesis (weeks 3–8 in humans) represents the window of maximum susceptibility.
2. **Genotype Specificity**: Teratogenic response depends on the maternal and fetal genome (e.g., metabolic clearance capacity).
3. **Dose-Response Relationship**: Pathological manifestations increase in severity as the dose of the teratogenic agent increases.
4. **Specific Mechanisms**: Teratogens act via specific physiological mechanisms (e.g., inhibition of angiogenesis, induction of cell death, receptor antagonism).

Question 26

Q. Give an account of the process and mechanism of limb regeneration in amphibians.

1. Introduction

Urodele amphibians (salamanders, newts, axolotls) possess the unique ability among adult vertebrates to fully regenerate complex body structures like lost limbs via **epimorphic regeneration**. The regenerated limb is structurally and functionally identical to the lost appendage.

2. Stages of Limb Regeneration

Wound Healing → Epidermal Wound Cap → Blastema Formation → Differentiation & Redifferentiation

1. **Wound Healing Phase (0–24 Hours)**:
 - Upon amputation, epidermal cells adjacent to the wound rapidly migrate to cover the cut surface, forming a specialized **Wound Epidermis**.
 - Unlike mammal wound healing, scar formation (dermal collagen deposition) is suppressed.
2. **Formation of Apical Epidermal Cap (AEC)**:
 - The wound epidermis thickens into a multi-layered structure called the **Apical Epidermal Cap (AEC)** (analogous to the Apical Ectodermal Ridge in embryonic

limb buds).

3. **Dedifferentiation and Blastema Formation:**

- Tissues beneath the AEC (dermis, muscle, cartilage, nerve sheaths) undergo local histolysis driven by Matrix Metalloproteinases (MMPs).
- Differentiated cells lose their specialized features and regress into a population of pluripotent/multipotent mesenchymal stem cells, forming a **Regeneration Blastema**.

4. **Proliferation Phase:**

- AEC secretes Fibroblast Growth Factors (FGF-4, FGF-8), driving rapid mitotic proliferation of blastema cells.

5. **Redifferentiation and Morphogenesis:**

- Blastema cells condense and redifferentiate into cartilage, muscle, and vascular elements in a distal-to-proximal order, guided by HOX gene expressions and retinoic acid gradients.

3. Molecular Mechanism and Neural Dependence

- **Nerve Dependence:** Limb regeneration strictly requires intact nerve supply. Nerves supply **Anterior Gradient Protein (nAG)** and **Newt Glial Growth Factor (GGF)**, which stimulate blastema cell mitoses. If a limb is denervated prior to blastema formation, regeneration fails.
- **Patterning Factors:**
 - *Retinoic Acid (RA)*: Resets positional information. High RA concentrations respecify distal blastema cells into proximal identities (e.g., regenerating an entire arm from a wrist amputation).
 - *FGF & Wnt Signaling*: Maintain blastema outgrowth and cell survival.

Question 27

Q. Write notes on: (A) Parthenogenesis (B) Hormonal control of insect metamorphosis

A. Parthenogenesis

Parthenogenesis (Greek: *parthenos* = virgin, *genesis* = creation) is a form of asexual/monoparental reproduction in which an unfertilized egg develops into a complete functional individual without sperm involvement.

Classification:

1. **Natural Parthenogenesis:** Occurs normally in natural life cycles.
 - *Arrhenotoky*: Unfertilized eggs develop exclusively into males (e.g., Honeybee drones, wasps).
 - *Thelytoky*: Unfertilized eggs develop exclusively into females (e.g., *Aphids*, *Lacerta saxicola* lizards).
 - *Amphotoky*: Unfertilized eggs develop into individuals of either sex.
2. **Cytological Mechanism:**
 - *Ameiotic (Apomictic) Parthenogenesis*: Eggs do not undergo reductional meiosis; offspring are diploid clones of the mother.

- *Meiotic (Automictic) Parthenogenesis*: Meiosis occurs, but diploidy (2n) is restored via duplication of chromosomes or fusion of the egg nucleus with a polar body.
3. **Artificial Parthenogenesis**: Induced artificially in non-parthenogenetic eggs (e.g., Sea urchin, Frog) using chemical (dilute acids, urea), physical (needle puncture, electric shock), or osmotic stimuli.

B. Hormonal Control of Insect Metamorphosis

Insect metamorphosis is regulated by the balanced interplay of three primary hormones:

[Brain: Intercerebralis NSCs] → PTTH → [Prothoracic Gland] → Ecdysone

|

▼

Reprogramming

Epidermal

▲

[Corpora Allata] —————> Juvenile Hormone (JH)

1. **Prothoracicotropic Hormone (PTTH)**: Peptide hormone secreted by neurosecretory cells of the brain, stored in Corpora Cardiac. It stimulates the Prothoracic Gland to produce ecdysone.
2. **Ecdysone (Moulting Hormone)**: Steroid hormone produced by Prothoracic Gland. Converts to active 20-hydroxyecdysone, triggering apolysis, epicuticle deposition, and cell division in epidermal cells.
3. **Juvenile Hormone (JH)**: Sesquiterpenoid synthesized by Corpora Allata.
 - *High JH + Ecdysone*: Inhibits adult gene expression; maintains larval characteristics during moulting ($L_1 \rightarrow L_2$).
 - *Low JH + Ecdysone*: Permits expression of pupal genes ($L_{\text{last}} \rightarrow \text{Pupa}$).
 - *Absence of JH + Ecdysone*: Activates adult-specific genes, resulting in metamorphosis into the mature adult imago ($\text{Pupa} \rightarrow \text{Adult}$).

Question 28

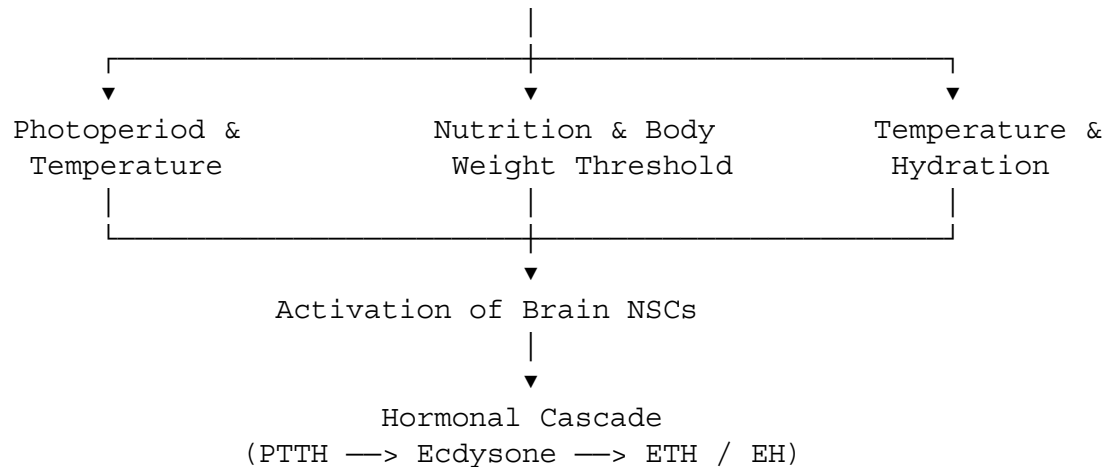
Q. Describe the factors controlling moulting in insects.

1. Definition

Moulting (Ecdysis) is the periodic shedding and replacement of the rigid chitinous exoskeleton (cuticle) to permit growth and body differentiation in insects.

2. External and Physiological Factors Controlling Moulting

Environmental & Internal Cues



A. Environmental / External Factors

- **Photoperiod:** Light-dark cycles influence circadian rhythms that regulate the release of neurosecretory factors from the protocerebrum.
- **Temperature:** Elevated ambient temperature increases metabolic rates, accelerating hormonal secretion and shorten the inter-moult duration (within physiological limits).
- **Tactile and Stretch Receptors:** Ingestion of food stretches the abdominal wall (e.g., in blood-feeding bugs like *Rhodnius prolixus*), firing stretch receptors that signal the brain to release PTTH.

B. Physiological / Hormonal Factors

1. **PTTH (Prothoracicotropic Hormone):** Initiates the endocrine cascade by activating prothoracic glands via cAMP second-messenger pathways.
2. **Ecdysteroids (20-hydroxyecdysone):** Directs epicuticle formation, secretion of moulting fluid containing chitinases/proteases, and digestion of the old endocuticle.
3. **Ecdysis-Triggering Hormone (ETH):** Released from Inka cells into the hemolymph; initiates pre-ecdysis motor programs (abdominal contractions).
4. **Eclosion Hormone (EH):** Secreted by central neurosecretory cells; acts synergistically with ETH to trigger swallowing of air/water and body inflation to split the old exoskeleton along the ecdysial line.
5. **Bursicon:** Neurohormone released post-ecdysis that controls sclerotization (tanning and hardening) and melanization of the newly formed cuticle.

Question 29

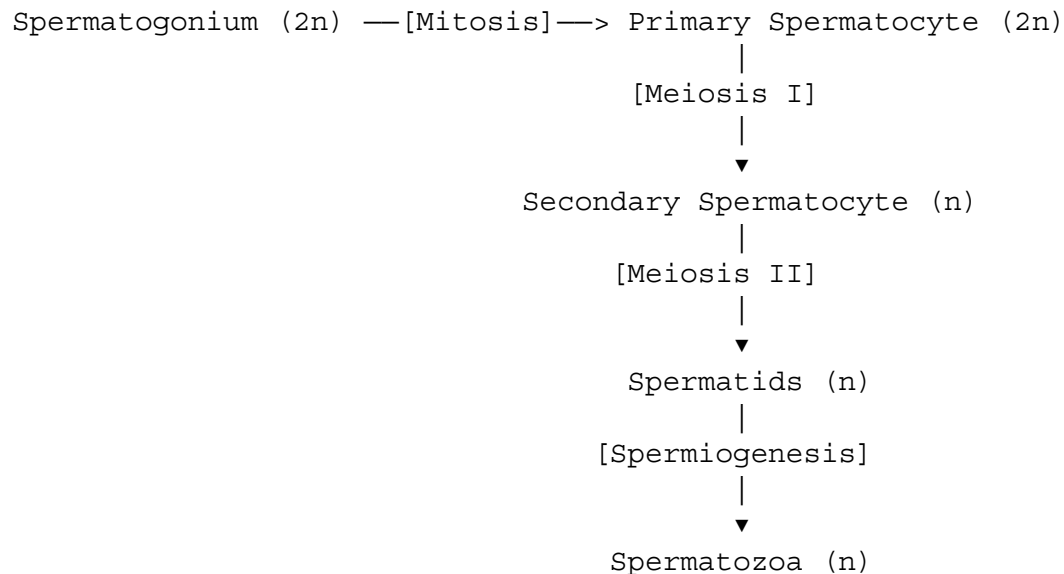
Q. Give an account of spermatogenesis.

1. Definition

Spermatogenesis is the complete developmental process by which primordial germ cells (spermatogonia) transform into mature, motile spermatozoa within the seminiferous tubules of

the male testes.

2. Detailed Phases of Spermatogenesis



1. Multiplication Phase (Spermatocytogenesis):

- Spermatogonia ($2n$) located at the basement membrane of seminiferous tubules undergo repeated mitotic divisions.
- *Type A Spermatogonia*: Function as stem cells.
- *Type B Spermatogonia*: Enter differentiation pathway and divide to yield primary spermatocytes.

2. Growth Phase:

- Primary Spermatocytes ($2n$) undergo cytoplasmic growth, DNA replication, and enter Prophase-I of meiosis.

3. Maturation Phase (Reduction Divisions):

- *Meiosis-I*: Primary Spermatocyte ($2n$) undergoes reduction division, dividing into two equal haploid **Secondary Spermatocytes** (n).
- *Meiosis-II*: Each Secondary Spermatocyte undergoes equational division to yield four equal, haploid **Spermatids** (n).

3. Spermiogenesis (Spermatid Metamorphosis)

Non-motile spherical spermatids transform into specialized spermatozoa through structural remodeling:

- **Acrosome Formation**: Golgi vesicles coalesce to form the acrosomal cap containing hydrolytic enzymes (hyaluronidase, acrosin) covering the anterior nucleus.
- **Nuclear Condensation**: Histones are replaced by protamines, tightly packing nuclear DNA into an osmiophilic dense mass.
- **Flagellar Development**: Distal centriole gives rise to the axial filament (9+2 microtubule pattern).
- **Mitochondrial Spiral**: Mitochondria align tightly around the anterior axial filament to form the middle piece power plant (**Nebenkern** sheath).

- **Cytoplasmic Reduction:** Excess cytoplasm is extruded as residual bodies and phagocytosed by **Sertoli cells**.

4. Hormonal Control

- **GnRH** (Hypothalamus) → **FSH & LH** (Anterior Pituitary).
- **LH:** Stimulates Leydig cells to produce **Testosterone**.
- **FSH:** Acts on Sertoli cells to release **Androgen-Binding Protein (ABP)**, concentrating testosterone within seminiferous tubules to drive spermatogenesis.

Question 30

Q. Differentiate between blastogenesis and embryogenesis.

(Note: Detailed tabular breakdown presented under Question 19A. Summary provided below for complete examination coverage).

Comparison Summary

Feature	Blastogenesis	Embryogenesis
Parental Source	Asexual development from somatic bud/blastema.	Sexual development from a fertilized diploid zygote.
Genetic Makeup	Clonal (100% identical to parent organism).	Genetically unique (recombination via meiosis/fertilization).
Cleavage	Absent; proceeds via somatic cell mitosis.	Present; highly organized cleavage patterns (holoblastic/meroblastic).
Blastula/Gastrula	Typical blastula and gastrula stages are bypassed.	Obligatory passing through blastula, gastrula, and germ layer formation.
Morphogenetic Movements	Minimal or highly modified morphogenetic sheet movements.	Well-defined morphogenetic movements (invagination, involution, epiboly).
Examples	Budding in <i>Hydra</i> , Gemmule growth in <i>Spongilla</i> .	Embryonic development of Frog, Chick, Human.

Question 31

Q. Write short notes on: (A) Oogenesis (B) Budding (C) Concept of growth

(A) Oogenesis

Oogenesis is the process of differentiation of female primordial germ cells into mature ova within the ovary. It features an extended growth phase and unequal meiotic divisions:

- **Multiplication:** Oogonia (2n) divide mitotically in fetal life.
- **Growth Phase:**
 - *Pre-vitellogenesis:* Amplification of rDNA, gene transcription, and organelle enrichment.

- *Vitellogenesis*: Accumulation of nutrient yolk proteins (vitellogenin) in cytoplasm.
- **Maturation Phase**: Meiosis-I produces one large **Secondary Oocyte** (n) and one small **First Polar Body** (n). Meiosis-II produces one functional **Ovum** (n) and a **Second Polar Body** (n). This unequal division ensures the ovum retains nearly all cytoplasm to sustain early embryogenesis.

(B) Budding

Budding is a form of asexual reproduction in which a new individual arises as an outgrowth (bud) from the parent body due to localized cell proliferation:

- **Exogenous Budding**: Outgrowth forms on the external body wall (e.g., *Hydra*). The bud develops a gut cavity, mouth, and tentacles continuous with the parent, eventually constricting at its base to detach as an independent individual.
- **Endogenous Budding**: Internal buds (gemmules) form within the parent tissue (e.g., freshwater sponge *Spongilla*). Archeocytes collect inside a protective chitinous envelope with spicules, surviving harsh environmental conditions before germinating through the micropyle.

(C) Concept of Growth

- Growth represents a fundamental, irreversible quantitative expansion of an organism's living mass, volume, and cell count over time.
- **Components**:
 1. *Hyperplasia*: Growth via cell proliferation (mitosis).
 2. *Auxesis*: Growth via expansion of individual cell volume.
 3. *Accretion*: Growth via secretion of extracellular matrix (e.g., bone matrix, cartilage).
- **Sigmoid Growth Curve**: Natural population/individual growth follows an S-shaped curve comprising: \text{Lag Phase (Slow initial growth)} \longrightarrow \text{Log Phase (Exponential growth)} \longrightarrow \text{Stationary Phase (Equilibrium)}
- Controlled by balanced metabolic synthesis (anabolism > catabolism) and governed by systemic growth hormones, thyroid hormones, and cell cycle checkpoints.

Question 32

Q. Differentiate between: (A) Ingression and Delamination (B) Induction and Competence

(A) Ingression vs. Delamination

Feature	Ingression	Delamination
Definition	Migration of individual cells from an epithelial sheet into an blastocoelic cavity.	Splitting of a single cellular layer into two parallel, distinct epithelial sheets.
Cell Behavior	Cells undergo Epithelial-to-Mesenchymal Transition (EMT) and move as solitary units.	Cells remain in sheets/layers and do not undergo individual mesenchymal migration.
Structural Output	Forms mesenchyme or diffuse	Forms continuous parallel

Feature	Ingression	Delamination
	mesodermal layers.	tissue sheets (e.g., hypoblast formation).
Example	Primary mesenchyme cell migration in Sea Urchin gastrula.	Hypoblast formation in chick blastoderm; mammalian hypoblast segregation.

(B) Induction vs. Competence

Feature	Embryonic Induction	Developmental Competence
Definition	Process by which one tissue (Inductor) signals an adjacent tissue to alter its fate.	Ability of a responder tissue to recognize, process, and respond to an inductive signal.
Agent / Role	Signal source (secretes signaling proteins like FGFs, BMPs, Wnts).	Signal recipient (expresses membrane receptors and transduction factors).
Mechanism	Release of chemical morphogens or direct cell-cell contact.	Presence of specific cell-surface receptors, signal cascades, and open chromatin loci.
Temporal Nature	Active signaling event.	Time-bound, stage-specific physiological window (competence can be lost over time).
Classic Example	Dorsal lip of blastopore inducing secondary neural tube formation.	Presumptive head ectoderm responding to optic vesicle signals to form lens.

Question 33

Q. Give a detailed account of growth curves and their mechanism with suitable examples.

1. Concept of Growth Curves

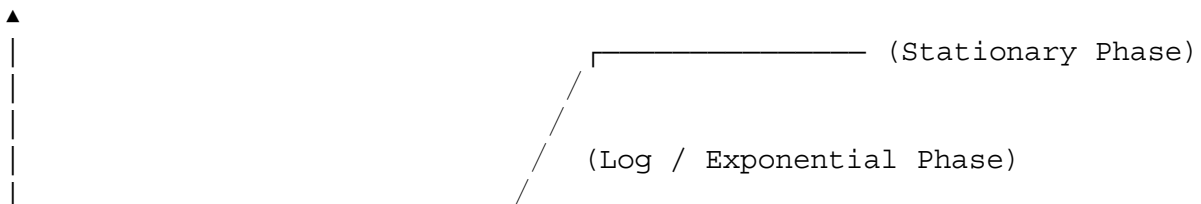
A growth curve is a graphical representation depicting the quantitative changes in an organism's size, mass, length, or cell number plotted against time during its developmental life cycle.

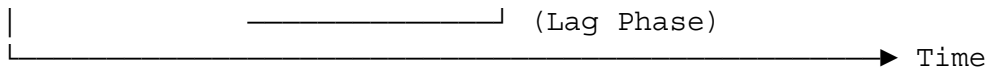
2. Types of Growth Curves

A. Sigmoid (S-Shaped) Growth Curve

Characteristic of whole organisms, organs, and population systems under controlled conditions.

Mass / Size



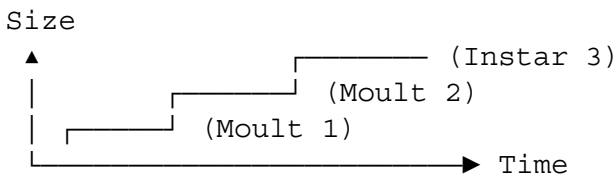


1. **Lag Phase:** Initial phase of slow growth. Cells adapt to the environment, synthesize enzymes, cytoplasm, and undergo preparation for division.
2. **Logarithmic (Exponential) Phase:** Phase of maximum growth velocity ($\Delta \log C$). Cells divide at a constant rapid rate under non-limiting nutrients.
3. **Decelerating Phase:** Growth slows down due to spatial restrictions, accumulation of metabolic wastes, or hormonal/genomic growth limits.
4. **Stationary Phase:** Growth reaches equilibrium ($\text{Anabolism} = \text{Catabolism}$); net size remains constant.

B. Staircase (Stepped) Growth Curve

Occurs in arthropods (e.g., Cockroach, Crab, Insects) undergoing ecdysis.

- **Mechanism:** Actual tissue mass increases continuously, but external linear dimensions (length/width) increase abruptly in steps during moulting (ecdysis) events, separated by flat inter-moult plateaus where the hard exoskeleton restricts physical expansion.



3. Mechanisms Governing Growth Curves

- **Cell Cycle Regulation:** Dynamic switching between G1, S, G2, and M phases regulated by Cyclins and CDKs.
- **Gompertz & Logistic Growth Models:** Mathematical models describing growth decay as mass increases.

Question 34

Q. Describe cellular and molecular mechanisms of regeneration in Hydra.

1. Introduction

Hydra (Phylum Cnidaria) represents a classical model organism for studying continuous pattern formation and morphallactic regeneration. It can regenerate a fully patterned body axis from small tissue fragments or even reaggregated cell suspensions.

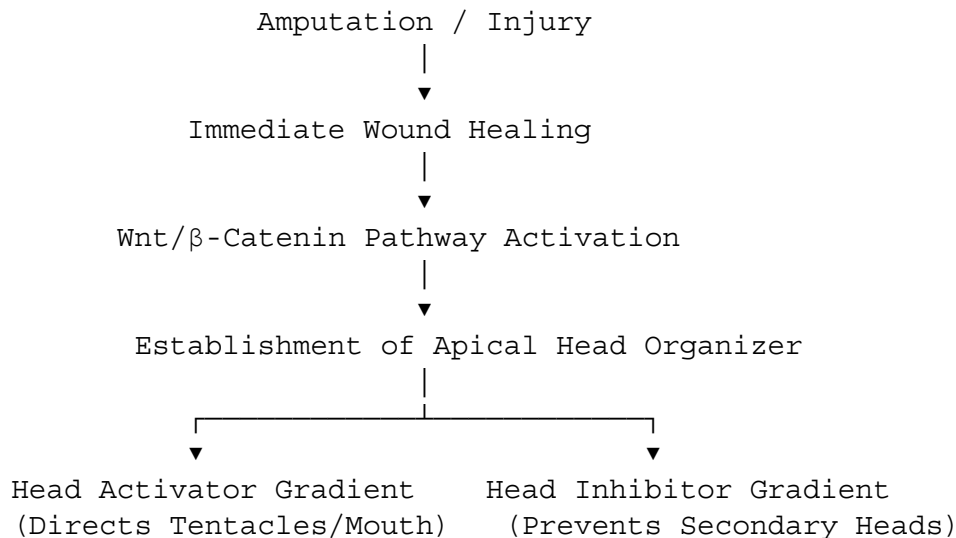
2. Cellular Mechanism: Morphallaxis

- **Stem Cell Lines:** *Hydra* tissue contains three non-senescent stem cell lineages:
 1. *Ectodermal Epithelial Stem Cells*
 2. *Endodermal Epithelial Stem Cells*
 3. *Interstitial Stem Cells (i-cells):* Multipotent stem cells residing in the ectodermal

layer that yield neurons, secretory cells, nematocytes, and germ cells.

- **Morphallactic Remodeling:** Regeneration does not rely on local blastema formation or widespread cell proliferation. Instead, existing body wall tissues undergo direct cellular transdifferentiation and structural spatial reorganization.

3. Molecular Mechanism and Signaling Centers



1. **Head Organizer Formation:**
 - Upon mid-gastric transection, the apical cut site rapidly activates the **Wnt/ β -catenin signaling pathway**.
 - High levels of nuclear β -catenin convert the cut edge into a **Head Organizer** center within hours.
2. **Head Activator vs. Head Inhibitor Morphogens:**
 - *Head Activator:* A 9-amino-acid peptide that stimulates head and tentacle differentiation at high local concentrations.
 - *Head Inhibitor:* A small, rapidly diffusible molecule secreted by the established head that suppresses head organizer formation in adjacent posterior tissues.
3. **Foot Patterning:** Governed by the **BMP/Smad signaling pathway** and **Peduncle Activator** factors at the basal terminal, inducing basal disc (foot) differentiation.

Question 35

Q. Write short notes on: (A) Senescence (B) Epimorphic regeneration (C) Gemmule formation

(A) Senescence

- Senescence is the progressive, irreversible physiological deterioration of cellular structure, metabolic efficiency, and functional capacity over time, ultimately culminating in natural death.
- **Cellular / Molecular Causes:**
 1. *Telomere Attrition:* Progressive shortening of chromosome telomeric ends during

successive mitoses acts as a biological clock, causing replicative arrest (**Hayflick Limit**).

2. *DNA Damage & Free Radicals*: Accumulation of Reactive Oxygen Species (ROS) from mitochondrial respiration causes oxidative damage to lipids, proteins, and nuclear/mtDNA.
3. *Accumulation of Waste (Lipofuscin)*: Indigestible "wear-and-tear" age pigments accumulate in lysosomes of non-dividing cells (e.g., neurons, cardiac muscle).
4. *Apoptosis dysregulation and stem cell exhaustion*.

(B) Epimorphic Regeneration

- Epimorphosis is a mode of regeneration involving active cell proliferation and the formation of a specialized stem cell mass called a **Blastema** prior to tissue replacement.
- **Process**: Amputation→Wound Epithelium→Dedifferentiation→Blastema} Morphogenesis
- **Characteristic Features**: Dedifferentiation of differentiated somatic tissues (muscle, bone, nerve) into pluripotent blastema cells, followed by redifferentiation and positional patterning.
- **Classic Examples**: Limb and tail regeneration in Urodele amphibians (salamanders, newts), starfish arm replacement.

(C) Gemmule Formation

- Gemmule formation is a specialized mode of internal asexual budding in freshwater sponges (e.g., *Spongilla*) adapted to survive unfavorable environmental conditions (drought, extreme cold).
- **Structure & Formation**:
 1. A mass of food-laden, totipotent **Archeocytes** aggregates in the mesohyl layer.
 2. Amoebocytes secrete a tough, protective inner and outer chitinous membrane surrounding the archeocyte mass.
 3. Scleroblasts deposit protective amphidisc **siliceous spicules** into the wall.
 4. The envelope leaves a tiny pore called a **Micropyle**.
- **Germination**: Upon restoration of favorable conditions, the inner archeocytes emerge through the micropyle, proliferate, and differentiate to establish a new sponge colony.

Question 36

Q. Describe the molecular mechanism of fertilization in animals.

1. Introduction

Fertilization is the species-specific cellular and molecular fusion of a haploid sperm with a haploid egg to restore diploidy and activate embryonic development.

2. Sequential Molecular Steps

A. Sperm Chemoattraction and Activation

- The egg jelly/cumulus layer releases species-specific chemoattractants (e.g., **Resact**, a

14-amino-acid peptide in sea urchins) that bind sperm membrane receptors, directing sperm swimming along a concentration gradient.

B. Primary Binding and Acrosome Reaction

- **Sperm-Zona Pellucida Binding:** Mammalian sperm bind species-specifically to the extracellular matrix (**Zona Pellucida**, specifically **ZP3 / ZP4 glycoproteins**).
- **Acrosome Reaction:** Binding triggers an influx of Ca^{2+} into the sperm head, causing exocytosis of the acrosomal vesicle. Hydrolytic enzymes (**Acrosin, Hyaluronidase**) digest a path through the Zona Pellucida.

C. Gamete Membrane Fusion

- Sperm membrane proteins (**IZUMO1**) interact specifically with egg plasma membrane receptors (**JUNO / CD9**), mediating plasma membrane coalescence.

Sperm-Egg Contact $\rightarrow$ Influx of Ca^{2+} $\rightarrow$ Cortical Reaction $\rightarrow$
Hardening of ZP (Slow Block)

D. Prevention of Polyspermy (Blocks to Polyspermy)

1. **Fast Block to Polyspermy (Electrophysiological):**
 - Binding triggers a rapid influx of Na^{+} ions into the egg, shifting resting membrane potential from -70mV to $+20\text{mV}$. Positive potential prevents further sperm fusion (predominant in sea urchins).
2. **Slow Block to Polyspermy (Cortical Reaction):**
 - Sperm entry triggers a wave of intracellular Ca^{2+} release from the endoplasmic reticulum.
 - Ca^{2+} surge causes **Cortical Granules** to fuse with the egg plasma membrane, releasing **Cortical Granule Protease** and **Peroxidase**.
 - Proteases cleave ZP3 receptors and destroy ZP2 links, converting the Zona Pellucida into a hardened **Fertilization Envelope** that mechanically prevents accessory sperm penetration.

E. Egg Activation and Pronuclear Fusion

- Intracellular Ca^{2+} wave reactivates the cell cycle by destroying Cyclin B, driving completion of Meiosis-II.
- Male and female pronuclei replicate DNA, migrate along microtubule asters, lose their nuclear envelopes, and intermingle chromatin (**Amphimixis**) to form the diploid zygotic genome.

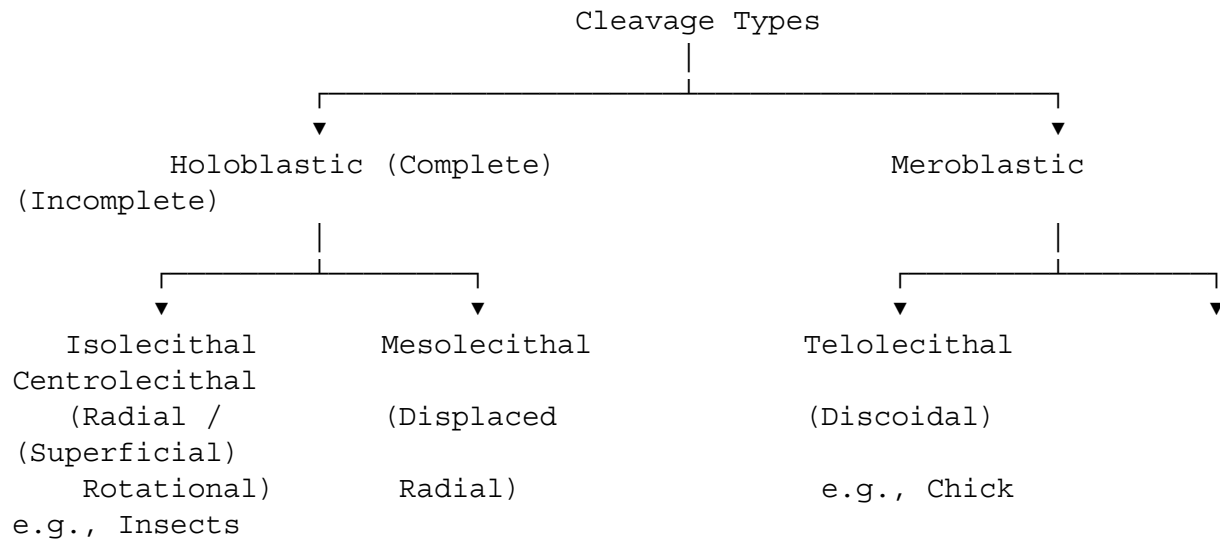
Question 37

Q. Write short notes on: (A) Cleavage patterns in eggs (B) Primary organizer

(A) Cleavage Patterns in Eggs

Cleavage consists of a rapid series of mitotic cell divisions transforming the unicellular zygote

into a multicellular blastula. The cleavage pattern is governed primarily by the quantity and distribution of yolk (**Lecithality**):



1. **Holoblastic (Complete) Cleavage:** Cleavage furrows slice completely through the egg:
 - *Isolecithal Eggs (Microlecithal):* Equal division. *Radial* (Sea urchin), *Spiral* (Mollusks), *Rotational* (Mammals).
 - *Mesolecithal Eggs:* Unequal holoblastic division producing micromeres and macromeres (Amphibians).
2. **Meroblastic (Incomplete) Cleavage:** Yolk prevents complete cleavage furrow penetration:
 - *Discoidal Cleavage:* Cleavage restricted to a small blastodisc at the animal pole (Birds, Reptiles, Fish).
 - *Superficial Cleavage:* Cleavage restricted to peripheral cytoplasm around a central yolk mass (Insects).

(B) Primary Organizer

- **Concept:** Discovered by **Hans Spemann** and **Hilde Mangold** (1924) in newt embryos (*Triturus*). The Primary Organizer refers to a region of early embryonic tissue that dictates the developmental fate and spatial organization of surrounding tissues.
- **Location:** In amphibians, it corresponds to the **Dorsal Lip of the Blastopore** (derived from the Nieuwkoop Center).
- **Classical Experiment:** Transplantation of the dorsal lip of a pigmented newt blastopore into the ventral region of an unpigmented host gastrula induced the host tissue to form a secondary, fully integrated neural tube and body axis ("siamese twin" larva).
- **Molecular Identity:** The organizer acts by secreting neuralizing factors (**Noggin**, **Chordin**, **Follistatin**) that bind and block **BMP-4** (Bone Morphogenetic Protein-4). Blocking BMP-4 prevents ectoderm from becoming epidermis, causing it to default into neural tissue.

Question 38

Q. Describe the fate maps of amphibian blastula with suitable diagrams.

1. Definition

A fate map is a topographical developmental diagram mapped onto the surface of an early embryo (blastula) that displays the specific adult or tissue regions that each group of blastomeres will normally give rise to during subsequent gastrulation and organogenesis.

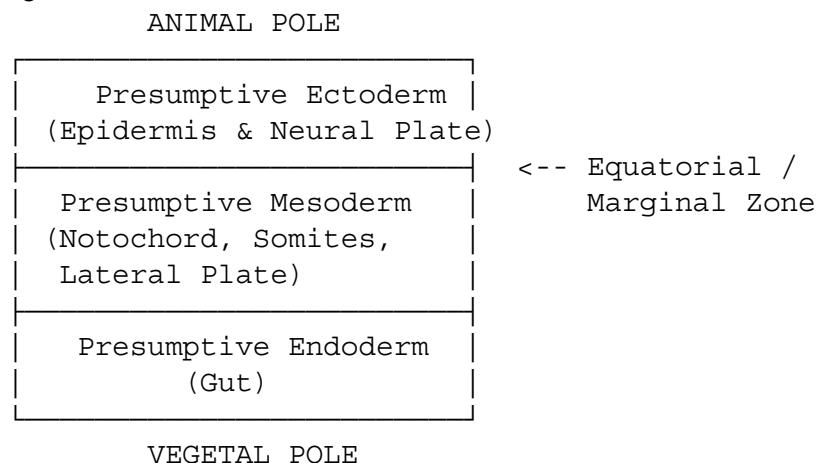
2. Construction Techniques

Fate maps are established using:

- *Vital Dye Marking (Vogt, 1929)*: Using Nile blue or Neutral red agar chips.
- *Fluorescent Lineage Tracers*: Injection of Rhodamine-dextran or GFP constructs into single blastomeres.
- *Carbon Particle Marking*.

3. Topography of Amphibian Blastula Fate Map

In an amphibian blastula (e.g., *Xenopus*, Frog), the fate map exhibits three fundamental zonal regions along the Animal-Vegetal Axis:



1. Animal Hemisphere (Presumptive Ectoderm):

- *Anterior/Apical Ectoderm*: Forms the outer skin **Epidermis** and sensory structures.
- *Dorsal Ectoderm (above dorsal lip)*: Forms the **Neural Plate** (Brain, Spinal Cord, Neural Crest).

2. Equatorial Zone / Marginal Zone (Presumptive Mesoderm):

- *Dorsal Marginal Zone*: Presumptive **Chordamesoderm** (Notochord).
- *Paraxial Marginal Zone*: Presumptive **Somitic Mesoderm** (Skeletal muscle, dermis, vertebrae).
- *Ventral & Lateral Marginal Zone*: Presumptive **Lateral Plate Mesoderm** (Splanchnic/Somatic layers, cardiovascular system) and **Intermediate Mesoderm** (Urogenital system).

3. Vegetal Hemisphere (Presumptive Endoderm):

- Large yolk-rich cells surrounding the vegetal pole give rise to the **Endodermal lining** of the digestive tract, liver, pancreas, and respiratory systems.

Question 39

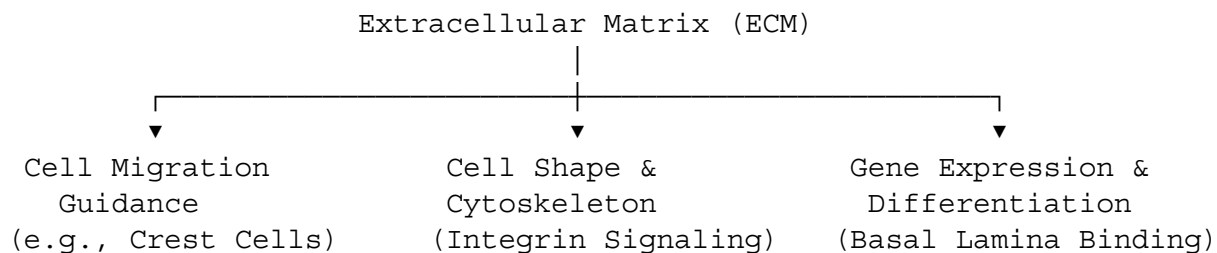
Q. Give a detailed account of extracellular matrix (ECM) and its role in morphogenesis.

1. Composition of the Extracellular Matrix (ECM)

The ECM is an intricate 3D network of extracellular macromolecules secreted locally by cells that provides structural scaffolding and biochemical cues. Its key components include:

- **Structural Fibrous Proteins:** *Collagen* (provides tensile strength) and *Elastin* (provides resilience).
- **Cell Adhesion Glycoproteins:**
 - *Fibronectin*: Dimeric glycoprotein with binding sites for cell-surface integrins, heparin, and collagen; guides cell migration paths.
 - *Laminin*: Primary component of the basal lamina; directs epithelial cell polarization and tissue sheet assembly.
- **Proteoglycans and Glycosaminoglycans (GAGs):**
 - *Hyaluronic Acid (Hyaluronan)*: High-molecular-weight GAG that hydrates tissue spaces, creating cell-free zones that facilitate cell migration.
 - *Chondroitin Sulfate & Heparan Sulfate*: Bind and present growth factors (e.g., FGFs, TGF- β) to cell surface receptors.

2. Role of ECM in Morphogenesis



1. Substrate for Cell Migration:

- Migrating embryonic cells (e.g., **Neural Crest Cells, Primary Mesenchyme Cells**) move along oriented fibronectin and laminin tracks in the ECM.
- Hyaluronic acid hydration expands intercellular spaces, creating physical pathways for ingressing and invaginating cell sheets during gastrulation.

2. Control of Cell Shape and Cytoskeletal Polarity:

- ECM molecules bind transmembrane cell receptors (**Integrins**).
- Integrins connect externally to ECM and internally to actin microfilaments via focal adhesion complexes (talin, vinculin), transmitting mechanical tension that alters cell shape, orientation, and tissue folding.

3. Regulation of Gene Expression and Cell Differentiation:

- Interaction of cell membrane integrins with laminin in the basement membrane delivers survival and differentiation signals (preventing **Anoikis**—detachment-induced apoptosis).
- *Example:* Mammary gland epithelial cells require basal lamina interaction to synthesize milk proteins (Casein).

4. Sequestration and Presentation of Morphogens:

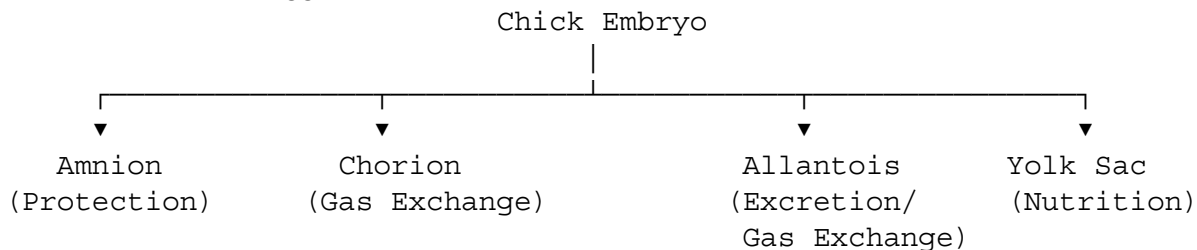
- Heparan sulfate proteoglycans in the ECM bind morphogens like FGF, Wnt, and Hedgehog, modulating their diffusion rates and establishing local concentration gradients essential for pattern formation.

Question 40

Q. Write short notes on: (A) Extra-embryonic membranes in chick (B) Role of calcium in fertilization

(A) Extra-Embryonic Membranes in Chick

Extra-embryonic membranes are four specialized extra-embryonic tissue structures formed by the chick blastoderm to support embryonic protection, respiration, nutrition, and waste storage inside the terrestrial egg:



1. Amnion:

- *Structure:* Inner somatopleure layer (Ectoderm + Somatic Mesoderm).
- *Function:* Encloses the fluid-filled **Amniotic Cavity**. Protects embryo from mechanical shocks, desiccation, and cell adhesion.

2. Chorion (Serosa):

- *Structure:* Outer somatopleure layer (Ectoderm + Somatic Mesoderm).
- *Function:* Encloses all internal extra-embryonic structures; fuses with allantois to form the **Chorioallantoic membrane** for respiratory gas exchange (O_2 / CO_2) through shell pores.

3. Allantois:

- *Structure:* Splanchnopleure outgrowth (Endoderm + Splanchnic Mesoderm) from the hindgut.
- *Function:* Acts as an embryonic urinary bladder storing toxic nitrogenous waste (**Uric Acid**). Highly vascularized for respiratory gas exchange and calcium absorption from the shell.

4. Yolk Sac:

- *Structure:* Splanchnopleure layer (Endoderm + Splanchnic Mesoderm) enclosing the yolk mass.
- *Function:* Highly vascularized with vitelline veins; secretes hydrolytic enzymes to digest yolk and transport nutrients to the growing embryo.

(B) Role of Calcium in Fertilization

Calcium (Ca^{2+}) acts as the central intracellular second messenger driving sperm activation, the block to polyspermy, and egg metabolic activation during fertilization:

1. Sperm Activation & Acrosome Reaction:

- Contact with egg jelly triggers an influx of extracellular Ca^{2+} into the sperm cytoplasm, driving membrane fusion and acrosomal enzyme exocytosis.

2. Cortical Reaction (Slow Block to Polyspermy):

- Sperm fusion stimulates Phospholipase C (PLC- γ), generating IP3.
- IP3 opens calcium channels on the endoplasmic reticulum, releasing a sweeping wave of free Ca^{2+} across the egg cytoplasm from the point of sperm entry.
- Elevated Ca^{2+} triggers exocytosis of cortical granules into the perivitelline space, hardening the vitelline envelope to block accessory sperm penetration.

3. Egg Metabolic Activation:

- The Ca^{2+} wave degrades Cyclin B via calmodulin activation, breaking meiotic arrest at Metaphase-II.
- Stimulates maternal mRNA translation, protein synthesis, DNA replication, and initiates first embryonic cleavage.

Question 41- Explain various causes for bird migration. How do they navigate during migration?

1. Introduction

Bird migration is a periodic, seasonal, and two-way journey of birds between their breeding grounds (northern or higher latitudes) and wintering/non-breeding feeding grounds (southern or lower latitudes). It represents a survival strategy shaped by evolutionary pressures to optimize foraging, reproductive success, and physiological maintenance.

2. Causes of Bird Migration

Bird migration is driven by a combination of ultimate (evolutionary) and proximate (physiological and environmental) factors:

- **Scarcity of Food and Water:** During winter, lower temperatures, snow cover, and frozen water bodies severely reduce food availability, such as insects, nectar, fruits, and small vertebrates. Birds migrate south to regions with abundant food reserves.
- **Reproductive Demands:** Northern breeding grounds offer long daylight hours during summer, rich insect blooms, abundant nesting space, and lower predator density, maximizing offspring survival.
- **Climatic Extremes:** Cold weather increases metabolic costs for thermoregulation in birds. Migration allows them to avoid harsh winters.
- **Photoperiodic and Hormonal Changes:** Increasing day length in spring stimulates the hypothalamus-pituitary axis. Pituitary hormones induce hyperphagia (excessive eating), leading to subcutaneous fat deposition and migratory restlessness (*Zugunruhe*).
- **Evolutionary Memory:** Historical glaciations forced bird species southward. As ice sheets receded, birds returned north annually, fixing migratory behavior into their genetic architecture.

3. Mechanisms of Navigation and Orientation

Birds use a multimodal orientation system combining environmental cues, sensory receptors, and internal clocks to navigate thousands of kilometers along established flyways.

Visual Landmarks → Topographic feature tracking (Rivers, Coasts)

Celestial Cues → Sun Compass (Azimuth + Internal Clock) & Star Compass (Constellations)

Geomagnetic Field → Cryptochromes in Retina & Magnetite Receptors in Beak

Olfactory Cues → Atmospheric Odor Map tracking

- **Visual Landmarks:** Diurnal migrants (birds traveling by day) rely on major topographic features such as river valleys, mountain ranges, coastlines, and islands as visual corridors.
- **Sun Compass:** Birds determine direction using the position of the sun (azimuth). Because the sun moves 15 degrees per hour from east to west, birds compensate for time using an internal circadian biological clock.

- **Star Compass:** Nocturnal migrants (birds traveling by night) navigate using star patterns around the northern center of rotation (Polaris). Experiments in planetariums show that young birds learn star configurations relative to rotation during early development.
- **Geomagnetic Orientation:** Birds possess magnetoreceptors capable of detecting the Earth's magnetic field strength and inclination angle. This mechanism operates via light-sensitive proteins called cryptochromes located in retinal cells, and tiny magnetite (iron oxide) particles embedded in the upper beak tissue connected to the trigeminal nerve.
- **Olfactory Navigation:** Pigeons and seabirds use spatial odor maps, recognizing regional atmospheric chemical profiles carried by winds to build navigational maps over familiar territories.
- **Infrasound Sensing:** Birds hear low-frequency acoustic waves generated by ocean waves, wind over mountain ranges, and seismic activity, using these infrasound patterns as acoustic land markers.

Question 42-Explain with examples the innate and learnt behaviour in animals.

1. Introduction

Animal behavior represents the coordinated responses of an organism to internal physiological changes and external environmental stimuli. Ethologically, behavior is categorized into two major classes: innate (instinctive) behavior and learned behavior.

2. Innate Behavior

Innate behavior refers to genetically programmed, non-learned response patterns that are stereotyped, species-specific, and fully functional upon the first exposure to a stimulus, even in complete isolation from conspecifics.

- **Characteristics:** Inherited through genetic code, resistant to environmental changes, uniform across members of a species, and developmentally fixed.
- **Components:**
 - **Kinesis:** Non-directional movement where stimulus intensity alters speed or turning frequency. *Example:* Woodlice (*Porcellio scaber*) move rapidly in dry areas and slow down in humid areas, causing them to aggregate in moist environments.
 - **Taxis:** Directional movement toward (positive) or away from (negative) a stimulus. *Example:* Positive phototaxis in *Euglena* swimming toward light; negative phototaxis in earthworms moving away from light.
 - **Fixed Action Pattern (FAP):** Invariable sequence of acts triggered by a specific sign stimulus or releaser. *Example:* The egg-retrieval behavior of the female Greylag Goose (*Anser anser*). If an egg rolls out of the nest, she extends her neck and rolls it back using precise bill movements. Even if the egg is removed midway, she completes the motion sequence.

3. Learnt Behavior

Learnt behavior involves adaptive modifications of individual behavior resulting from experience,

practice, or environmental interactions. It allows animals to adapt flexibly to changing environments.

- **Characteristics:** Acquired during an individual's lifetime, variable among individuals of a species, dependent on memory and neural plasticity, and reversible or subject to extinction.
- **Forms of Learning:**
 - **Habituation:** The simplest form of learning involving a progressive decrease in response to a repeated, neutral stimulus that carries neither reward nor punishment. *Example:* A snail (*Helix*) withdraws into its shell when its surface is tapped. Repeated tapping at regular intervals causes the snail to stop responding as it learns the stimulus is harmless.
 - **Classical Conditioning (Pavlovian Learning):** Learning by association, where a novel neutral stimulus becomes associated with an unconditioned stimulus to produce a conditioned response. *Example:* Ivan Pavlov's dogs associated the ringing of a bell (conditioned stimulus) with meat powder (unconditioned stimulus), eventually salivating at the sound of the bell alone.
 - **Operant Conditioning (Trial-and-Error Learning):** An animal learns to associate its voluntary behavior with a consequence (reward or punishment). *Example:* B.F. Skinner's experiments where a hungry rat accidentally presses a lever inside a box, receives a food pellet reward, and rapidly learns to press the lever intentionally.
 - **Imprinting:** A specialized, irreversible form of rapid learning occurring during a brief, restricted phase in early development known as the critical or sensitive period. *Example:* Konrad Lorenz demonstrated that newly hatched goslings follow the first moving object they see and hear (human, wooden box) and treat it as their mother.
 - **Insight Learning (Reasoning):** The most complex cognitive learning, where an animal solves a novel problem instantly without prior trial-and-error, using mental processing of spatial and causal relationships. *Example:* Wolfgang Köhler's chimpanzee (Sultan) stacked scattered wooden crates inside a room to reach bananas hanging high from the ceiling.

Question 43. Describe various methods adopted by animals for mate location and courtship.

1. Introduction

Reproductive success in animals depends on locating an appropriate mate of the same species, assessing genetic quality, synchronizing physiological readiness, and overcoming territorial or aggressive defenses. Animals have evolved diverse sensory mechanisms for mate location and courtship.

2. Methods Adopted for Mate Location

Mate location relies on long-range signals produced across visual, chemical, auditory, and electrical communication channels:

- **Chemical Signals (Sex Pheromones):** Many species release volatile chemical signals into air or water to attract mates across long distances. *Example:* Female Silk Moths (*Bombyx mori*) release a sex pheromone named **Bombykol**. Male moths use feather-like

antennae containing specialized chemoreceptors to detect minute concentrations of bombykol, flying upwind along the odor plume to find the female.

- **Auditory and Acoustic Calls:** Vocalizations travel through dense foliage and operate effectively in dark environments. *Example:* Male frogs and toads inflate vocal sacs to produce species-specific calls during breeding season. Female frogs evaluate call pitch, duration, and frequency to locate calling males around water bodies.
- **Visual Signals and Bioluminescence:** Visual attraction operates over moderate distances in open habitats or dark conditions. *Example:* Fireflies (*Photinus*) use species-specific bioluminescent flash codes produced by the enzyme luciferase in abdominal light organs. Flying males emit specific flash sequences, and perched females respond with synchronized flash signals.
- **Electrical Signals:** Weakly electric fish (*Gymnotiformes*) generate high-frequency electric organ discharges (EODs) creating electric fields. Differences in discharge frequency allow individuals to detect species identity, sex, and physical location in turbid river waters.

3. Courtship Mechanisms

Courtship encompasses short-range behaviors exchanged between potential mates prior to copulation. It serves three main functions: species recognition, suppression of aggression, and assessment of mate quality.

Visual Displays → Morphological features, plumage coloration, and dances

Nuptial Gifts → Pre-copulatory food offerings reducing predatory response

Acoustic Duets → Vocal synchronization confirming species and readiness

Tactile Stimulation → Allogrooming and physical contact cementing pair bonds

- **Visual Displays and Dances:** Animals display bright colors, specialized structures, and coordinated body movements. *Example:* Male Peacocks (*Pavo cristatus*) erect and vibrate their colorful tail coverts in a train display before peahens. Male Three-Spined Sticklebacks (*Gasterosteus aculeatus*) perform a zigzag courtship dance to lead females into their constructed nests.
- **Nuptial Gifts:** Males present food items to females during courtship. *Example:* Male Hangingflies (*Bittacus*) present captured prey insects to females. The female feeds on the gift during copulation; larger gifts allow longer copulation times and higher sperm transfer rates.
- **Acoustic Dueting:** Males and females engage in precise vocal exchanges. *Example:* In many bird species (Gibbons, Barbets), paired partners sing synchronized duets that maintain pair bonds, confirm physiological synchronization, and defend joint territories.
- **Tactile and Tactile-Chemical Courtship:** Physical contact reduces aggressive territorial responses. *Example:* Snakes engage in mutual body rubbing and chin pressing; male mammals lick female genital regions to analyze estrus-specific non-volatile pheromones via the vomeronasal organ.

Question 44. Write short notes on: (A) Social behaviour in honey bees

(B) Pheromones

(A) Social Behaviour in Honey Bees

Honey bees (*Apis mellifera*) are eusocial insects characterized by division of labor, overlapping generations, and cooperative brood care. A single hive contains tens of thousands of individuals divided into three morphological and functional castes:

- **Queen (Fertile Female):** One per colony. Possesses a large abdomen and functional ovaries. Her primary roles are laying eggs (up to 2,000 eggs per day) and secreting **Queen Substance** (9-hydroxy-2-decenoic acid), a pheromone that suppresses ovarian development in worker bees and maintains colony cohesion.
- **Drones (Fertile Males):** A few hundred per colony. Developed parthenogenetically from unfertilized eggs (haploid, n). Their sole function is to mate with a virgin queen during a nuptial flight. They lack stings and foraging structures, and are driven out of the hive before winter.
- **Workers (Sterile Females):** Form the vast majority of the colony (diploid, 2n). They exhibit age-dependent polyethism (division of labor based on age):
 - *Days 1–3:* Clean hive cells.
 - *Days 4–10:* Serve as nurse bees, feeding larvae with royal jelly secreted by hypopharyngeal glands.
 - *Days 11–20:* Build wax cells (using abdominal wax glands), process honey, and guard hive entrances.
 - *Days 21+:* Forage outside for nectar, pollen, water, and propolis.
- **Communication via Dance Behavior:** Discovered by Karl von Frisch. Forager bees communicate the location of food sources to hive mates using stylized dances on vertical honeycomb surfaces:
 - *Round Dance:* Performed if food is located within 50 meters of the hive. Indicates distance and food type without directional information.
 - *Waggle Dance:* Performed if food is further than 50 meters. The bee runs in a straight line while shaking its abdomen side-to-side, then loops back in a figure-eight pattern. The angle of the straight run relative to vertical gravity equals the angle between the food source and the sun. The frequency of abdominal waggles indicates exact distance.

(B) Pheromones

Pheromones are ectohormones—chemical substances synthesized and released by an animal into the environment that elicit specific behavioral or physiological responses in receiving individuals of the same species.

- **Classification by Function:**
 - **Sex Pheromones:** Attract members of the opposite sex for mating across distances. *Example:* Bombykol released by female silk moths; androstenol present in boar saliva.
 - **Alarm Pheromones:** Secreted when an animal faces predation or injury, warning conspecifics to flee or attack. *Example:* Honey bees release isopentyl acetate from sting glands during defense, drawing worker bees to sting the intruder.
 - **Trail Pheromones:** Deposited along substrate paths by foraging social insects to

guide nestmates to food resources or back to the nest. *Example:* Ants (*Formica*) deposit formic acid and specialized hydrocarbon trails between food sources and anthills.

- **Aggregation Pheromones:** Attract individuals of both sexes to a specific locality for feeding, mating, or overwintering shelter. *Example:* Bark beetles release aggregation pheromones to coordinate attacks on host trees.
- **Territorial / Marking Pheromones:** Used to establish home ranges and deter intruders. *Example:* Felids and canids deposit urine containing glandular scent markers along territory boundaries.
- **Mechanisms of Action:**
 - *Releaser Pheromones:* Produce immediate, rapid behavioral changes in the receiver via direct central nervous system signaling (e.g., alarm or trail pheromones).
 - *Primer Pheromones:* Trigger slow, long-term physiological and developmental changes in the receiver by altering endocrine pathways (e.g., Queen substance inhibiting worker ovary maturation).

Question 45. Write short notes on: (A) Imprinting (B) Territorial behaviour

(A) Imprinting

Imprinting is a specialized form of rapid, stable learning that occurs during a restricted, genetically determined period early in an animal's life, known as the **critical period** or **sensitive period**.

- **Key Characteristics:**
 - Occurs rapidly without explicit reinforcement or rewards.
 - Irreversible; information acquired during the sensitive window persists into adulthood.
 - Directs behaviors toward a broad class of stimuli rather than a single individual.
- **Types of Imprinting:**
 - **Filial Imprinting:** Process where young animals learn the social and structural characteristics of their parent. *Example:* Konrad Lorenz showed that newly hatched precocial birds (goose chicks, ducklings) form an attachment to the first moving object they observe (e.g., human boots, wooden models) between 13 and 16 hours post-hatching, following that object as their mother figure.
 - **Sexual Imprinting:** Early experience with parents determines future mate choices in adulthood. Young birds learn species-specific characteristics from their parents, directing their courtship toward similar-looking individuals later in life.
 - **Habitat / Host Imprinting:** Anadromous salmon imprint on the unique chemical profile of their home stream during early developmental stages, allowing them to navigate back from the open ocean years later to spawn in the same river.

(B) Territorial Behaviour

Territorial behavior encompasses the active defense of a fixed, bounded geographic area (territory) by an individual, breeding pair, or social group against conspecific intruders.

- **Functions of Territory:**
 - Secures exclusive access to limited food resources within the area.
 - Provides safe nesting, shelter, and offspring-rearing sites protected from predators.
 - Attracts potential mates by demonstrating higher territory quality and owner fitness.
 - Reduces population density pressures and controls disease transmission.
- **Costs vs. Benefits:** Defense requires significant energy expenditure and carries risk of injury. A territory is maintained only when the economic benefits of resource access outweigh the energetic costs of defense (Economic Defendability Model).
- **Defence Mechanisms:**
 - *Visual Displays:* Erecting crests, flashing bright plumage, or adopting threat postures at territory borders.
 - *Vocalizations:* Singing in male songbirds advertises territory ownership over long distances.
 - *Scent Marking:* Depositing glandular secretions, urine, or feces along boundaries to mark spatial ownership without requiring constant physical presence.

Question 46. What do you understand by territory marking? How do animals defend them?

1. Concept of Territory Marking

Territory marking is the strategic deposition of visual, acoustic, or chemical signals across a geographic region to advertise ownership, define boundaries, and deter potential conspecific intruders without engaging in physical combat. Marking communicates the owner's presence, species identity, sex, and competitive status.

2. Modes of Territory Marking

Animals use three main sensory channels to mark their territories:

Chemical Marking → Deposition of scent signals (Urine, Feces, Glandular Secretions)

Acoustic Marking → Long-range vocalizations (Bird song, Howler monkey calls)

Visual Marking → Environmental modifications (Tree scrapes, Antler rubs, Bowers)

- **Chemical Scent Marking:** Common among mammals.
 - *Urine & Feces:* Dogs, wolves, and tigers deposit urine at elevated landmarks along border trails. Hippopotamuses spread feces using tail flicking to mark river territories.
 - *Specialized Glandular Secretions:* Lemurs possess brachial skin glands; antelopes have preorbital glands near their eyes to rub scent onto twigs; cats use facial glands to scent-mark objects.
- **Acoustic Marking:**
 - Male songbirds sing from high perches around their territory borders, creating acoustic boundaries that reduce physical trespass.
 - Howler monkeys (*Alouatta*) use enlarged hyoid bones to produce loud communal

vocalizations, establishing spatial spacing between neighboring troops in dense forest canopies.

- **Visual Marking:**
 - Bears scratch tree bark and rub hair onto trunks to signal body size to rivals.
 - Male antelopes scrape topsoil with their horns and hooves; male Bowerbirds construct elaborate visual bowers decorated with colored objects to define display grounds.

3. Mechanisms of Territory Defence

Territory defense follows a progressive, ritualized escalation gradient to minimize energetic costs and injury risks:

- **Step 1: Passive Advertising (Scent and Acoustic Signals):** Initial boundary defense relies on established scent posts or vocalizations. Intruders often turn back upon encountering fresh owner scent marks.
- **Step 2: Threat Displays and Posturing:** If an intruder enters, the owner approaches and engages in threat displays:
 - Pilomotor reaction (raising fur or feathers to appear larger).
 - Displaying weapons (bared canine teeth, raised horns, opened bills).
 - Engaging in stereotyped ritualized walks or lateral displays along boundary lines.
- **Step 3: Escalated Vocalizations and Mock Charges:** The owner issues aggressive calls (growls, roars) and performs short, rapid charges stopping just short of contact.
- **Step 4: Physical Combat:** Used as a last resort when rivals are equally matched or when territory value is extremely high. Involves biting, goring, wrestling, or ramming. Fights end when the subordinate individual adopts submission postures (e.g., exposing the throat or belly) and retreats from the area.

Question 47. Define migration. Write a brief essay on fish migration.

1. Definition of Migration

Migration is the regular, mass, directional, and seasonal movement of animal populations from one habitat to another, typically driven by reproductive requirements, food availability, or seasonal climate shifts, followed by a return journey to the original home habitat.

2. Types of Fish Migration

Fish migrations are classified based on the aquatic environments involved and the biological purpose of the movement:

Diadromous → Migrates between Freshwater and Ocean

└─ Anadromous : Marine → Spawning in Freshwater (e.g., Salmon)

└─ Catadromous : Freshwater → Spawning in Ocean (e.g., Anguilla)

└─ Amphidromous: Moves between environments for non-reproductive reasons

Potamodromous → Migrates entirely within Freshwater systems (Rivers → Lakes)

Oceanodromous → Migrates entirely within Marine systems (Open Ocean → Coasts)

A. Diadromous Migration

Involves crossing the saline gradient between freshwater and marine environments:

- **Anadromous Migration:** Fish live and grow in oceanic saltwater but migrate into freshwater rivers to spawn.
 - *Example:* Atlantic Salmon (*Salmo salar*). Adult salmon live in the ocean for several years, then swim upstream into rivers, navigating rapids and waterfalls to reach gravel spawning beds. After spawning, most adults die, while hatched larvae (**parr** and **smolt** stages) return downstream to the sea.
- **Catadromous Migration:** Fish live and mature in freshwater rivers and lakes, but migrate down to deep ocean waters to spawn.
 - *Example:* European Freshwater Eel (*Anguilla anguilla*). Adult eels live in rivers for 10–15 years. Upon reaching sexual maturity, they turn silver, migrate thousands of kilometers across the Atlantic Ocean to the **Sargasso Sea**, spawn at great depths, and die. Transparent leaf-like larvae (**leptocephali**) drift along ocean currents back to European river mouths, metamorphosing into **glass eels** and **elvers** that ascend freshwater streams.
- **Amphidromous Migration:** Fish move between fresh and salt water during specific developmental stages, but not for reproductive purposes (e.g., Gobies).

B. Potamodromous Migration

Migratory movements restricted entirely within freshwater systems, such as moving between main river channels, tributaries, and lakes for feeding or overwintering (e.g., *Carps*, *Trout*).

C. Oceanodromous Migration

Migratory movements restricted entirely within marine environments, often traveling thousands of kilometers across ocean basins following warm currents and plankton blooms (e.g., *Tuna*, *Herring*, *Cod*).

3. Factors Influencing Fish Migration

- **Physiological Factors:** Pituitary gland secretions, thyroid hormones (T₃/T₄), and gonadal steroid accumulation stimulate migratory drive and induce osmoregulatory adaptations (adjusting gill chloride cells) needed to transition between fresh and salt water.
- **Physical & Environmental Factors:** Water temperature shifts, water current direction (**rheotaxis**), photoperiod variations, salinity gradients, and water pH levels initiate migration.

4. Navigational Mechanisms in Fish

Fish orient using multiple sensory inputs:

- **Olfaction:** Salmon detect the unique chemical signature of their natal stream down to parts-per-billion levels using sensitive olfactory organs.
- **Rheotaxis:** Orientation relative to water flow; fish swim against currents (**positive**

- **rheotaxis**) during upstream spawning runs.
- **Ocean Currents & Temperature:** Eels use deep ocean currents and thermal gradients to navigate across open sea basins.
- **Geomagnetic Sensitivity:** Magnetite particles in sensory tissues allow fish to navigate along Earth's magnetic field lines across open ocean waters.

Question 48. Describe briefly the methods used in ethological studies.

1. Introduction

Ethology is the scientific study of animal behavior under natural or controlled conditions. Observing, recording, and analyzing complex behavioral sequences requires systematic, objective, and quantitative methodologies.

2. Observational Methods in Natural Environments

Direct field observation minimizes artificial interference, capturing natural behavioral routines:

- **Ethogram Construction:** An ethogram is a comprehensive inventory of all distinct, standardized behavioral patterns displayed by a species. Researchers define behavioral categories objectively (e.g., "head-bob," "tail-flick") to ensure consistent scoring.
- **Sampling Strategies:**
 - *Focal Animal Sampling:* The observer focuses on a single individual for a specified time block, recording all its behavioral actions and social interactions.
 - *Instantaneous (Scan) Sampling:* The observer scans an entire group of animals at regular intervals (e.g., every 5 minutes) and records the current activity of each individual. Useful for estimating time budgets.
 - *Ad Libitum Sampling:* Informal, non-systematic recording of noticeable or rare events within a group (e.g., sudden predator attacks).
- **Field Equipment:** High-resolution video cameras, telephoto lenses, night-vision infrared scopes, thermal imaging, and parabolic directional microphones for acoustic recording.

3. Experimental and Laboratory Methods

Laboratory setups isolate environmental variables to test specific physiological or environmental hypotheses:

- **Choice Boxes and Mazes:** T-mazes, Y-mazes, and radial arm mazes evaluate learning, memory, spatial navigation, and stimulus preferences under controlled conditions.
- **Controlled Stimulus Presentation (Dummy/Model Tests):** Artificial models vary specific visual features to identify sign stimuli triggering fixed action patterns. *Example:* Niko Tinbergen used cardboard models of sticklebacks with painted red undersides to identify the visual trigger for aggressive territorial displays in male sticklebacks.
- **Olfactometers:** Dual-choice or multi-choice air chambers evaluate animal responses to volatile chemical stimuli and pheromones.

4. Neurophysiological and Marking Techniques

Advanced methods track individual identity, spatial movement, and underlying neural mechanisms:

- **Individual Marking:** Banding/tagging birds, applying non-toxic paint codes on insect thoraxes, freeze-branding mammals, and inserting Passive Integrated Transponder (PIT) microchips under the skin.
- **Telemetry and GPS Tracking:** Attaching lightweight radio transmitters, satellite collars, or geolocators to track migration routes and home-range movements across vast geographic areas.
- **Neuroendocrine Interventions:**
 - *Hormone Administration:* Injecting hormones or hormone receptor blockers to analyze endocrine influences on aggressive or reproductive behavior.
 - *Stereotaxic Brain Stimulation & Lesioning:* Micro-electrodes stimulate or lesion specific brain nuclei (e.g., hypothalamus) to identify neural circuits controlling specific behaviors.
 - *Optogenetics:* Using light to activate or inhibit genetically modified neurons in real time to observe direct behavioral changes.

Question 49. Write short notes on: (A) Nesting behaviour (B) Chemical defence

(A) Nesting Behaviour

Nesting behavior encompasses the specialized construction of structures (nests) by animals to protect eggs, developing offspring, and parents from physical weather hazards and predation.

- **Widespread Occurrence:**
 - *Insects:* Termites construct complex clay mounds featuring internal ventilation channels; solitary wasps build mud chambers; leaf-cutter ants construct subterranean fungus-growing chambers.
 - *Fishes:* Male Three-spined Sticklebacks use kidney-secreted **mucoprotein (spiggin)** to bind aquatic vegetation into a tunnel nest. Male Gouramis blow floating air-bubble nests coated with mucus to hold eggs.
 - *Amphibians & Reptiles:* Foam-nesting frogs whip skin secretions into protective foam masses over water. Sea turtles dig pit nests in sandy beaches to incubate buried eggs.
 - *Birds:* Exhibit high structural diversity, including simple ground scrapes (Terns), mud cup nests (Swallows), woven suspended baskets (Weaver birds), and tree cavity excavations (Woodpeckers).
 - *Mammals:* Rodents dig multi-chambered underground burrows; Orangutans construct fresh leafy sleeping platforms daily in the forest canopy.
- **Functions:** Provides microclimate control (regulating temperature and humidity), reduces egg and offspring mortality from predators, and serves as a visual focus for courtship and parental care.

(B) Chemical Defence

Chemical defense involves the synthesis, sequestration, or release of toxic, noxious, or irritating chemical compounds by prey species to deter predators.

- **Mechanisms and Types:**
 - **Endogenous Toxins:** Synthesized internally by specialized glands. *Example:*

Toads (*Bufo*) possess enlarged **parotoid glands** behind their eyes that secrete **bufotoxin**, a cardiac poison that causes severe mucosal irritation and cardiac arrest in predators.

- **Exogenous Sequestration:** Toxins acquired from food sources and stored safely within body tissues. *Example:* Monarch Butterfly larvae (*Danaus plexippus*) feed on milkweed plants (*Asclepias*), sequestering toxic **cardiac glycosides**. Adult butterflies retain these toxins, causing violent vomiting in birds that attempt to eat them.
- **Repellent Spray Systems:** Directional discharge of noxious chemicals. *Example:* The Bombardier Beetle (*Brachinus*) stores hydroquinones and hydrogen peroxide in separate abdominal reservoirs. When threatened, it mixes them with catalytic enzymes inside a reaction chamber, firing a boiling (100°C), toxic benzoquinone spray toward the attacker.
- **Emanating Odors:** Skunks (*Mephitis*) spray sulfur-containing compounds (**thiols**) from anal glands, causing temporary blindness and intense nausea in predators up to several meters away.
- **Aposematism:** Animals with chemical defenses often display bright, highly visible warning coloration (red, yellow, black patterns) to communicate their unpalatability to visual predators.

Question 50.. Describe in detail the various methods of nest building in animals. Also add a note on nesting behaviours.

1. Introduction

Nest building is an adaptive behavioral pattern that creates a micro-environment optimized for egg incubation, offspring survival, and protection against environmental extremes and predators. Nest building ranges from simple ground scrapes to intricate architecture constructed using specialized materials and instinctive behavioral programs.

2. Methods of Nest Building in Different Animal Groups

A. Invertebrates

- **Burrowing and Excavation:** Solitary mining bees and digger wasps excavate subterranean shafts in loose soil, forming terminal brood chambers packed with paralyzed prey insects.
- **Use of Secreted Materials:** Paper wasps (*Polistes*) chew wood fiber, mix it with salivary enzymes, and construct hexagonal paper-pulp cells. Honey bees secrete lipid wax from abdominal sternal glands, molding it into vertical honeycomb grids.
- **Leaf Sewing:** Weaver Ants (*Oecophylla*) construct nests by pulling live tree leaves together. Worker ants carry silkworm-like larvae in their mandibles, gently squeezing them to produce liquid silk threads that bind leaf edges together.

B. Fishes

- **Bubble Nests:** Surface-dwelling fishes (e.g., *Betta splendens*, Siamese Fighting Fish) engulf air and release saliva-coated air bubbles that coalesce into a floating bubble raft.

Eggs are placed inside the bubble matrix and guarded by the male.

- **Excavated Depressions:** Salmon and trout dig gravel depressions called **redds** using tail-flicking motions, laying eggs in the gravel before covering them with stones.
- **Woven Plant Nests:** Male Sticklebacks collect fine algae, binding them with sticky kidney secretions (**spiggin**) to construct hollow, tubelike nests.

C. Amphibians and Reptiles

- **Foam Nests:** Tropical tree frogs (*Rhacophorus*) whip cutaneous secretions and fluid into a firm foam mass attached to leaves hanging over water bodies. Hatched tadpoles drop directly into the water below.
- **Buried Nest Chambers:** Crocodilians construct large mounds of rotting vegetation and mud. The heat generated by plant decomposition incubates the eggs inside the chamber.

D. Birds

Birds demonstrate sophisticated nest-building techniques:

Scrape Nests → Simple ground depressions lined with pebbles
(Terns, Plovers)

Mud Cup Nests → Pellets of wet mud reinforced with grass
(Swallows)

Cavity Nests → Excavated tree trunks (Woodpeckers) or dirt banks
(Kingfishers)

Woven Nests → Intricately knotted grass strips suspended from
branches (Weaver birds)

- **Scrape Nests:** Minimalist depressions made on bare soil or gravel, often camouflaged by surrounding stones (e.g., Ostrich, Plovers).
- **Mud Nests:** Barn Swallows collect wet mud pellets in their beaks, building cup-shaped nests on vertical walls, reinforced with grass and saliva.
- **Excavated Cavities:** Woodpeckers use their chisel-like bills to carve nest chambers inside live or dead tree trunks.
- **Woven Hanging Nests:** Male Baya Weaver Birds (*Ploceus philippinus*) strip long fibers from palm leaves, weaving intricate helmet-shaped or flask-shaped hanging nests featuring entrance tubes.

E. Mammals

- **Subterranean Burrows:** Rabbits dig extensive subterranean tunnel systems called warrens, featuring specialized grass-lined birthing chambers.
- **Arboreal Nests (Dreys):** Tree squirrels build spherical leaf and twig nests (dreys) in high tree forks. Gorillas construct fresh sleeping nests on the ground or in tree branches each night using folded branches.

3. Ethological Note on Nesting Behaviour

Nesting behavior is driven by an interaction between innate Fixed Action Patterns (FAPs) and hormonal states. Rising levels of prolactin and estrogen induce nest-building behaviors in birds and mammals.

- Nest construction follows rigid, stereotyped motor routines (e.g., tucking grass blades under the body).
- However, experience refines construction efficiency; older birds select sturdier materials and build structurally stronger nests than first-time builders.
- Nesting sites are selected based on microclimate features (shelter from prevailing winds) and camouflage, minimizing detection by predators.

Question 51. Write short notes on: (A) Camouflage (B) Fixed action pattern (C) Social hierarchy

(A) Camouflage

Camouflage (Crypsis) is an antipredator adaptation where an animal uses visual matching of color, pattern, shape, or behavior to blend into its surrounding environment, preventing detection by predators or prey.

- **Mechanisms of Camouflage:**
 - **Concealing Coloration (Background Matching):** Animal color matches the dominant visual backdrop. *Example:* Arctic foxes and ptarmigans turn pure white in winter to match snow, and brown during summer tundra conditions.
 - **Disruptive Coloration:** High-contrast patterns break up the continuous outline of an animal's body shape, confusing predator visual perception. *Example:* Striped coat patterns of Zebras and spots on Leopards.
 - **Countershading (Obliterative Coloration):** The dorsal surface is darkly pigmented while the ventral surface is light. Light falling from above makes the animal appear flat rather than three-dimensional. *Example:* Sharks and penguins (dark back matches deep water from above; white belly matches sunlit surface from below).
 - **Masquerade (Mimesis):** The animal morphologically resembles an inedible environmental object rather than blending into the background. *Example:* Stick insects (*Carausius*) resemble twigs; Kallima butterflies resemble dead leaves complete with midrib patterns.

(B) Fixed Action Pattern (FAP)

A Fixed Action Pattern (FAP) is an unlearned, innate, highly stereotyped behavioral sequence initiated by a specific external stimulus called a **Sign Stimulus** or **Releaser**. Coined by Konrad Lorenz and Niko Tinbergen, FAPs form the foundation of classical ethology.

- **Key Characteristics:**
 - *Stereotyped:* The behavioral sequence is uniform in movement pattern every time it is performed.
 - *Universal:* Shared by all normal members of a species of the same sex and age class.
 - *Independent of Learning:* Performed correctly on the very first exposure without prior practice.
 - *Self-Exhausting / Ballistic:* Once triggered by a sign stimulus, the FAP continues to completion even if the triggering stimulus is removed or modified midway.
- **Classic Example:** Egg-retrieval in the female Greylag Goose (*Anser anser*). If an egg

rolls outside her nest, she extends her neck, places her bill behind the egg, and rolls it back into the nest using precise side-to-side balancing movements. If the egg is pulled away by an experimenter mid-movement, the goose continues the neck-retracting and bill-balancing motor sequence all the way back to the nest.

(C) Social Hierarchy

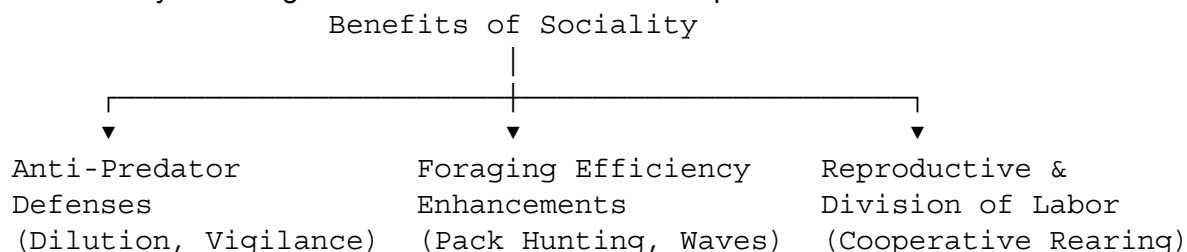
A social hierarchy (Dominance Hierarchy) is a structured organization within a social group where individuals are ranked based on competitive encounters, establishing a clear order of access to limited resources like food, mates, and resting sites.

- **Pecking Order:** First described by Schjelderup-Ebbe (1921) in domestic hens. The top-ranked individual (α / alpha) dominates all others; the second-ranked (β / beta) yields only to the alpha; down to the lowest-ranked (ω / omega) individual, which yields to all group members.
- **Types of Hierarchies:**
 - *Linear Hierarchy:* A straightforward chain of command ($A \rightarrow B \rightarrow C \rightarrow D$).
 - *Despotic Hierarchy:* One single individual dominates all other group members, which share equal subordinate status below it (e.g., Wolf packs).
- **Adaptive Value:** Reduces energy costs and injury risks associated with continuous fighting. Once the hierarchy is established through initial aggressive interactions, rank is maintained using subtle visual or vocal threat displays, maintaining group stability.

Question 52.. Write short notes on: (A) Advantages of being social (B) Mating groups

(A) Advantages of Being Social

Sociality involves living in structured, cooperative groups. Group living provides several evolutionary advantages that offset the increased competition for resources:



- **Enhanced Anti-Predator Defenses:**
 - *Dilution Effect:* An individual's probability of being targeted by a predator decreases as group size increases ($\text{Risk} = 1/N$).
 - *Many-Eyes Hypothesis (Increased Vigilance):* Larger groups detect approaching predators faster because multiple individuals scan the environment, allowing each member to spend more time feeding.
 - *Selfish Herd & Confusion Effect:* Swarming, schooling, or herding movements visually confuse predators, making it difficult to isolate a single prey target.
 - *Communal Defense:* Groups mob larger predators collectively (e.g., Crows)

attacking an owl).

- **Increased Foraging Efficiency:**
 - Social predators engage in cooperative pack hunting to capture larger prey that an individual could not subdue alone (e.g., Lions, Wolves).
 - Information sharing at roosts or hives guides group members toward rich, patchy food sources.
- **Division of Labor and Cooperative Care:** Specialized castes handle reproductive, defensive, and foraging tasks (e.g., Eusocial ants, bees), increasing total colony efficiency.
- **Thermal Advantages:** Huddling in dense clusters conserves metabolic body heat during severe cold (e.g., Emperor Penguins).

(B) Mating Groups

Mating groups (Mating Systems) describe the structural social arrangements formed between males and females for reproduction. Mating systems are shaped by resource distribution and parental investment:

- **Monogamy:** One male pairs exclusively with one female during a breeding season or lifetime.
 - *Adaptive Context:* Favored when raising offspring requires intensive parental care from both parents (e.g., 90% of bird species).
- **Polygyny:** One male mates with multiple females during a breeding season.
 - *Resource-Defense Polygyny:* Males defend rich territories containing resources needed by females.
 - *Female-Defense Polygyny (Harems):* Males directly guard groups of females from rival males (e.g., Elephant Seals, Lions).
 - *Lek Polygyny:* Males aggregate in display arenas (leks), performing courtship displays. Females visit the lek, choose the top-performing male, mate, and leave to raise young alone (e.g., Black Grouse).
- **Polyandry:** One female mates with multiple males during a breeding season.
 - *Adaptive Context:* Rare system where sex roles are reversed. Females are larger and brightly colored, competing for territories while males incubate eggs and rear offspring (e.g., Jacanas, Phalaropes).
- **Promiscuity / Polygynandry:** Multiple males mate with multiple females without stable pair bonds.

Question 53.. Give an account of courtship behaviour.

1. Definition

Courtship behavior consists of a series of ritualized, species-specific behavioral interactions exchanged between a male and female prior to mating. It functions as a reproductive communication system that facilitates mate selection and fertilization.

2. Primary Biological Functions of Courtship

- **Species Recognition:** Ensures mating occurs exclusively between conspecifics, preventing energetic waste and sterile hybridization.

- **Gender and Readiness Assessment:** Confirms sex identity and synchronizes male and female physiological readiness (gamete maturation).
- **Reduction of Hostility:** Suppresses territorial aggressive or predatory responses, allowing partners to approach closely.
- **Mate Quality Assessment (Sexual Selection):** Allows females (which make a higher parental investment) to evaluate the genetic quality, health, and vigor of males based on display intensity, coloration, or gift size.

3. Sensory Channels in Courtship

Visual Displays	→ Bright breeding plumage, dances, and anatomical ornaments
Acoustic Songs	→ Species-specific calls advertising territory and quality
Chemical Signals	→ Pheromonal release indicating estrus and species identity
Tactile Contact	→ Rubbing, preening, and nudging to build pair tolerance
Nuptial Gifting	→ Providing food items to signal foraging capability

- **Visual Courtship Displays:** Involves bright breeding colors, structural ornaments, and stylized movements. Male birds-of-paradise display colorful plumage while performing complex acrobatic dances on cleared display grounds.
- **Acoustic Courtship:** Male songbirds sing complex, multi-phrase songs. Females prefer males with larger song repertoires, which correlate with age, brain development, and survival ability.
- **Chemical Courtship:** Female mammals release volatile estrus pheromones in urine, triggering the **Flehmen response** (lip curling) in males to sample chemical cues using the vomeronasal organ.
- **Tactile Courtship:** Mutual preening (allopreening) in birds or gentle head-rubbing in mammals lowers defensive anxiety, strengthening pair bonds.

4. Classical Example: The Three-Spined Stickleback

Described by Niko Tinbergen, the courtship of the Three-Spined Stickleback (*Gasterosteus aculeatus*) follows a chain of release reactions:

1. In spring, the male's belly turns bright red, and he constructs a tubelike plant nest.
2. When a gravid female with a swollen silvery belly enters his territory, her appearance (sign stimulus) triggers the male to perform a **zigzag dance**.
3. The female responds by swimming toward him in a head-up posture.
4. This posture stimulates the male to swim toward his nest, leading her to the entrance.
5. He taps her tail base with his snout (**tremble-thrusting**), triggering her to enter the nest and deposit eggs.
6. The female leaves the nest, and the male enters to fertilize the eggs, remaining to guard and fan them.

Question 54. Write short notes on: (A) Warning signals (B)

Neurophysiological methods in ethology

(A) Warning Signals

Warning signals (Aposematism) are conspicuous visual, acoustic, or olfactory signals displayed by unpalatable, toxic, or dangerous prey animals to advertise their defenses to potential predators.

- **Types of Warning Signals:**
 - **Aposematic Coloration:** High-contrast color combinations—typically red, yellow, orange, and black—that stand out against green foliage. *Example:* Poison Dart Frogs (*Dendrobatidae*) display vivid bright colors advertising skin alkaloids.
 - **Acoustic Warning Signals:** Sound warnings emitted when threatened. *Example:* Rattlesnakes vibrate tail rattles to produce a loud buzzing sound, warning large mammals to avoid stepping on them.
 - **Chemical Warning Signals:** Pungent odors released when disturbed. *Example:* Skunks raise their tails, displaying black-and-white stripes while releasing thiols.
- **Evolutionary Basis:** Predators learn to associate high-contrast warning patterns with bad taste or illness after a single unpleasant experience, avoiding similar-looking prey in the future.
- **Mimicry Complexes:**
 - *Batesian Mimicry:* A harmless, palatable species mimics the warning coloration of a genuinely toxic species (e.g., Harmless Viceroy butterfly mimicking the toxic Monarch).
 - *Müllerian Mimicry:* Two or more unpalatable, toxic species share a similar warning color pattern, reinforcing predator avoidance learning across both populations (e.g., Heliconius butterfly species).

(B) Neurophysiological Methods in Ethology

Neurophysiological methods analyze the neural circuits, brain structures, and physiological mechanisms that receive environmental stimuli, process information, and generate behavioral responses.

- **Key Techniques:**
 - **Stereotaxic Microelectrode Recording:** Fine electrodes implanted into specific brain regions record the electrical activity (action potentials) of single neurons while the animal performs specific behaviors. *Example:* Identifying "place cells" in the hippocampus that fire when an animal enters a specific spatial area.
 - **Electrical Brain Stimulation (EBS):** Micro-currents passed through implanted electrodes stimulate specific brain regions to evoke complete behavioral routines. *Example:* Stimulating the lateral hypothalamus in cats triggers predatory attack behavior, while stimulating the ventromedial nucleus induces rage displays.
 - **Brain Lesioning and Ablation:** Surgical removal, chemical destruction (using neurotoxins like kainic acid), or temporary inactivation (using local anesthetics) of specific brain nuclei to observe resulting behavioral deficits, helping map brain function.
 - **Neurochemical Microdialysis:** Collects extracellular fluid from specific brain regions in freely moving animals to measure real-time changes in neurotransmitter

- levels (e.g., dopamine, serotonin) during behavioral events.
- **Optogenetics:** Light-sensitive ion channel proteins (e.g., Channelrhodopsin) are inserted into targeted neuron populations using viral vectors. Fiber-optic cables then deliver precise pulses of blue light to activate or inhibit specific neural circuits instantaneously, directly controlling behavioral outputs.

Question 55

Q. Describe fixed action patterns.

1. Definition

A Fixed Action Pattern (FAP) is an innate, unlearned, highly stereotyped sequence of coordinated motor movements executed by an animal in response to a specific external triggering stimulus known as a **Sign Stimulus** or **Releaser**. The concept was formulated by ethologists Konrad Lorenz and Niko Tinbergen.

2. Core Properties of Fixed Action Patterns

Sign Stimulus (Releaser) → Innate Releasing Mechanism (IRM) → Fixed Action Pattern (FAP)

- **Stereotyped Structural Uniformity:** The motor sequence follows a fixed spatio-temporal pattern, showing minimal variation across performances by the same individual or across different members of a species.
- **Innate and Universal:** Fully expressed on the first exposure without prior learning, observation, or practice. Shared across all individuals of a particular sex and developmental stage within a species.
- **Triggered by Sign Stimuli:** Initiated by specific key stimuli (color, shape, sound, movement pattern).
- **Independence from Continuous Feedback (Ballistic Nature):** Once initiated by the Innate Releasing Mechanism (IRM), the motor sequence continues to completion even if the triggering stimulus is removed or modified midway.
- **Self-Exhausting:** Once completed, the threshold for re-triggering the same FAP rises temporarily due to a temporary reduction in action-specific energy.

3. Classic Examples of Fixed Action Patterns

A. Egg Retrieval in Greylag Geese

When a nesting female goose (*Anser anser*) sees an egg outside her nest (Sign Stimulus), she extends her neck, lowers her bill over the egg, and rolls it back into the nest using precise, balancing bill movements (FAP). If the egg is removed midway by an experimenter, she continues the bill-retracting motion all the way back to the nest.

B. Territorial Attack in Three-Spined Sticklebacks

During the breeding season, male sticklebacks establish territories. A male attacks any

approaching rival male. Experiments by Niko Tinbergen revealed that the red belly of the rival male acts as the sign stimulus. Males attack crude, misshapen wooden models with red undersides, while ignoring realistic stickleback models lacking red bellies.

C. Yawning in Mammals

In humans and other mammals, yawning is an involuntary FAP triggered by internal physiological cues (fatigue, oxygen levels) or visual sign stimuli (seeing another individual yawn). Once initiated, the inhalation, jaw stretch, and exhalation sequence runs to completion.

4. Neurophysiological Basis

FAPs operate through a conceptual neural architecture:

- **Sign Stimulus:** The environmental trigger.
- **Innate Releasing Mechanism (IRM):** A neuro-sensory filter within the central nervous system that recognizes the sign stimulus and removes tonic inhibition from motor centers.
- **Action Specific Energy:** Physiological drive or motivation that accumulates over time, lowering the stimulus threshold required to release the FAP.

Question 56

Q. Give detailed account of fighting behaviour in animals.

1. Introduction

Fighting behavior (Agonistic Behavior) encompasses all behavioral interactions associated with conflict between individuals, including aggressive displays, physical combat, threats, appeasement, and escape behaviors. Fighting rarely aims at destroying an opponent; instead, it serves as a mechanism to resolve competition over limited resources while minimizing injury risks.

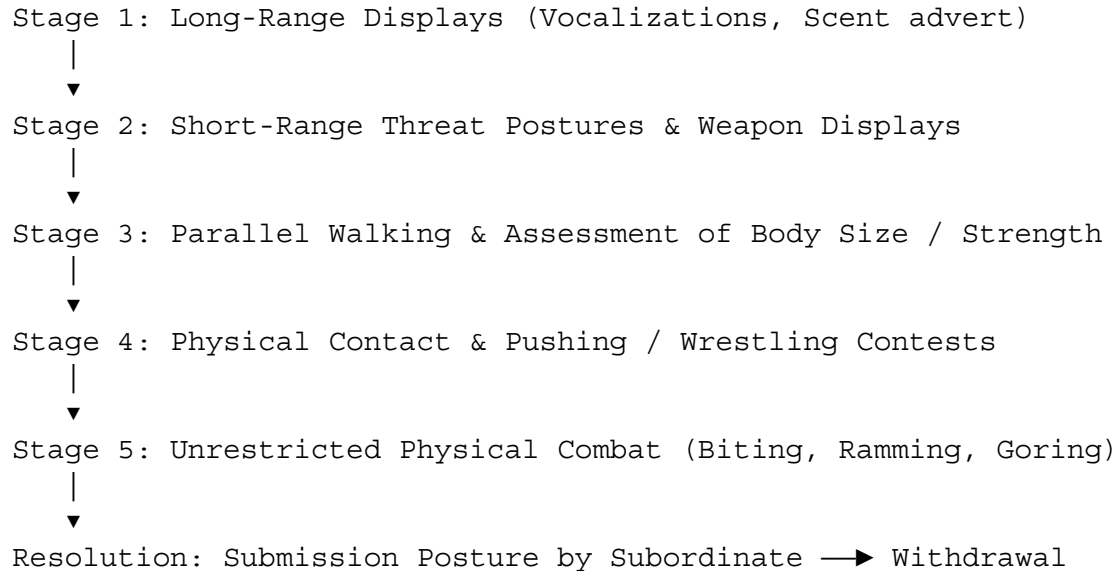
2. Causes of Fighting Behaviour

Animals engage in agonistic interactions over vital ecological and social resources:

- **Territorial Defense:** Defending home boundaries against conspecific intruders.
- **Resource Access:** Securing high-quality food patches or water holes during periods of scarcity.
- **Mating Rights:** Male-male competition to secure access to receptive females or harem control.
- **Establishment of Social Hierarchy:** Determining social rank within a newly formed or destabilized social group.
- **Parental Defense:** Protecting offspring from infanticidal conspecifics or predators.

3. Sequential Escalation Model of Fighting

Ethological studies show that fighting behavior follows a structured, ritualized sequence of stages. Contests escalate only if rivals are closely matched in resource holding power (RHP) or when resource value is extremely high:



- **Stage 1: Assessment via Threat Displays:** Competitors display anatomical weapons (canines, antlers, spurs) and increase apparent body size by raising fur, fins, or feathers. *Example:* Male Siamese fighting fish (*Betta splendens*) flare their opercular gill covers and extend fins to display body size.
- **Stage 2: Parallel Walking and Mutual Evaluation:** Rivals walk side-by-side along straight paths, evaluating body mass, muscle condition, and stamina without physical contact. *Example:* Red Deer stags (*Cervus elaphus*) engage in prolonged roar duets followed by parallel walking.
- **Stage 3: Ritualized Physical Contests:** Physical contact occurs using specialized, non-lethal locking mechanisms that measure physical strength rather than causing immediate injury. *Example:* Red Deer stags interlock antlers and push against each other; male Marine Iguanas engage in head-pushing contests.
- **Stage 4: Unrestricted Violent Combat:** Occurs when neither individual backs down during ritualized stages. Involves biting, goring, or slashing. Fights end when one contestant sustains an injury or becomes exhausted.
- **Stage 5: Resolution via Submission:** The losing individual adopts a stereotyped **submission posture** that inhibits further aggression from the victor. *Example:* A defeated wolf rolls onto its back, exposing its throat and belly; a defeated bird lowers its head and flattens its feathers. The subordinate then retreats, ending the contest.

Question 57. Give detailed account of social behaviour in animals.

1. Introduction

Social behavior encompasses all cooperative or structured interactions occurring between two or more individuals of the same species. It ranges from temporary aggregations around resources to complex, highly organized societies characterized by division of labor and reproductive altruism.

2. Levels of Social Organization

Ethologists categorize social structures based on complexity, group stability, and interaction levels:

- **Asocial / Solitary:** Individuals live independently, meeting conspecifics only briefly for mating (e.g., Tigers, Leopards).
- **Aggregations:** Temporary groupings formed passively around a localized environmental resource, lacking internal social structure (e.g., Moths around a light source, vultures at a carcass).
- **Simple Anonymous Groups:** Group members lack individual recognition but react collectively to alarm cues (e.g., Fish schools, bird flocks, wildebeest herds).
- **Individualized Societies:** Group members recognize each other individually, establishing dominance hierarchies, stable pair bonds, and social coalitions (e.g., Wolf packs, Elephant matriarchies, Primate troops).
- **Eusocial Societies:** The highest level of social organization, defined by three strict criteria:
 1. Cooperative care of immature offspring.
 2. Overlapping generations living within the same colony.
 3. Division of labor into reproductive and non-reproductive castes (e.g., Ants, Termites, Honey bees, Naked Mole-rats).

3. Evolutionary Mechanisms Driving Sociality

Living in groups carries costs (increased disease transmission, resource competition). Social behavior evolved because specific genetic and ecological mechanisms yield net fitness benefits:

A. Kin Selection and Inclusive Fitness (Hamilton's Rule)

Formulated by W.D. Hamilton. An individual can increase its total genetic representation in future generations not only through direct reproduction, but also indirectly by helping close genetic relatives survive and reproduce. Hamilton's Rule states that an altruistic trait evolves when:

$$r \cdot B > C$$

(Where B = Benefit to recipient, C = Cost to altruist, r = Coefficient of genetic relatedness). In Hymenoptera (bees, ants), females are haplodiploid ($r = 0.75$ between full sisters), predisposing them to help raise sisters rather than reproduce directly.

B. Reciprocal Altruism

Proposed by Robert Trivers. Altruistic actions evolve between non-relatives if the favor is returned in the future under three conditions: long group association, high memory capacity, and active punishment of cheaters. *Example:* Vampire bats (*Desmodus rotundus*) regurgitate blood meals to feed starving nestmates that failed to forage successfully, expecting reciprocal feeding when roles are reversed.

4. Communication Systems in Social Groups

Social cohesion depends on multi-modal communication channels:

- *Chemical:* Pheromones maintain caste order in ant colonies.
- *Acoustic:* Complex alarm call systems (e.g., Vervet monkeys use distinct alarm calls for

- eagles, snakes, and leopards).
- *Tactile*: Allogrooming in primates cleans parasites, lowers heart rates, and reinforces social alliances.

Question 58. What are important characteristics of instinctive and learned behaviour? Explain briefly with suitable examples.

1. Introduction

Animal behavior represents the functional output of sensory processing and neural integration. Behaviors are broadly divided into instinctive (innate) behaviors, which are genetically hardwired, and learned behaviors, which are modified through individual experience.

2. Characteristics and Examples of Instinctive Behaviour

Instinctive behavior is non-learned, genetically programmed behavior that develops independently of specific environmental experience.

- **Key Characteristics:**
 - *Heritable*: Transmitted from parents to offspring via DNA code.
 - *Stereotyped*: Performed in a highly uniform, predictable manner across all individuals of a species.
 - *Intrinsic*: Expressed fully even in animals raised in complete isolation from conspecifics (deprivation experiments).
 - *Inflexible*: Resistant to modification; executed even in inappropriate contexts.
 - *Species-Specific*: Universal among members of a species of the same age and sex.
- **Examples:**
 - *Web Spinning in Spiders*: Young orb-weaver spiders (*Araneus*) construct complex, geometrically precise webs on their very first attempt without observing adult spiders.
 - *Pecking in Chicks*: Newly hatched herring gull chicks peck instinctively at the red spot on their parent's bill to induce food regurgitation.
 - *Cocoons in Moths*: Silkworm caterpillars spin intricate protective silk cocoons before pupation without prior training.

3. Characteristics and Examples of Learned Behaviour

Learned behavior involves persistent modifications of behavior resulting from individual experience, observation, practice, or trial-and-error interactions with the environment.

- **Key Characteristics:**
 - *Acquired*: Developed during an individual's lifetime through environmental interaction.
 - *Flexible and Adaptive*: Modified as environmental conditions change.
 - *Extrinsic*: Dependent on external stimuli, practice, and social exposure.
 - *Variable*: Differs among individuals within the same population based on distinct life experiences.
 - *Subject to Extinction*: Learned associations fade if not reinforced over time.
- **Examples:**

- *Habituation in Birds*: Scarecrows placed in crop fields initially frighten crows. Over time, as crows learn the scarecrow presents no actual harm, they habituate and ignore it.
- *Tool Use in Chimpanzees*: Young chimpanzees learn to strip leaves off twigs to "fish" for termites from mounds by observing older group members over years of practice.
- *Song Dialects in White-Crowned Sparrows*: Young male sparrows possess an innate template for bird song, but must hear the specific acoustic song dialect of their local adult population during a critical window to sing a complete, correct adult song.

Question 60. Describe in detail the important methods used for study of animal behaviour in wild environment.

1. Introduction

Studying animal behavior in natural environments (field ethology) provides ecologically valid data on how behaviors function in real-world contexts. Observing wild animals requires non-invasive, systematic, and quantitative methodologies designed to record behaviors without altering natural animal routines.

2. Direct Observational Sampling Methods

Developed by ethologists such as Jeanne Altmann (1974), standardized sampling techniques eliminate observational bias:

- **Focal Animal Sampling:**
 - *Method*: The observer selects a single individual (focal subject) and records all its behavioral actions, vocalizations, and social interactions for a pre-determined duration (e.g., 30 minutes).
 - *Use*: Provides detailed individual time budgets and social network data.
- **Instantaneous (Scan) Sampling:**
 - *Method*: The observer scans an entire group of animals at fixed time intervals (e.g., every 10 minutes) and records the immediate activity state of each individual at that exact moment.
 - *Use*: Measures group activity patterns (e.g., percentage of time spent foraging versus resting).
- **All-Occurrences (Behavioral) Sampling:**
 - *Method*: The observer monitors an entire group and records every instance of a specific, rare behavioral event (e.g., fights, matings, infanticide), noting the identities of participating individuals.

3. Remote Sensing and Technological Tracking Methods

Modern technology enables continuous observation of wild animals across dark, subterranean, or vast geographical landscapes:

Camera Trapping → Infrared motion-sensor capture of elusive wildlife
 GPS Telemetry → Real-time satellite tracking of territorial &

migratory paths

Acoustic Monitoring ► Sub-surface / Canopy bioacoustic recording arrays

Drones & UAVs → High-resolution aerial monitoring of herd dynamics

- **Infrared Camera Trapping:** Motion- and heat-activated digital cameras installed along game trails or waterholes record natural nocturnal behaviors, social structure, and population densities without human presence.
- **GPS Telemetry and Satellite Tracking:** Lightweight collars, leg bands, or backpacks equipped with GPS receivers transmit spatial coordinates to satellites. Used to map home ranges, daily movement velocities, and continental migratory flyways in mammals, birds, and marine turtles.
- **Bioacoustic Recording Arrays:** Omnidirectional microphone arrays installed in rainforest canopies or marine hydrophone networks record vocalizations (e.g., whale calls, primate duets), identifying species presence, vocal repertoire, and spatial locations.
- **Unmanned Aerial Vehicles (Drones):** Drones equipped with high-definition and thermal cameras capture group dynamics, herd movements, and nesting colony distributions from above without disturbing ground vegetation.

4. Non-Invasive Biomarker and Marking Methods

- **Individual Identification Systems:**
 - *Natural Markings:* Using unique natural color patterns, scar shapes, or fin notches (e.g., stripe patterns in zebras, fluke patterns in whales) for non-invasive individual tracking via photographic identification databases.
 - *PIT Microchips:* Inserting tiny radio-frequency identification (RFID) transponders under the skin of wild animals captured briefly, allowing automatic identification when individuals pass near antennae-equipped nest boxes or feeding stations.
- **Non-Invasive Endocrinology:** Collecting fresh feces, urine, or shed hair from wild animals to measure excreted steroid metabolite levels (cortisol for stress; estrogen/testosterone for reproductive state), correlating physiological states directly with observed behaviors.

Question 61.. Give a detailed account on migratory behaviour of birds. Add a note on theories proposed for bird navigation.

1. Introduction

Bird migration is an adaptive, seasonal, and two-way directional journey undertaken by bird species between their northern breeding grounds and southern non-breeding wintering grounds. Migration allows birds to exploit seasonal peaks in food availability and long daylight hours while avoiding winter weather extremes.

2. Types of Bird Migration

Bird migration varies based on geographical vectors, distance, and timing patterns:

- **Latitudinal Migration:** North-South movement between cold temperate/arctic regions

and warm tropical/equatorial regions (e.g., Arctic Tern traveling between Arctic breeding grounds and Antarctic wintering seas).

- **Longitudinal Migration:** East-West movement between continental interiors and coastal zones (e.g., Starling migrations in Europe).
- **Altitudinal (Vertical) Migration:** Movement up and down mountain slopes. Birds breed at high elevations during summer and descend to sheltered valleys during winter (e.g., Himalayan Monal, Wallcreeper).
- **Loop Migration:** Birds follow different geographical routes during their autumn and spring journeys to take advantage of seasonal prevailing wind patterns (e.g., Red-backed Shrike).
- **Partial Migration:** Some populations within a species migrate while others remain resident, depending on local resource availability.

3. Physiological Preparations for Migration

Migration is preceded by complex physiological changes regulated by photoperiod and neuroendocrine activity:

- **Hyperphagia and Fat Deposition:** Stimulated by prolactin and corticosterone, birds consume vast quantities of food, storing fat under their skin and around visceral organs. Fat serves as an energy-dense fuel (9.3 kcal/g) compared to 4.1 kcal/g for protein/carbohydrate).
- **Zugunruhe (Migratory Restlessness):** Caged migratory birds display intense nocturnal restlessness and orientation fluttering toward their migratory path, driven by elevated pituitary hormones.

4. Theories Proposed for Bird Navigation

Birds navigate thousands of kilometers across unfamiliar terrain using an integrated, redundant orientation system. Several scientific theories explain bird navigation:

Sun Compass Theory → Azimuth tracking with Circadian Clock compensation
Star Compass Theory → Rotation center (Polaris) & Constellation learning
Geomagnetic Theory → Cryptochrome Radical Pairs & Beak Magnetite receptors
Olfactory Map Theory → Atmospheric trace odor gradient mapping

A. Sun Compass Theory (Gustav Kramer)

Diurnal migrants determine compass direction using the horizontal position of the sun (azimuth). Because the sun moves 15° per hour across the sky, birds use their internal circadian biological clock to compensate for time changes, maintaining a constant compass heading throughout the day.

B. Star Compass Theory (Stephen Emlen)

Nocturnal migrants navigate using night star patterns. Experiments in planetarium tents (Emlen

Funnels) demonstrated that young birds learn the rotational center of the night sky (Polaris) and surrounding constellations during early development, using these star patterns as a directional orientation compass.

C. Geomagnetic Navigation Theory

Birds detect the Earth's magnetic field strength and inclination angle. Two complementary biophysical mechanisms explain this ability:

1. **Radical Pair Mechanism (Photoreception):** Blue light entering the bird's eye excites **Cryptochrome** proteins in retinal ganglion cells, creating radical pair electrons. This generates visual patterns of light and shade overlaid on the bird's visual field, corresponding to magnetic field lines.
2. **Magnetite Receptor Mechanism:** Microscopic iron oxide (**magnetite**) crystals embedded in the tissue of the upper beak connect to the ophthalmic branch of the trigeminal nerve, acting as a magnetic intensity map reader.

D. Olfactory Map Theory (Florian Papi)

Pigeons and seabirds construct an olfactory map of their home region by learning trace atmospheric odor gradients carried by regional winds. When displaced into unfamiliar territory, they smell local air currents and fly in the direction that returns them to familiar scent profiles.

E. Infrasound and Visual Landmark Theories

Birds supplement celestial and magnetic orientation by tracking low-frequency acoustic waves (infrasound generated by ocean waves and mountain wind currents) and major topographic features like rivers, coastlines, and mountain ranges.